

Aerodynamic Considerations in Tactical Missile Design

2.1 Introduction

This chapter addresses aerodynamic considerations for tactical missile design. The conceptual design methods, design tradeoffs, design criteria, and technologies emphasize low-aspect ratio wing and wingless configurations. Consideration is given to missile diameter tradeoffs; nose fineness tradeoffs; boattail considerations; lifting body versus axisymmetric body; calculation of body and surfaces normal force, lift-to-drag ratio, zero-lift drag coefficient, and center-of-pressure location; wings versus no wings; surface planform geometry alternatives; flight control alternatives; maneuver alternatives; roll orientation; static stability; tail area sizing; stability and control conceptual design criteria; and body buildup aerodynamics.

As shown in Fig. 2.1, aerodynamic configuration synthesis follows the initial activities of defining mission requirements and establishing a baseline missile. Aerodynamic configuration synthesis requires consideration of alternative configurations, aerodynamic technology, and the process for resizing the missile. The output of aerodynamic configuration synthesis includes the missile diameter, length, nose geometry, wing size and geometry, stabilizer and control surfaces size and geometry, and the aerodynamic coefficients and derivatives. The output of the aerodynamic configuration synthesis is an input to the propulsion system design.

2.2 Missile Diameter Tradeoff

Missile diameter tradeoff has drivers toward a small diameter as well as drivers toward a large diameter. These must be harmonized. The drivers toward a small diameter missile include lower drag and smaller lateral dimensions for improved launch platform compatibility. However these must be harmonized with the other drivers toward a large diameter missile. These include 1) increased range, higher resolution, better tracking, and higher signal-to-noise for the seeker; 2) increased warhead effectiveness from higher fragment velocity and higher blast overpressure for a blast fragmentation warhead or a longer and faster jet for a shaped charge warhead; 3) increased body bending frequency, which reduces aeroelasticity and improves flight control performance; 4) improved subsystem packaging for diameter limited subsystems; and 5) shorter length for launch platform compatibility.

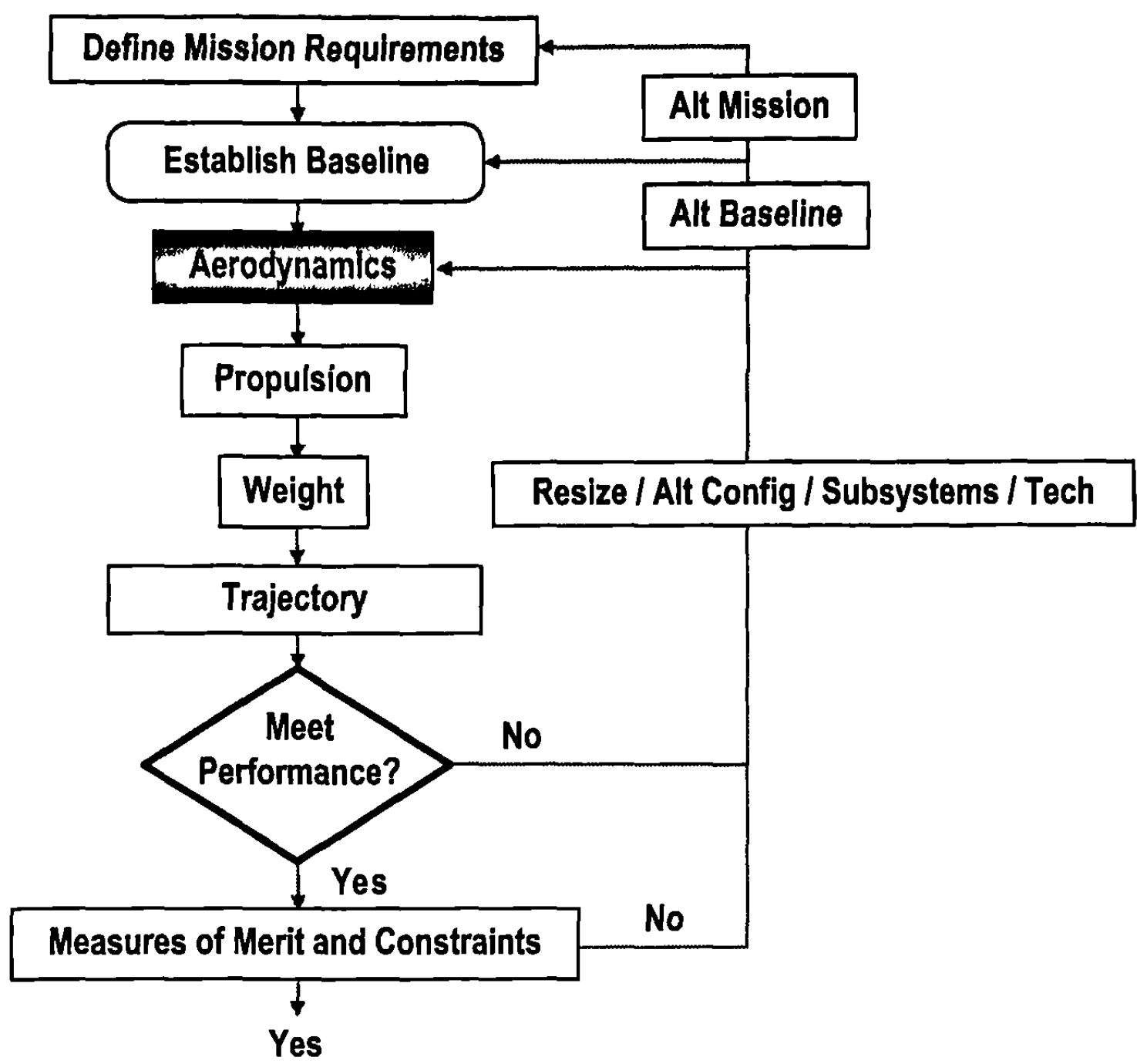


Fig. 2.1 Missile concept synthesis requires evaluation of alternatives and iteration.

The typical range in missile body fineness ratio (length-to-diameter ratio) is

$$5 < l/d < 25 \tag{2.1}$$

An example of a low-fineness missile is the Javelin man-portable anti-armor missile, which has an $l/d = 42.6 \text{ in}/5 \text{ in} = 8.5$. Javelin is a small subsonic missile with relatively low drag. An example of a high-fineness missile is the AIM-120 AMRAAM supersonic air-to-air missile, which has an $l/d = 144 \text{ in}/7 \text{ in} = 20.5$.

Drag is a major design parameter in satisfying the flight range requirement of tactical missiles, especially supersonic missiles. It is a function of drag coefficient, dynamic pressure, and reference area, given by the equation

$$D = C_D q S_{\text{Ref}} \tag{2.2}$$

For a missile configuration, the reference area is the body cross-sectional area. Substituting missile diameter for a typical circular cross section gives

$$D = C_D q (\pi/4) d^2 = 0.785 C_D q d^2 \tag{2.3}$$

Shown in Fig. 2.2 is D/C_D as a function of diameter and dynamic pressure for a typical range in the values of these parameters for tactical missiles.

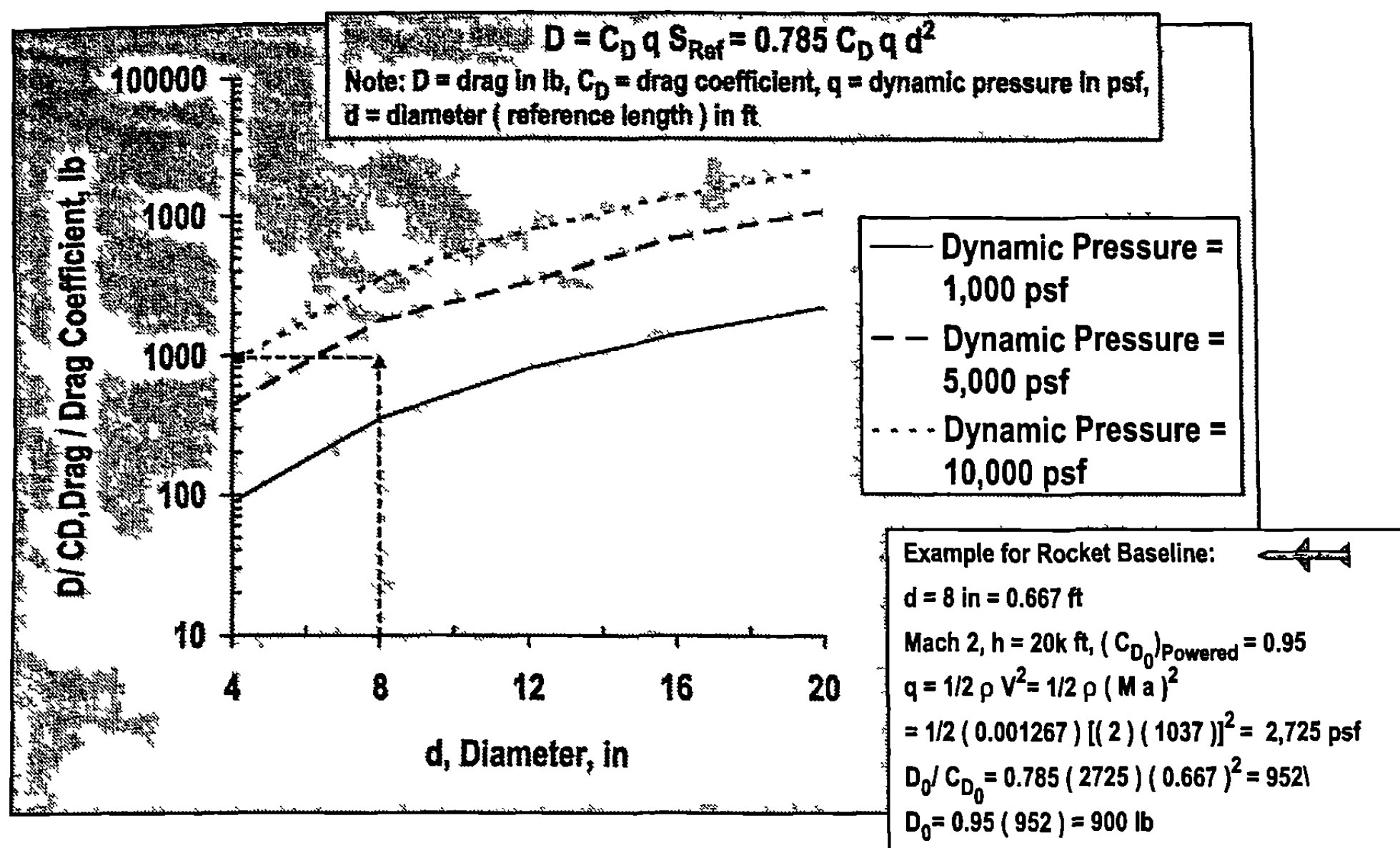


Fig. 2.2 Small diameter missiles have low drag.

As an example of missile drag, consider the rocket baseline missile. The rocket baseline missile has a reference area of 0.349 ft^2 and a reference length (diameter) of 8 in. At a typical flight condition of Mach 2 and flight altitude of 20,000 ft, the dynamic pressure $q = 2725 \text{ psf}$. Also, at Mach 2, the rocket baseline zero-lift drag coefficient for powered flight $C_{D_0} = 0.95$. Substituting into the drag equation, the zero-lift powered flight drag of the rocket 8-in diameter baseline is $D_0 = 900 \text{ lb}$. A missile with configuration geometry similar to the rocket baseline, but with one-half the diameter (4 in) would have one-fourth the drag (225 lb). A similar configuration with twice the diameter (16 in) would have four times the drag (3600 lb).

Figure 2.3 shows the benefit of a large diameter in enhancing seeker detection range and resolution. It is based on the two-way radar range equation for an active radfar seeker, given by

$$R_D = \{ \pi \sigma n^{3/4} / [64 \lambda^2 k T B F L (S/N)] \}^{1/4} P_T^{1/4} d \quad (2.4)$$

Assumptions are a uniformly illuminated circular aperture, noncoherent radar with integration of only the pulse signal amplitude, negligible clutter, negligible interference, negligible atmospheric attenuation, and thermal noise-limited receiver sensitivity. In regard to pulse integration, the efficiency of integrating the number of pulses (n) can vary from $n^{1/2}$ to n . Coherent radar, which integrates both the signal amplitude and phase, can have an integration efficiency approaching an ideal value of n^1 . For a noncoherent radar, which integrates only the signal amplitude, a typical value of the integration efficiency is $n^{3/4}$. Although a coherent radar seeker is usually more efficient, most tactical radar missiles use noncoherent seekers, which integrate only the pulse amplitude. A noncoherent

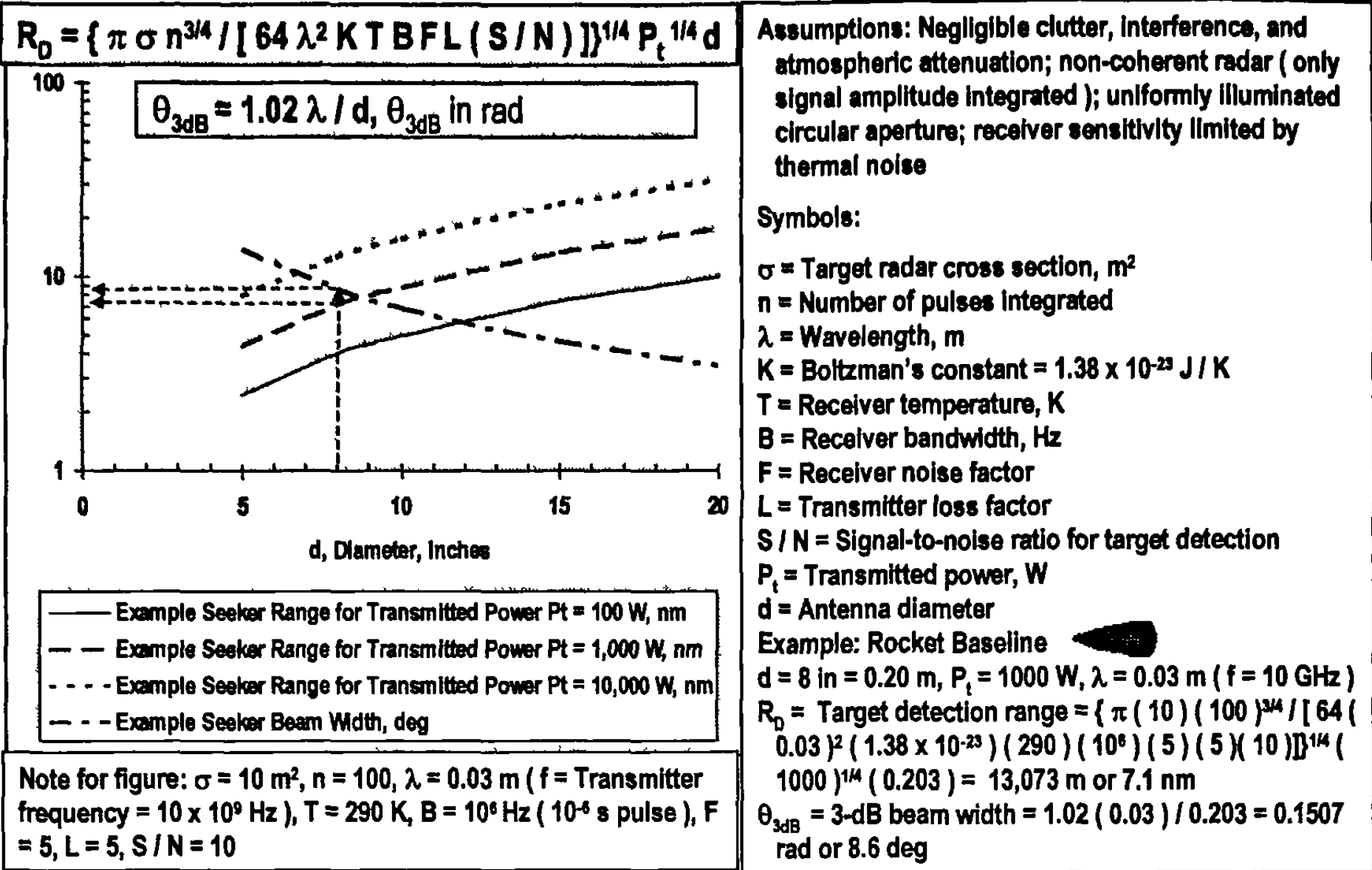


Fig. 2.3 Large diameter radar seeker provides longer detection range and better resolution.

seeker is usually preferred for tactical missiles because it is lower cost. Referring back to the radar range equation of the figure, the receiver threshold sensitivity P_r (in Watts) is given by the equation $P_r = k T B F$. Assumptions are a nominal receiver temperature $T = 290 \text{ K}$, bandwidth $B = 10^6 \text{ Hz}$ (pulse width of about 10^{-6} s), and noise factor $F = 5$. Boltzman's constant is the term $k = 1.38 \times 10^{-23} \text{ J/K}$. Substituting,

$$P_r = 1.38 \times 10^{-23} (290) (10^6) (5) = 2.0 \times 10^{-14} \text{ W}$$

(2.5)

Other assumptions for the example plotted in the figure are a target radar cross section $\sigma = 10 \text{ m}^2$, number of pulses integrated $n = 100$, transmitter frequency $f = 10 \text{ GHz}$, transmitter loss factor $L = 5$, and target detection signal-to-noise ratio $S/N = 10$. The assumed value of $S/N = 10$ provides a false alarm of about 1% for a probability of detection of 90%. For a typical transmitter duty cycle of 20% and a loss factor of 5, the average output power is about 20% of the peak power and the input power is comparable to the peak power. The maximum allowable transmitted power by the seeker is limited by the transmitter technology, power available from the battery, maximum allowable temperature because of heating, and signal processing technology (such as pulse compression).

As an example, assume that the rocket baseline missile has an antenna diameter of about $d = 8 \text{ in}$ and a transmitted power $P_T = 1000 \text{ W}$. Detection range is computed to be $R_D = 13,073 \text{ m}$ or 7.1 n miles.

Beamwidth of a radar seeker is given by the equation

$$\theta_{3\text{ dB}} = 1.02\lambda/d \quad (2.6)$$

The 3-dB beamwidth for a frequency of 10 GHZ and a diameter of 8 in is

$$\theta_{3\text{ dB}} = 1.02(0.03)/0.203 = 0.151 \text{ rad or } 8.6 \text{ deg} \quad (2.7)$$

A larger antenna diameter or a higher transmission frequency provides longer range, more precise tracking, and finer resolution. For example, an increase of 100% in diameter (to 16 in) provides an increase in the seeker range of 100% (to 14.1 n miles) and a decrease in the seeker beamwidth of 50% (to 4.3 deg). Also, a longer pulse width would provide longer detection range, with disadvantages of degraded range measurement accuracy, higher false alarm rate, and potentially receiving overlapping pulses. Pulse modulation (e.g., chirp) can be used to avoid range ambiguity for a long duration pulse. Pulse compression is another approach to improve range. Processing provides the equivalence of an increase in the pulse width. Pulse compression has advantages of not degrading the range measurement accuracy, the pulse nonambiguity, or the false alarm rate.

There is a similar benefit of a large diameter in enhancing the detection range and resolution of a passive infrared (IR) seeker. It is based on the one-way IR range equation

$$R_D = \{(I_T)_{\Delta\lambda} \eta_a A_o \{D^*/(\Delta f_p^{1/2} A_d^{1/2})\} (S/N)^{-1}\}^{1/2} \quad (2.8)$$

As shown in the equation, there are seven drivers for IR seeker range: radiant intensity of the target within the upper and lower wavelength cutoffs of the seeker $(I_T)_{\Delta\lambda}$, atmospheric transmission efficiency for the total distance from the target to the seeker η_a , seeker optics aperture area A_o , pixel specific detectivity D^* , pixel bandwidth Δf_p , detector total area A_d , and signal-to-noise ratio required for detection S/N .

Referring to the seeker range equation, the second term is atmospheric transmission efficiency η_a . It accounts for the attenuation over the total distance from the target to the seeker. The transmission can range from perfect transmission ($\eta_a = 1$) for exo-atmospheric conditions to almost no transmission ($\eta_a \approx 0$) through clouds. The effect of atmospheric transmission on detection range is shown in Fig. 2.4. A typical value of atmospheric humidity of 7.5 g/m³ results in an atmospheric attenuation of -0.2 dB per km for an IR signal, or an atmospheric transmission of 0.955 per km. Increasing the range reduces the received signal. As an example, an atmospheric attenuation of 0.955 per km results in 95% transmission at a range of 1 km but only 63% transmission at a range of 10 km ($0.955^{10} = 0.63$). Adverse weather results in less transmission. As an example, moderate rainfall (4 mm/h) has an atmospheric attenuation of -2 dB per km for an IR signal, or an atmospheric transmission of 0.631 per km. It reduces the transmission to 63% at a range of 1 km. There is only 1% transmission at a range of 10 km.

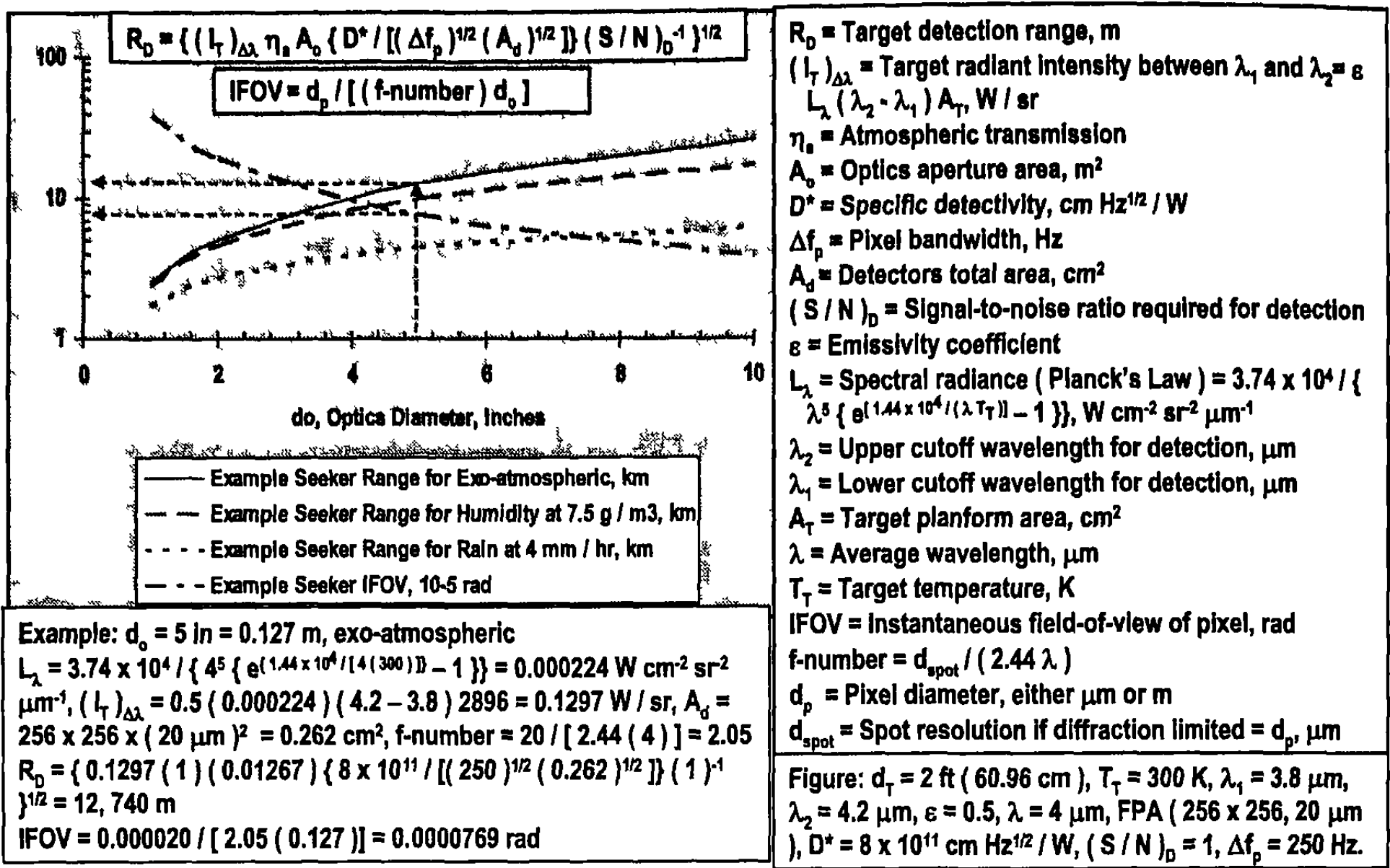


Fig. 2.4 Large diameter IR seeker provides longer detection range and better resolution.

The third term in the seeker range equation is seeker optics aperture area, which is a major configuration design parameter. A larger aperture delivers more optical energy to the detectors, providing longer detection range. It also provides better resolution. For a strap-down seeker, the seeker optics diameter can be almost as large as the diameter of the missile. For a gimbaled seeker, the seeker optics diameter is limited to about one-half of the missile diameter. Shown in the figure is the effect of seeker optics diameter on detection range. A small optics diameter of 1 in, representative of a small diameter missile such as a Stinger, has a detection range of only a few km and the detection range can be less than 1 km against a low-signature target in adverse weather. A larger IR sensor diameter of 10 in, such as a FLIR on an aircraft or unmanned air vehicle (UAV), has ranges that can greatly exceed 10 km against moderate signature targets in favorable weather.

As an example, the seeker detection range will be calculated for a 5 in diameter aperture in an exo-atmospheric intercept. The seeker is assumed to operate over a wavelength range of 3.8 to 4.2 μm, with an average wavelength of $\lambda = 4 \mu\text{m}$. The target characteristics are assumed to be 2 ft diameter ($A_T = 2896 \text{ cm}^2$), temperature $T_T = 300 \text{ K}$, and emissivity $\epsilon = 0.5$. The spectral radiance (Planck's law) of the target is

$$\begin{aligned} L_\lambda &= 3.74 \times 10^4 / \{ \lambda^5 \{ e^{[1.44 \times 10^4 / (\lambda T_T)]} - 1 \} \} \\ &= 3.74 \times 10^4 / \{ 4^5 \{ e^{(1.44 \times 10^4 / [4(300)])} - 1 \} \} \\ &= 0.000224 \text{ Wcm}^{-2} \text{ sr}^{-2} \mu\text{m}^{-1} \end{aligned} \tag{2.9}$$

Substituting into the equation for target radiant intensity gives

$$\begin{aligned}(I_T)_{\Delta\lambda} &= \varepsilon L_\lambda(\lambda_2 - \lambda_1)A_T = 0.5(0.000224)(4.2 - 3.8)(2896) \\ &= 0.1297 \text{ Wsr}^{-1}\end{aligned}\quad (2.10)$$

The detector is assumed to be a diffraction limited 256×256 focal plane array with $20 \mu\text{m}$ pixels. Detector area is therefore $A_d = 256(256)(0.002) = 0.262 \text{ cm}^2$. A $\text{Hg}_{0.67}\text{Cd}_{0.33}\text{Te}$ detector is assumed, which has a specific detectivity $D^* = 8 \times 10^{11} \text{ cmHz}^{1/2}\text{W}^{-1}$ at a detector temperature of 77 K (liquid nitrogen) when it is optimized for the $\lambda = 4 \mu\text{m}$ application. The pixel detector bandwidth for best signal-to-noise ratio (S/N) is assumed to be $\Delta f_p = 250 \text{ Hz}$ and the S/N for detection is assumed to be $(S/N)_D = 1$. Finally, substituting into the IR seeker range equation gives

$$\begin{aligned}R_D &= \{(I_T)_{\Delta\lambda} \eta_a A_o \{D^*/(\Delta f_p^{1/2} A_d^{1/2})\} (S/N)^{-1}\}^{1/2} \\ &= \{0.1297(1)(0.01267)\{8 \times 10^{11}/[(250)^{1/2}(0.262)^{1/2}]\}(1)^{-1}\}^{1/2} \\ &= 12,740 \text{ m}\end{aligned}\quad (2.11)$$

The figure also shows the benefit of optics diameter on instantaneous field of view (IFOV), which is the field of view of a single pixel. The equation is

$$\text{IFOV} = d_p/[(f\text{-number})d_o] \quad (2.12)$$

Note from the preceding equation that the IFOV becomes smaller (better resolution) as the diameter increases.

A diffraction limited system is assumed, with the minimum spot size equal to the size of a single pixel detector. The Rayleigh criterion for the limiting angular resolution of the optics results in the equation

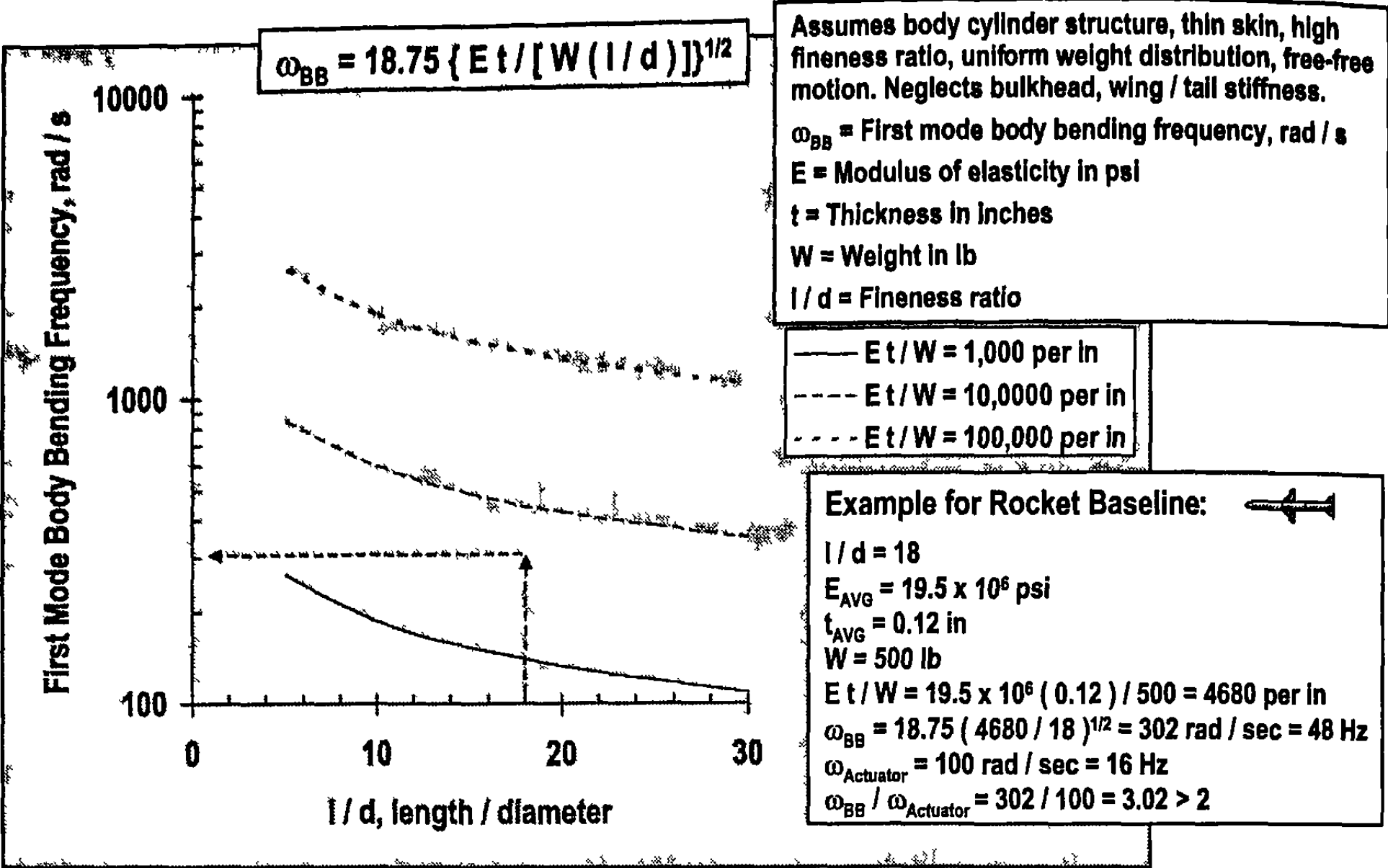
$$f\text{-number} = d_p/(2.44 \lambda) \quad (2.13)$$

As an example, we will assume an optics aperture diameter $d_o = 5 \text{ in}$ (12.7 cm) diameter, a system that is diffraction limited, pixel size $d_p = 20 \mu\text{m}$, and wavelength $\lambda = 4 \mu\text{m}$. The f-number is given by the equation $f\text{-number} = 20 \mu\text{m}/[2.44(4 \mu\text{m})] = 2.05$. Substituting into the equation for instantaneous field of view,

$$\text{IFOV} \approx 20 \times 10^{-6}/[2.05(0.127)] = 0.0000769 \text{ rad} \quad (2.14)$$

Missile fineness ratio is a design consideration that may limit the missile diameter. Figure 2.5 illustrates that high-fineness missiles may be limited by aeroelasticity problems from the body bending frequency. Shown is the nondimensional first mode body bending frequency as a function of body fineness ratio. The equation is

$$\omega_{BB} = 18.75\{Et/[W(l/d)]\}^{1/2} \quad (2.15)$$



Adding a stiff bulkhead, such as a welded interface between rocket motor case and the airframe, usually adds stiffness and increases the body bending frequency. However, a bulkhead interface using fittings may have a poor mechanical attachment and could decrease the stiffness, resulting in lower body bending frequency.

2.3 Nose Fineness Tradeoff

Shown in Fig. 2.6 are the nose fineness tradeoff drivers toward high-fineness ratio versus low-fineness ratio. These must be harmonized. A high-fineness nose, such as $l_N/d = 5$, is ideal aerodynamically and has low observables. A low-fineness nose, such as an $l_N/d = 0.5$ spherical nose, is ideal electromagnetically and allows more propellant length and volume for a length-limited missile. For supersonic missiles, a moderate nose fineness compromise dome is a nose length-to-diameter ratio of about 2.

Alternative dome geometry includes tangent ogive, faceted or pyramidal, flat window, and multilens domes. A tangent ogive dome has a circular arc surface. Although a tangent ogive dome has lower supersonic drag and low radar cross section, the curved surface results in high error slope for a seeker tracking the target. An alternative dome is a faceted dome. A faceted dome is typically composed of six to eight triangular surfaces that form a pyramid. A faceted dome has the advantage of low error slope because of the flat surfaces. The bond line between the facets must be small to minimize optical distortion. A disadvantage of a faceted dome for an IR seeker is the solar energy reflection off the facets into the seeker. A third type of dome is a window dome composed of a single flat surface. The window dome also has an advantage of low dome error slope.

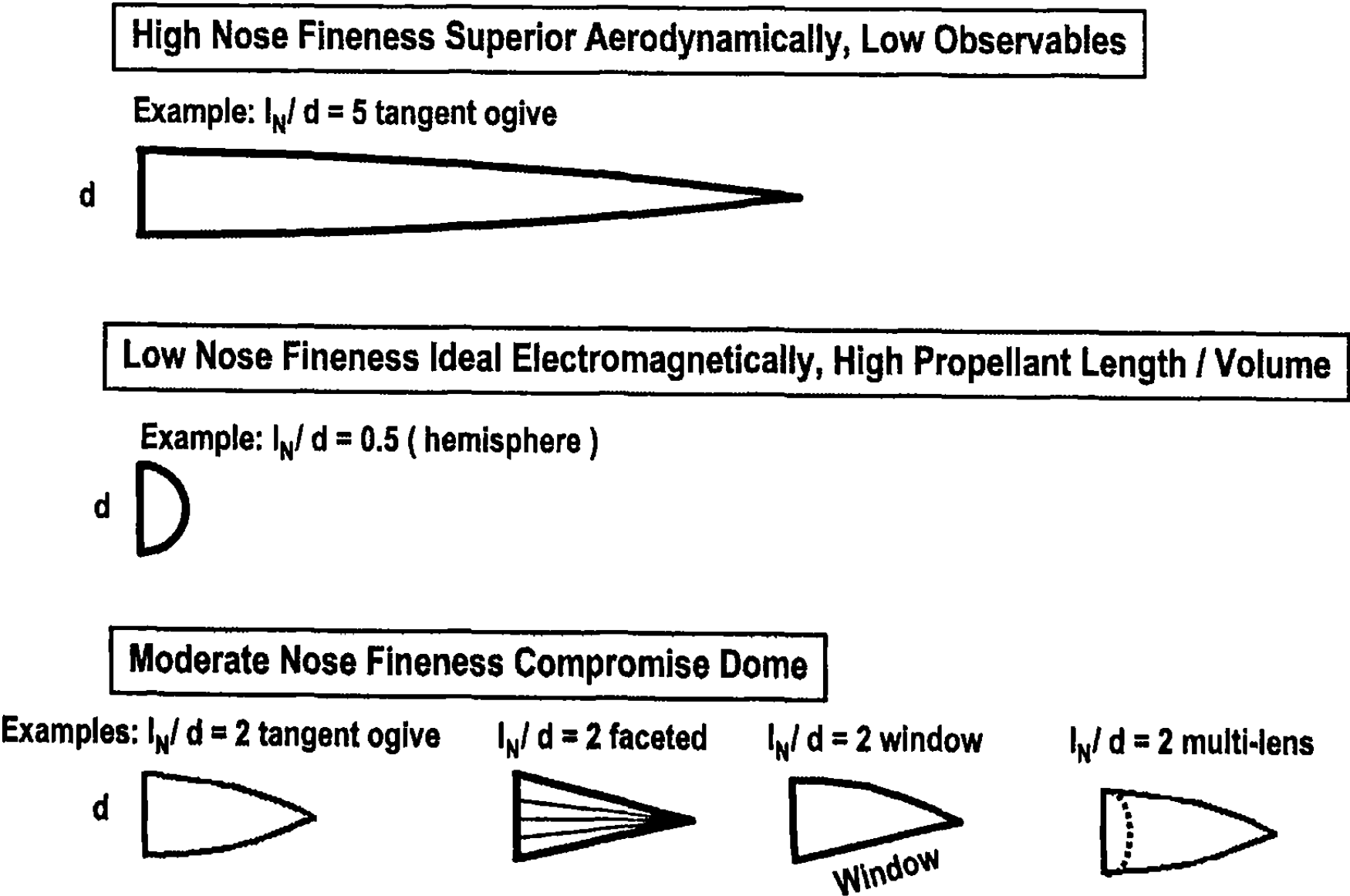


Fig. 2.6 Nose fineness tradeoff.

A disadvantage of a window dome is that it limits the field of regard of the seeker, which may limit the ability of the seeker to acquire an off-boresight target. Finally, a multilens dome is composed of two or more concentric domes, with a high-fineness outer dome. A multilens dome can provide compensation that reduces the dome error slope. The interior dome(s) provide optical correction to the dome error slope induced by the low drag/low radar cross section (RCS) exterior dome.

Shown in Fig. 2.7 are current missiles with IR domes that are high-fineness ratio and also have low dome error slope. Faceted dome technology has been applied to supersonic surface-to-air and air-to-air missiles to provide low drag combined with low dome error slope. Two supersonic missiles that have a high-fineness faceted dome are the United Kingdom Firestreak and the French Mistral. A concern with faceted domes is multiple reflections of the facets. Background reflections into the detector increase the background clutter, reducing target contrast. Solar reflections off the facets can result in false “hot spots.” A video of a sea-launched application of Mistral is in the upper right section of the figure. Faceted and window dome technology has also been applied to cruise missiles to provide low radar cross section combined with low dome error slope. Cruise missiles that have high-fineness IR domes are the U.S. SLAM-ER missile and the JASSM missile. SLAM-ER has a two-facet IR dome. JASSM has a window dome. The JASSM seeker has a limited field of regard and is biased downward to look through the window. Finally, the U.S. THAAD anti-missile missile dome is a high-fineness window for low drag and low dome error slope. Because the THAAD seeker is biased to look through the side window, there is a limited field of regard. Shown in the bottom right

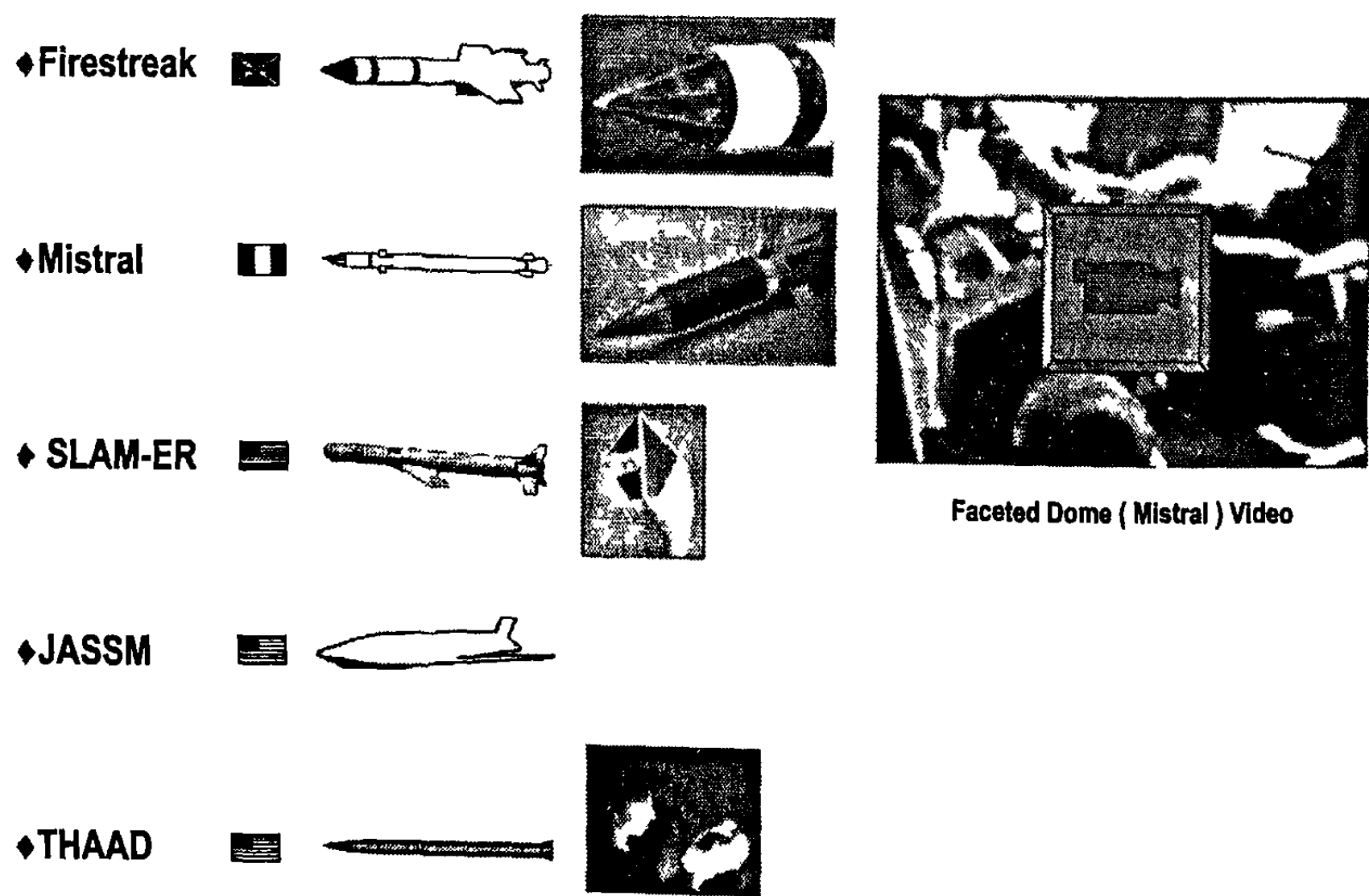


Fig. 2.7 Faceted and flat window domes can have low dome error slope, low drag, and low RCS. Figure also in color section. Video is in the CD included with this textbook.

section of the figure is a photograph of developmental hardware for a cooled window dome similar to THAAD.

The total zero-lift drag coefficient of a body is the combination of the skin friction drag, base drag, and wave drag (if supersonic). It is given by the equation

$$(C_{D_0})_{\text{Body}} = (C_{D_0})_{\text{Body,Friction}} + (C_{D_0})_{\text{Base}} + (C_{D_0})_{\text{Body,Wave}} \tag{2.16}$$

In the case of wave drag, Fig. 2.8 shows the benefit of a high-fineness nose to reduce supersonic drag. For supersonic missiles, the drag from the shock wave on the nose may be comparable to or even larger than skin friction drag and base drag. $(C_{D_0})_{\text{Body,Wave}}$ is a function of nose fineness and Mach number. The equation for body wave drag coefficient based on Bonney³ is

$$(C_{D_0})_{\text{Body,Wave}} = (1.586 + 1.834/M^2) \{ \tan^{-1} [0.5 / (l_N/d)] \}^{1.69} \tag{2.17}$$

In the above equation the value of the inverse tangent is in radians. Note from the figure that $(C_{D_0})_{\text{Body,Wave}}$ decreases with increasing nose fineness ratio. The figure also shows body base drag coefficient during coasting flight $(C_D)_{\text{Base,Coast}}$ as a function of Mach number. Body base drag can be a major contributor to the total drag during coasting flight because of the low pressure in the base. Base drag is caused by flow separation at the base, which forms a low pressure wake. For a high-fineness nose, the base drag can be larger than the wave drag. The base drag for coasting flight at supersonic Mach number can be approximately correlated as

$$(C_D)_{\text{Base,Coast}} = 0.25/M \tag{2.18}$$

$(C_{D_0})_{\text{Body}} = (C_{D_0})_{\text{Body,Friction}} + (C_{D_0})_{\text{Base}} + (C_{D_0})_{\text{Body,Wave}}$
 $(C_{D_0})_{\text{Body,Friction}} = 0.053 (l/d) [M/(q l)]^{0.2}$. Based on Jerger reference, turbulent boundary layer, q in psf, l in ft.
 $(C_{D_0})_{\text{Base,Coast}} = 0.25/M$, if $M > 1$ and $(C_{D_0})_{\text{Base,Coast}} = (0.12 + 0.13 M^2)$, if $M < 1$
 $(C_{D_0})_{\text{Base,Powered}} = (1 - A_e/S_{\text{Ref}}) (0.25/M)$, if $M > 1$ and $(C_{D_0})_{\text{Base,Powered}} = (1 - A_e/S_{\text{Ref}}) (0.12 + 0.13 M^2)$, if $M < 1$
 $(C_{D_0})_{\text{Body,Wave}} = (1.59 + 1.83/M^2) \{ \tan^{-1} [0.5 / (l_N/d)] \}^{1.69}$, for $M > 1$. Based on Bonney reference, $\tan^{-1}$ in rad.

Note: $(C_{D_0})_{\text{Body,Wave}}$ = body zero-lift wave drag coefficient, $(C_{D_0})_{\text{Base}}$ = body base drag coefficient, $(C_{D_0})_{\text{Body,Friction}}$ = body skin friction drag coefficient, $(C_{D_0})_{\text{Body}}$ = body zero-lift drag coefficient, l_N = nose length, d = missile diameter, l = missile body length, A_e = nozzle exit area, S_{Ref} = reference area, q = dynamic pressure, $\tan^{-1} [0.5 / (l_N/d)]$ in rad.

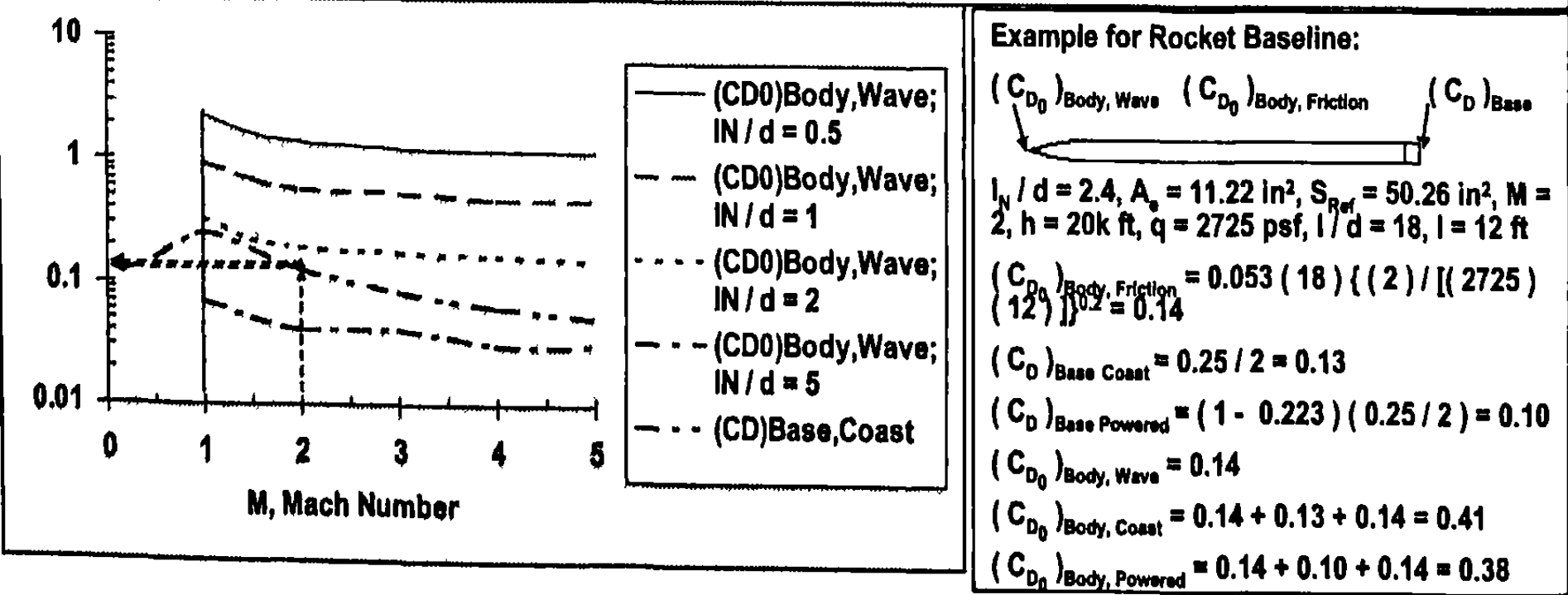


Fig. 2.8 Supersonic body drag driven by nose fineness while subsonic drag driven by wetted area.

A correlation for coasting flight at subsonic Mach number is

$$(C_D)_{\text{Base,Coast}} = 0.12 + 0.13 M^2 \quad (2.19)$$

During powered flight the base drag is reduced by the factor $(1 - A_e/S_{\text{Ref}})$. If the nozzle exit area is nearly as large as the missile base area, the base drag may be negligible during powered flight. Finally, skin friction drag is a major contributor to subsonic drag. $(C_{D_0})_{\text{Body,Friction}}$ is primarily driven by body fineness ratio. It is also a weak function of Mach number, dynamic pressure, and body length. The equation for the body skin friction drag coefficient, based on Jerger⁴, is

$$(C_{D_0})_{\text{Body,Friction}} = 0.053(l/d)[M/(ql)]^{0.2} \quad (2.20)$$

Assumptions in the above equation are that the body wetted area of a noncircular lifting body can be approximated by the wetted area of an equivalent circular cross-section cylinder, the variation in the free stream speed of sound and viscosity with altitude is relatively small, and a turbulent boundary layer. The assumption of a turbulent boundary layer is most in error for a short length missile at very low Mach number (e.g., $M < 0.2$) and very high altitude (e.g., $h > 80,000$ ft). Note that for the same flight conditions, a circular cross section body has less skin friction drag because it has less wetted area than a lifting body.

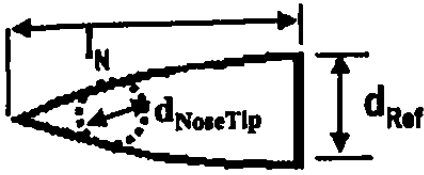
As an example, we will calculate the body zero-lift drag coefficient of the rocket baseline missile, which has a nose fineness ratio of $l_N/d = 2.4$, body fineness $l/d = 18$, and body length $l = 12$ ft. At a nominal Mach number $M = 2$, the predicted zero-lift wave drag coefficient is $(C_{D_0})_{\text{Body,Wave}} = 0.14$. The base drag coefficient during coasting flight at Mach 2 is $(C_D)_{\text{Base,Coast}} = 0.13$, which is comparable to $(C_{D_0})_{\text{Body,Wave}}$. During powered flight at Mach 2, the base drag coefficient is smaller, $(C_D)_{\text{Base,Powered}} = 0.10$. The skin friction drag coefficient at Mach 2 and a nominal altitude of 20,000 ft (corresponding to a dynamic pressure of $q = 2725$ psf) is $(C_{D_0})_{\text{Body,Friction}} = 0.14$. Note that this is also comparable to the value of $(C_{D_0})_{\text{Body,Wave}}$. Finally, the body total power-off drag coefficient is computed as the sum of the contributions from the wave drag plus the base drag plus the skin friction drag. At Mach 2 and 20,000 ft in altitude, the predicted body drag coefficient during coasting flight of the rocket baseline missile is $(C_{D_0})_{\text{Body,Coast}} = 0.41$. The body drag coefficient in powered flight is slightly lower, $(C_{D_0})_{\text{Body,Powered}} = 0.38$.

A small amount of nose tip bluntness is usually desirable for a tactical missile. Nose tip bluntness alleviates problems of localized stress concentration and nose tip heating, usually with a negligible change in drag. Figure 2.9 illustrates the process of calculating the effect of nose tip bluntness on zero-lift wave drag. First, the blunted nose geometry is described as a spherical nose tip. Next, the zero-lift wave drag coefficient is calculated for a sharp nose, based on the equation

$$(C_{D_0})_{\text{Wave,SharpNose}} = (1.59 + 1.83/M^2)\{\tan^{-1}[0.5/(l_N/d)]\}^{1.69} \quad (2.21)$$

◆ Steps to Calculate Wave Drag of a Blunted Nose

1. Relate blunted nose tip geometry to pointed nose tip geometry



2. Compute $(C_{D_0})_{Wave,SharpNose}$ for sharp nose, based on the body reference area

$$(C_{D_0})_{Wave,SharpNose} = (1.59 + 1.83 / M^2) \{ \tan^{-1} [0.5 / (l_N / d)] \}^{1.69}$$

3. Compute $(C_{D_0})_{Wave,Hemi}$ of the hemispherical nose tip ($l_{NoseTip} / d_{NoseTip} = 0.5$), based on the nose tip area

$$(C_{D_0})_{Wave,Hemi} = (1.59 + 1.83 / M^2) \{ [\tan^{-1} (0.5 / (0.5))] \}^{1.69} = 0.665 (1.59 + 1.83 / M^2)$$

4. Finally, compute $(C_{D_0})_{Wave,BluntNose}$ of the blunt nose, based on the body reference area

$$(C_{D_0})_{Wave,BluntNose} = (C_{D_0})_{Wave,SharpNose} (S_{Ref} - S_{NoseTip}) / S_{Ref} + (C_{D_0})_{Wave,Hemi} S_{NoseTip} / S_{Ref}$$

◆ Example Rocket Baseline ($d_{Ref} = 8$ in) with 10% Nose Tip Bluntness at Mach 2

- $(C_{D_0})_{Wave,SharpNose} = [1.59 + 1.83 / (2)^2] [\tan^{-1} (0.5 / 2.4)]^{1.69} = 0.14$
- $d_{NoseTip} = 0.10 (8) = 0.8$ in
- $S_{NoseTip} = \pi d_{NoseTip}^2 / 4 = 3.1416 (0.8)^2 / 4 = 0.503 \text{ in}^2 = 0.00349 \text{ ft}^2$
- $(C_{D_0})_{Wave,Hemi} = 0.665 [1.59 + 1.83 / (2)^2] = 1.36$
- $(C_{D_0})_{Wave,BluntNose} = 0.14 (0.349 - 0.003) / 0.349 + (1.36) (0.003) / (0.349) = 0.14 + 0.01 = 0.15$

Fig. 2.9 Moderate nose tip bluntness causes a negligible change in supersonic drag.

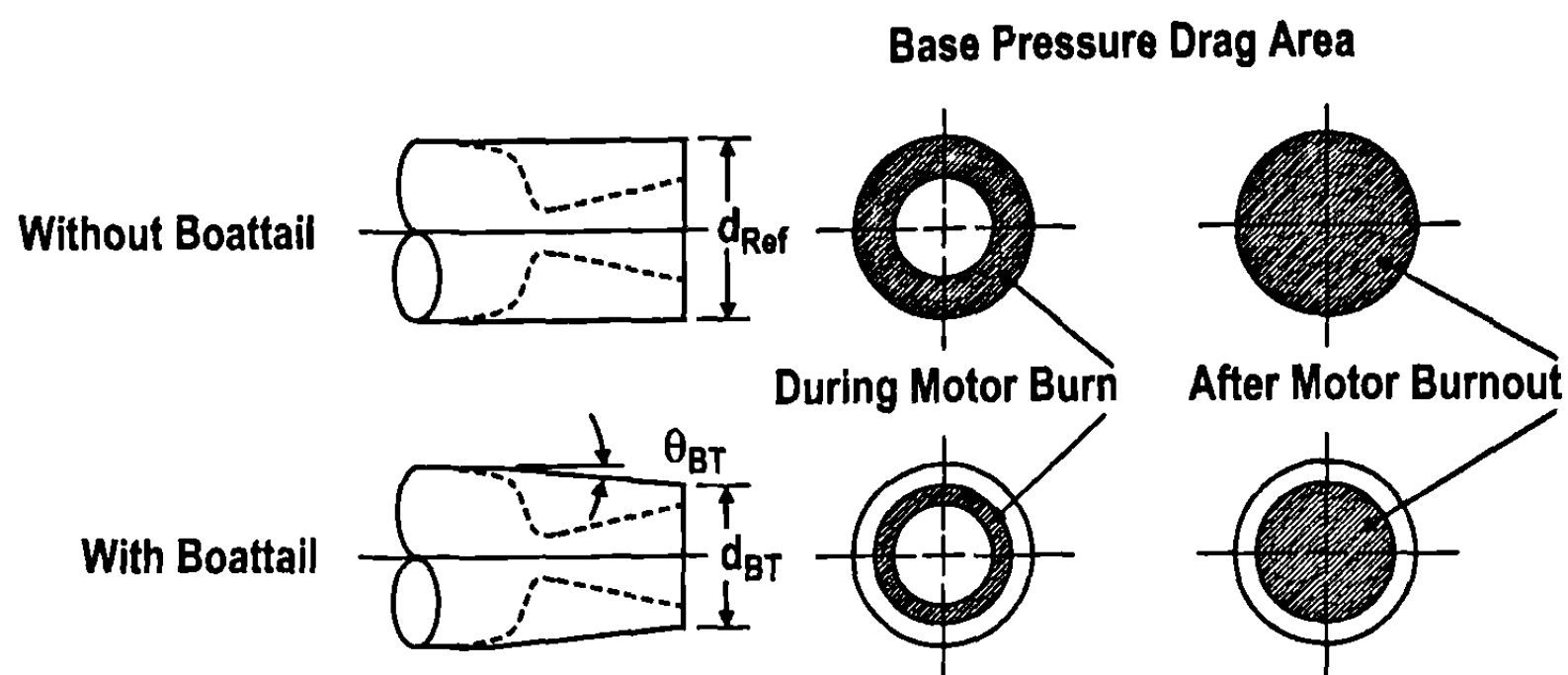
Next, the zero-lift drag wave coefficient of the hemispherical nose tip ($l_{NoseTip} / d_{NoseTip} = 0.5$) is calculated. The equation is

$$(C_{D_0})_{Wave,Hemi} = 0.665(1.59 + 1.83/M^2) \quad (2.22)$$

In the above equation $(C_{D_0})_{Wave,Hemi}$ is based on the nose tip cross-sectional area. Finally, the zero-lift wave drag coefficient of a missile with a blunt nose tip is the summation of the wave drag coefficient of the sharp nose plus the wave drag coefficient of the hemispherical nose tip, scaled by their areas. The equation is

$$(C_{D_0})_{Wave,BluntNose} = (C_{D_0})_{Wave,SharpNose} (S_{Ref} - S_{NoseTip}) / S_{Ref} + (C_{D_0})_{Wave,Hemi} S_{NoseTip} / S_{Ref} \quad (2.23)$$

As an example, the wave drag coefficient of the rocket baseline with a sharp nose of fineness ratio $l_N/d = 2.4$ and flying at Mach 2 has a nose zero-lift wave drag coefficient $(C_{D_0})_{Wave,SharpNose} = 0.14$. The wave drag coefficient of a hemispherical nose tip is given by $(C_{D_0})_{Wave,Hemi} = 1.36$, where $(C_{D_0})_{Wave,Hemi}$ is based on the nose tip cross-sectional area. Finally, the rocket baseline missile ($d = d_{Ref} = 8$ in) with an assumed 10% nose tip bluntness at Mach 2 has a predicted zero-lift wave drag coefficient of $(C_{D_0})_{Wave,BluntNose} = 0.15$, which is only slightly larger than that of a sharp nose.



Note: Boattail angle θ_{BT} and boattail diameter d_{BT} limited by propulsion nozzle packaging, tail flight control packaging, and flow separation

Reference: Chin, S. S., *Missile Configuration Design*, McGraw-Hill Book Company, New York, 1961

Fig. 2.10 Boattail decreases base pressure drag area.

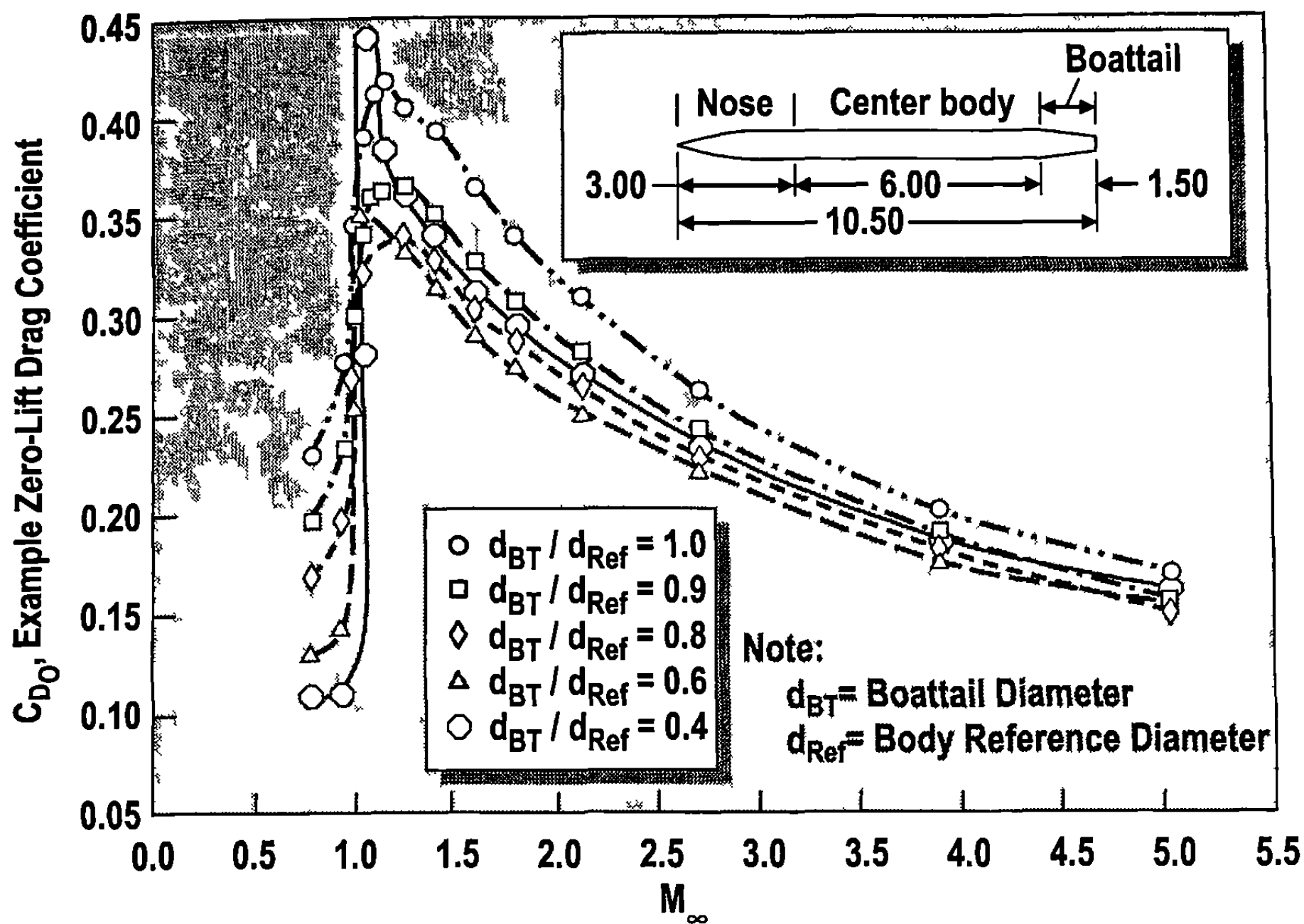
2.4 Boattail

Boattailing has its best payoff for subsonic missiles; supersonic missiles are susceptible to aft flow separation from boattailing. Figure 2.10 from Chin⁵ illustrates the benefit of a moderate amount of boattailing. Without a boattail, during motor burn the base area that is outside the nozzle has a pressure less than that of the free stream, causing base drag. After motor burnout, the entire base area causes base drag. However, with boattailing the area occupied by the boattail has a larger base pressure, reducing the base drag in the area of the boattail. A design consideration if the missile has a boattail is that there is less volume available to package subsystems such as tail control actuators.

Figure 2.11 from Ref. 6 shows that the payoff for boattailing is greatest for subsonic missiles. The example body has an overall fineness ratio of 10.5, a nose fineness ratio of 3.00, a center body fineness ratio of 6.00, and a boattail fineness ratio 1.50. Note that for subsonic Mach number, boattailing reduces the drag by more than 50%. However, at supersonic to hypersonic Mach number there is only a slight reduction in drag with moderate boattailing, and large boattailing actually causes an increase in drag. From the figure, note that at Mach 5.0 the C_{D_0} for $d_{BT}/d_{Ref} = 0.6$ (boattail half-angle = 7.6 deg) is larger than the C_{D_0} for $d_{BT}/d_{Ref} = 0.8$ (boattail half-angle = 3.8 deg). At high Mach number, a large boattail angle can cause flow separation over the boattail, resulting in increased drag. As a rule of thumb, the boattail half-angle should be less than 10 deg.

2.5 Lifting Body Versus Axisymmetric Body

As shown in Fig. 2.12, the maximum normal force of a lifting body is higher than that of an axisymmetric body. The equation for normal force



Note: Boattail half angle should be less than 10 deg, to avoid flow separation.
Source: Mason, L.A., Devan, L. and Moore, F.G., "Aerodynamic Design Manual for Tactical Weapons," NSWC TR 81-156, July 1981

Fig. 2.11 Boattailing reduces drag for subsonic missiles.

$|C_N| = [|(a/b) \cos^2 \phi + (b/a) \sin^2 \phi|] [|\sin(2\alpha) \cos(\alpha/2)| + 2(l/d) \sin^2 \alpha]$

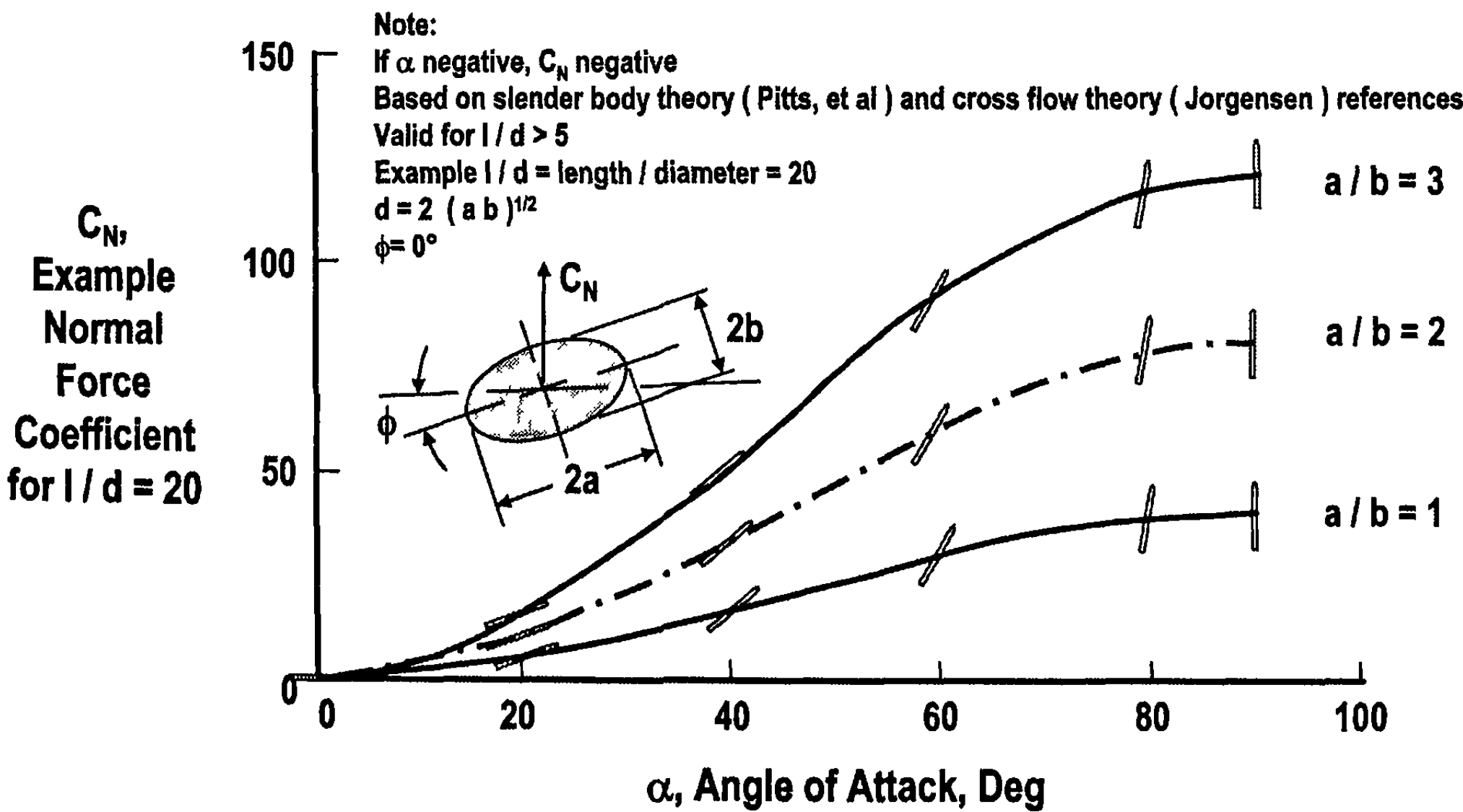


Fig. 2.12 A lifting body has higher normal force.

coefficient of a body is

$$|C_N| = [(a/b) \cos^2 \phi + (b/a) \sin^2 \phi][|\sin(2\alpha) \cos(\alpha/2)| + 2(l/d) \sin^2 \alpha] \quad (2.24)$$

The normal force coefficient of a slender body is a function only of angle of attack and body geometry and is independent of Mach number. The normal force prediction is based on combining slender body theory (Ref. 7) and body crossflow theory (Ref. 8). It is valid for a body fineness ratio $l/d > 5$. For an elliptical cross section, an equivalent diameter is based on a circular cross section of the same area. The figure shows normal force coefficient increases with α (up to $\alpha = 90$ deg) and a/b . As an example, at a 90-deg angle of attack, the normal force coefficient for an elliptical cross section with a major-to-minor axis ratio of $a/b = 2$ is twice that of a circular cross section.

The body normal force curve slope from angle of attack is used in sizing the tail (or in some cases a flare) to meet the static stability requirement. $(C_{N_\alpha})_{\text{Body}}$ is the derivative of the equation for body normal force coefficient. At low angle of attack, $(C_{N_\alpha})_{\text{Body}} = 2[(a/b)\cos^2 \phi + (b/a)\sin^2 \phi]$, with the units of per rad.

A measure of aerodynamic efficiency is the ratio of lift-to-drag (L/D). The equation for aerodynamic efficiency is

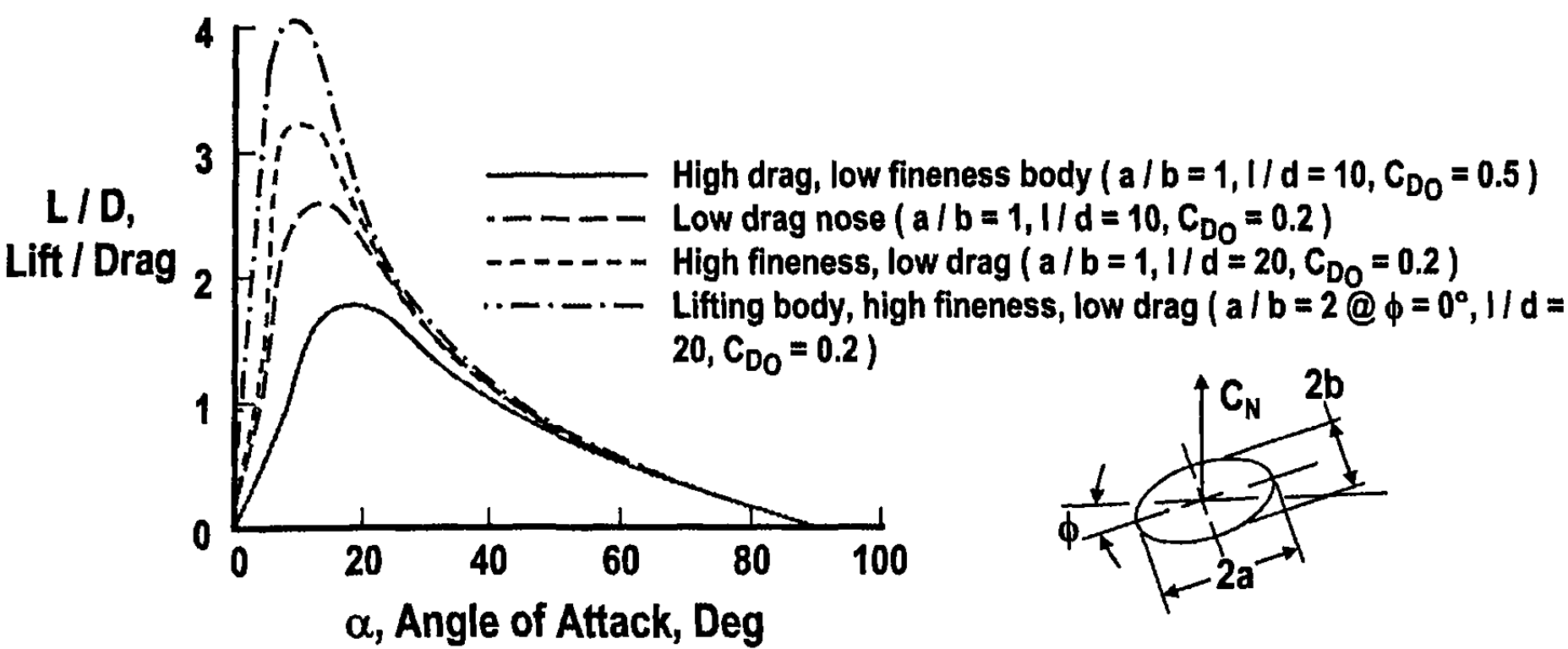
$$L/D = C_L/C_D = (C_N \cos \alpha - C_{D_0} \sin \alpha)/(C_N \sin \alpha + C_{D_0} \cos \alpha) \quad (2.25)$$

For a lifting body missile without wings, the equation for the normal force coefficient is $|C_N| = [(a/b)\cos^2 \phi + (b/a)\sin^2 \phi][|\sin(2\alpha) \cos(\alpha/2)| + 2(l/d) \sin^2 \alpha]$. Again, C_N for a body is based on combining slender body theory with crossflow theory. As shown in Fig. 2.13, an increase in L/D is achievable by reducing the zero-lift drag coefficient, increasing the body fineness, or by providing a lifting body configuration ($a/b > 1$). Also shown is that as a higher L/D is achieved, the angle of attack in which $(L/D)_{\text{Max}}$ is achieved is decreased. Note that other design considerations, such as launch platform lateral and length constraints, may limit the aerodynamic shaping that is practical. Also note that for 1-g constant altitude flight the angle of attack is usually much lower than the angle of attack for $(L/D)_{\text{Max}}$. As a result, the L/D during the flyout of most rocket powered missiles is usually much lower than $(L/D)_{\text{Max}}$.

Although a lifting body configuration has a higher $(L/D)_{\text{Max}}$ than a circular cross-sectional configuration, for 1-g flight at low angles of attack a circular configuration provides comparable L/D . Figure 2.14 compares the L/D of a lifting body configuration ($a/b = 2$) with that of a circular cross-sectional ($a/b = 1$) configuration. Typical values are given for a precision strike missile configuration of 2 ft² cross-sectional area and 2000 lb weight. As before, the L/D is based on combining slender body theory with crossflow theory. Maximum L/D for the lifting body configuration occurs at a dynamic pressure $q \approx 500$ psf. At $q = 500$ psf, the lifting body L/D is 40% higher than the circular cross-sectional body ($L/D = 3.37$ vs 2.40). However, note that the circular body cross-sectional configuration provides comparable L/D if the dynamic pressure is greater than 5000 psf. At $q = 5000$ psf, the lifting body L/D is only 5% higher than the circular cross-sectional body ($L/D = 0.96$ vs 0.91).

$$L/D = C_L / C_D = (C_N \cos \alpha - C_{D0} \sin \alpha) / (C_N \sin \alpha + C_{D0} \cos \alpha)$$

For a lifting body, $|C_N| = [|(a/b) \cos^2(\phi) + (b/a) \sin^2(\phi)|][|\sin(2\alpha) \cos(\alpha/2)| + 2(l/d) \sin^2 \alpha]$

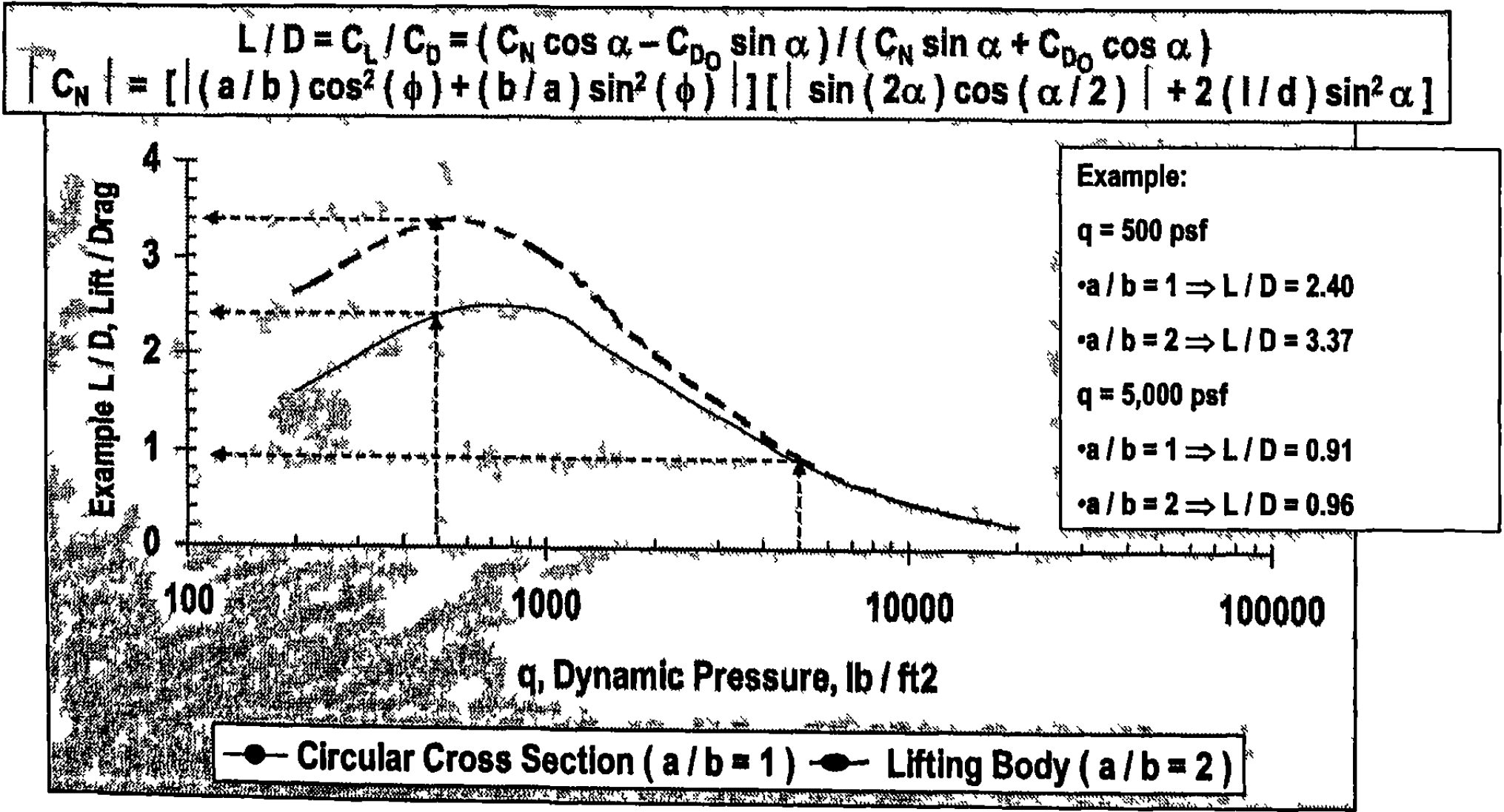


Note:

- If α negative, L / D negative
- $d = 2 (a b)^{1/2}$
- Launch platform span constraints (e.g., VLS launcher) and length constraints (e.g., aircraft compatibility) may limit missile aero configuration enhancements

Fig. 2.13 L/D is impacted by C_{D0}, body fineness, and lifting body cross section geometry.

A tailored lifting body (e.g., $a/b > 2$) or adding a wing increases $(L/D)_{Max}$, reduces $\alpha_{(L/D)Max}$, and reduces $q_{(L/D)Max}$. For example, the rocket baseline has a relatively high L/D because of its wing. At Mach 0.8, $(L/D)_{Max} = 6.2$, $\alpha_{(L/D)Max} = 4.5$ deg, and $q_{(L/D)Max} = 350$ psf. Supersonic missiles usually fly at a



Note. Example figure based on following assumptions:
Body lift only (no surfaces), cruise flight (lift = weight), $W = L = 2,000$ lb, $d = 2 (a b)^{1/2}$, $S = 2$ ft²,
 $l/d = 10$, $C_{D0} = 0.2$

Fig. 2.14 A lifting body requires flight at low dynamic pressure to achieve high aerodynamic efficiency.

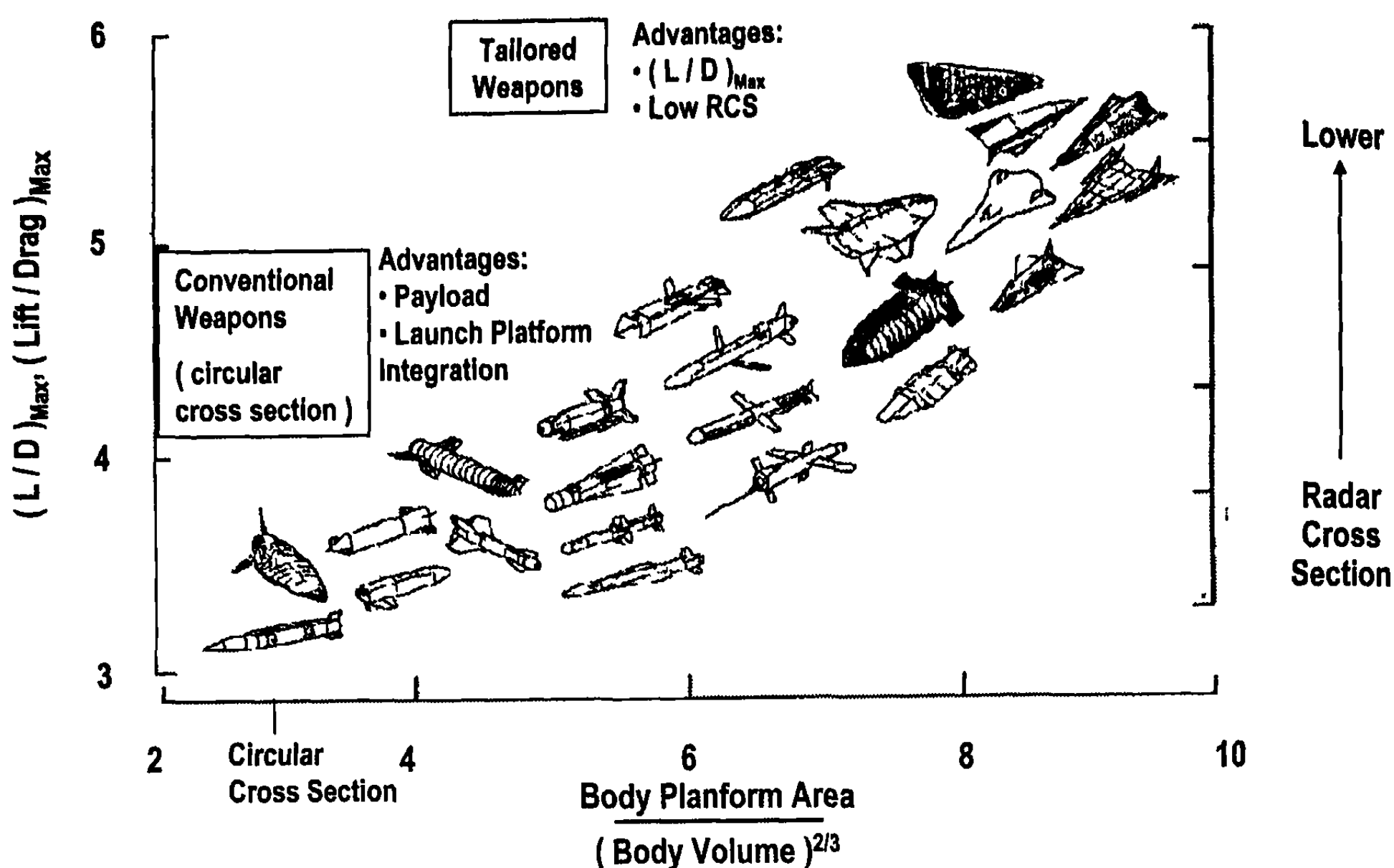


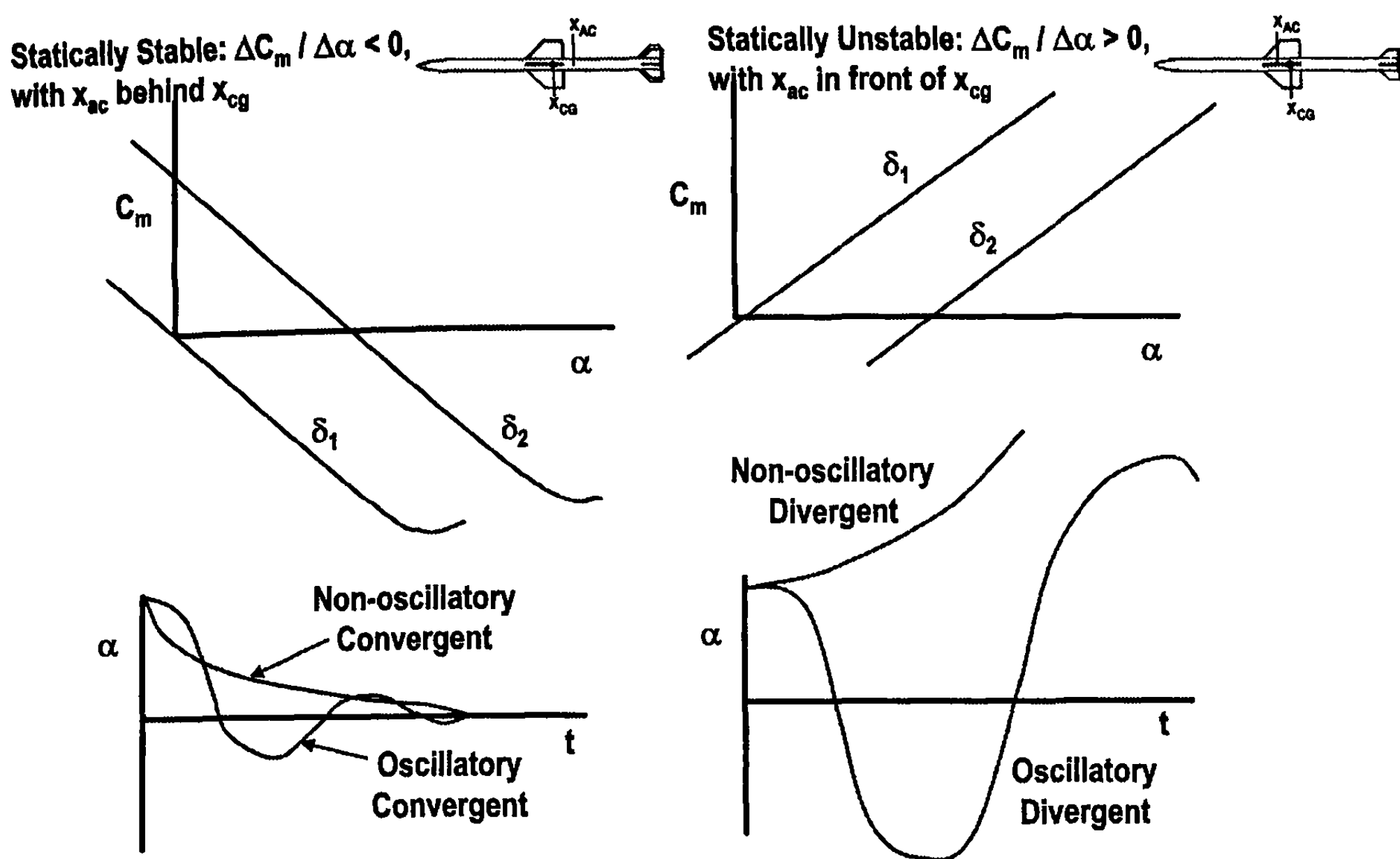
Fig. 2.15 Trade-off of low observables and $(L/D)_{\text{Max}}$ vs volumetric efficiency.

dynamic pressure greater than 1000 psf. The L/D for a supersonic missile in 1-g flight is usually much less than $(L/D)_{\text{Max}}$. As an example, the rocket baseline during 1-g powered flight at Mach 2, 20,000 ft altitude has an $L/D = 0.41$.

Figure 2.15 compares weapon configurations that have conventional cylindrical bodies of circular cross section to other weapons that are highly tailored, using aerodynamic shaping of their lifting body configurations. An indication of subsystems packaging efficiency is the ratio of body planform area to the $2/3$ power of the body volume ($S_P/V^{2/3}$). For a circular cross-sectional body, this parameter has a value of about 3. As shown in the figure, a highly tailored missile could have a value of the subsystems packaging efficiency parameter $S_P/V^{2/3} > 9$. An advantage of a tailored lifting body missile is higher aerodynamic efficiency L/D (lift-to-drag ratio) for extended range cruise performance and enhanced maneuverability. Also shown is the synergy of tailored missiles with reduced radar cross section. Tailored missiles are also synergistic with ramjets in areas such as inlet integration and liquid hydrocarbon fuel packaging. Disadvantages of tailored missiles include their relative inefficiency for solid subsystems packaging and an adverse impact on launch platform integration because of a larger span. Improved methods and tests are required for the prediction of the aerodynamics and the structural loads of nonaxisymmetric weapons. This includes more extensive wind tunnel tests, computational fluid dynamics (CFD) predictions, and finite element modeling (FEM) of structural integrity.

2.6 Static Stability

Static stability in pitch is defined by the slope of the pitching moment coefficient versus angle of attack $\Delta C_m / \Delta \alpha = C_{m_\alpha}$. Shown in the top left portion of Fig. 2.16 is an example of the pitching moment coefficient versus angle of



Note: Statically unstable missile requires high bandwidth autopilot.

Autopilot negative rate feedback provides stability augmentation.

Fig. 2.16 $\Delta C_m / \Delta \alpha$ and static margin define static stability.

attack curve for a statically stable missile. For a statically stable missile, the slope of the pitching moment coefficient versus angle of attack is negative, with the aerodynamic center aft of the center of gravity. An increase in angle of attack (nose up) causes a negative incremental pitching moment (nose down), which then tends to decrease the angle of attack. Control surface deflections allow the missile to trim ($C_m = 0$) at the desired angle of attack. The bottom left portion of the figure shows the decay of an incremental disturbance in the angle of attack as a function of time. It assumes a statically stable missile without an autopilot correction. For a highly stable, highly damped missile, the convergence is usually nonoscillatory. Most missiles have small tail surfaces, small static margin, and are lightly damped during free flight if there is no autopilot feedback (e.g., their unguided flight trajectory during store separation). The lightly damped missiles have an oscillatory convergence from a disturbance in angle of attack. The ideal damping for autopilot feedback is $\zeta \approx 0.7$.

As shown in the upper right of the figure, a statically unstable missile has the opposite trend. The slope of the pitching moment coefficient versus angle of attack is positive, with the aerodynamic center forward of the center of gravity. An increase in angle of attack causes an increase in pitching moment, which tends to cause a further increase in angle of attack. The rate of divergence is a function of dynamic pressure, with rapid divergence occurring at high dynamic pressure. If the autopilot has a high bandwidth and is sufficiently fast, it is possible to trim and control a statically unstable missile. Negative rate feedback from the gyros provides the required stability augmentation. A statically unstable missile might be desirable to improve the responsiveness and

maneuverability. The bottom right portion of the figure shows the divergence of a statically unstable missile without an autopilot correction. A highly unstable missile will diverge monotonically. A statically unstable missile that is slightly unstable will tend to oscillate with increasing divergence.

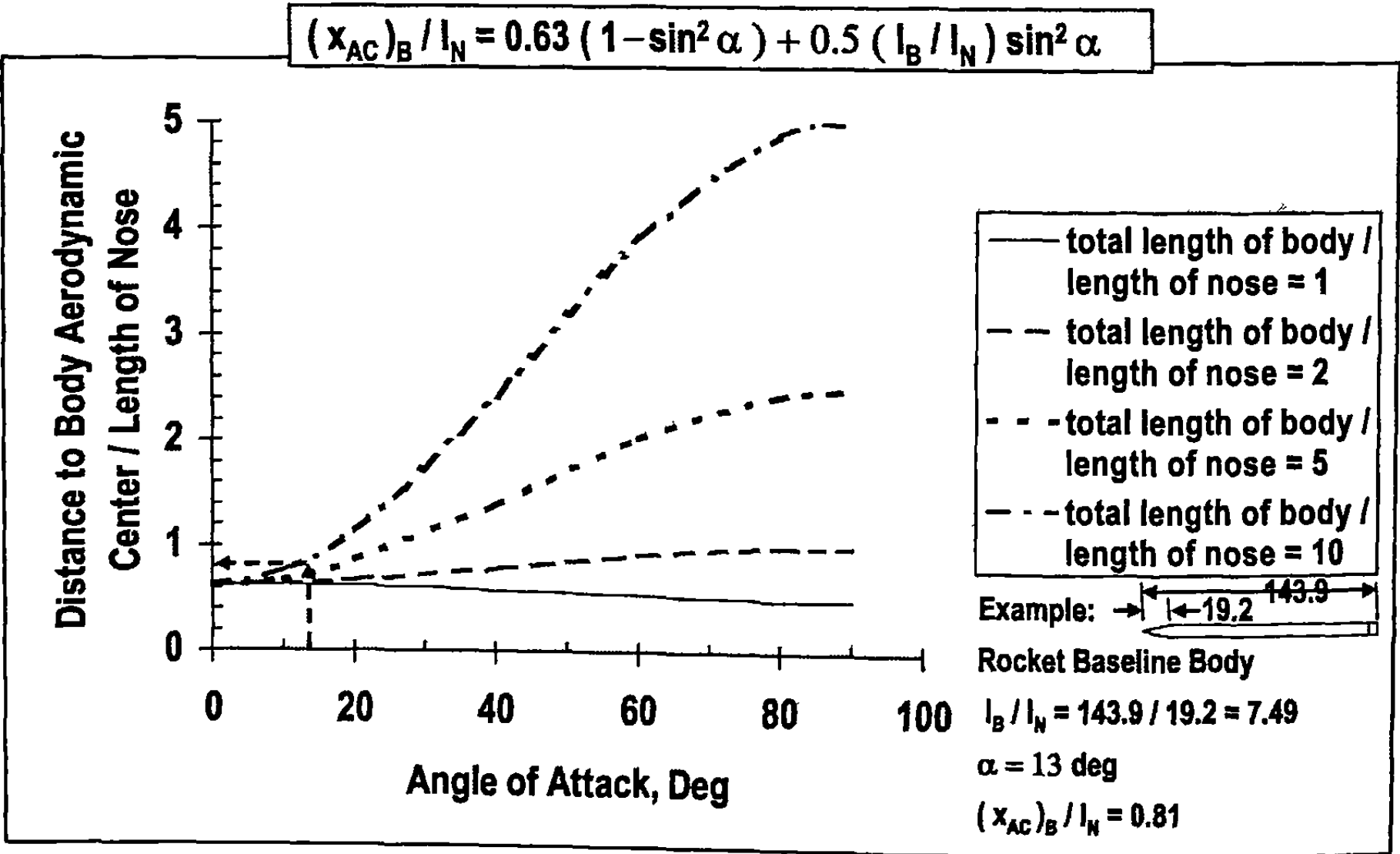
2.7 Body Aerodynamic Center

Aerodynamic center is defined as the location for neutral static margin. If the center of gravity is located at the aerodynamic center, the pitching moment does not change with angle of attack. The aerodynamic center location for the body $(x_{AC})_B$ depends mostly on three parameters, angle of attack, nose fineness, and aft body fineness. For conceptual design, the effect of Mach number on $(x_{AC})_B$ may be neglected. As shown in Fig. 2.17, for a body without a flare at low angle of attack, $(x_{AC})_B$ is located at about 63% of the nose length. At angles of attack approaching 90 deg, $(x_{AC})_B$ approaches the center of the body. Results in the figure are shown as a function of the body-length-to-nose-length ratio l_B/l_N and the angle of attack α . As before, results are based on combining slender body theory with crossflow theory, giving the equation

$$(x_{AC})_B/l_N = 0.63(1 - \sin^2 \alpha) + 0.5(l_B/l_N) \sin^2 \alpha$$

(2.26)

As an example, the rocket baseline missile has a nose length of 19.2 in and a total body length of 143.9 in. The ratio of body length to nose length is $l_B/l_N = 7.49$. For an angle of attack of $\alpha = 13$ deg, the center of pressure of the body $(x_{cp})_B$ is located at 81% of the length of the nose.



Note: Based on slender body theory (Pitts, et al) and cross flow theory (Jorgensen) references. No flare.
 $(x_{AC})_B$ = Location of body aerodynamic center, l_N = length of nose, α = angle of attack, l_B = total length of body.

Fig. 2.17 Body aerodynamic center is a function of angle of attack, nose fineness, and body length.

2.8 Flare Stabilizer

A flare may be added to the aft end of a body to increase the static stability. Typically the flare half-angle is less than 10deg to avoid flow separation. Figure 2.18a illustrates a process to calculate the static margin of a body-flare configuration. It is based on the locations of the normal force coefficient derivative contributors to pitching moment. As shown previously, at low angle of attack, the normal force coefficient derivative contribution and its location for a circular cross-sectional body, based on slender body theory are

$$(C_{N\alpha})_B = 2 \text{ per rad} \quad (2.27)$$

$$(x_{ac})_B = 0.63l_N \quad (2.28)$$

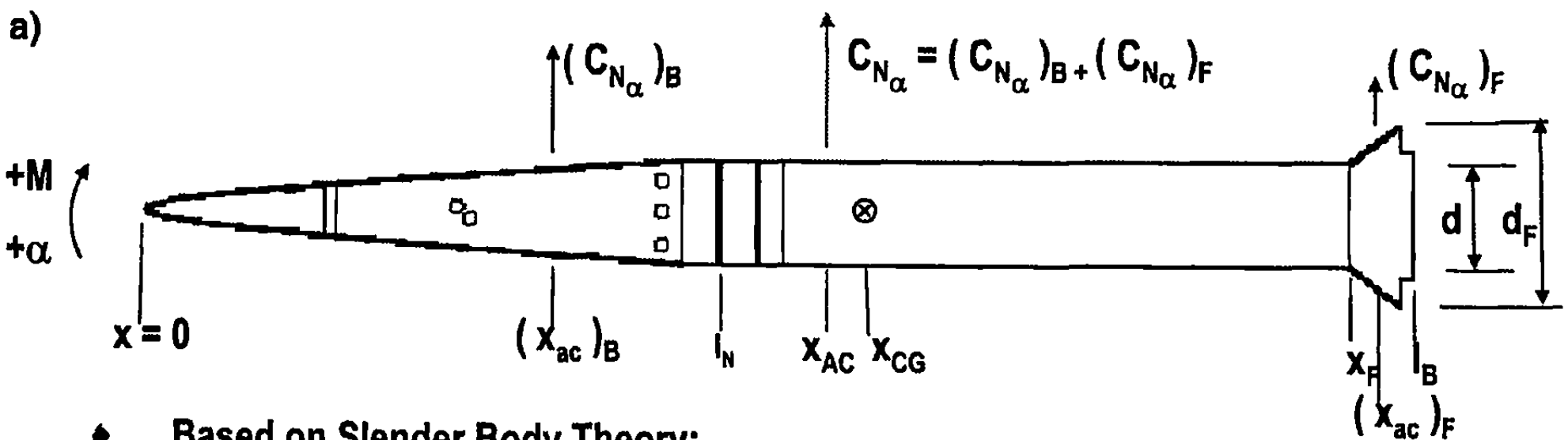
Also for low angle of attack, the flare normal force coefficient derivative contribution and its location, based on slender body theory⁸ are

$$(C_{N\alpha})_F = 2[(d_F/d)^2 - 1] \quad (2.29)$$

$$(x_{ac})_F = x_F + 0.33l_F[2(d_F/d) + 1]/[(d_F/d) + 1] \quad (2.30)$$

The static margin of a body-flare missile, such as THAAD, is given by the equation

$$(x_{AC} - x_{CG})/d = -\{(C_{N\alpha})_B\{[x_{CG} - (x_{AC})_B]/d\} + (C_{N\alpha})_F\{[x_{CG} - (x_{AC})_F]/d\}\} / [(C_{N\alpha})_B + (C_{N\alpha})_F] \quad (2.31)$$



◆ Based on Slender Body Theory:

- ◆ $(C_{N\alpha})_F = 2[(d_F/d)^2 - 1]$
- ◆ $(x_{ac})_F = x_F + 0.33l_F[2(d_F/d) + 1]/[(d_F/d) + 1]$
- ◆ $(C_{N\alpha})_B = 2 \text{ per rad}$
- ◆ $(x_{ac})_B = 0.63l_N$

◆ $\Sigma M = 0$ at Aerodynamic Center. For a Body-Flare:

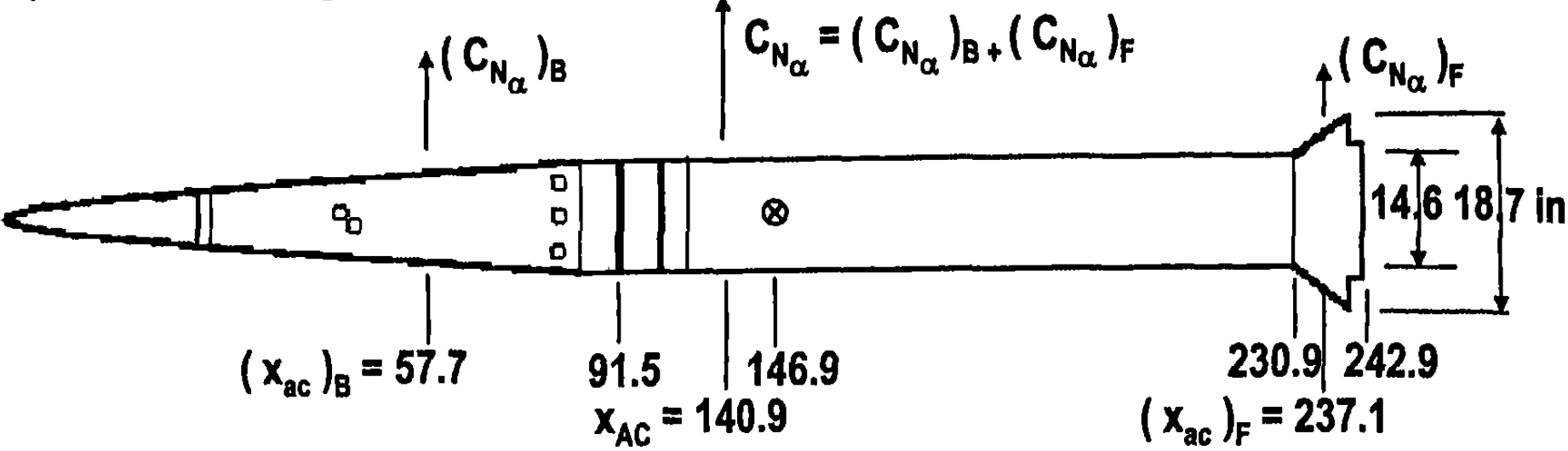
- ◆ $(C_{N\alpha})_B\{[x_{CG} - (x_{AC})_B]/d\} + (C_{N\alpha})_F\{[x_{CG} - (x_{AC})_F]/d\} = -[(C_{N\alpha})_B + (C_{N\alpha})_F]\{[x_{AC} - x_{CG}]/d\}$

◆ Static Margin for a Body-Flare

- ◆ $(x_{AC} - x_{CG})/d = -\{(C_{N\alpha})_B\{[x_{CG} - (x_{AC})_B]/d\} + (C_{N\alpha})_F\{[x_{CG} - (x_{AC})_F]/d\}\} / [(C_{N\alpha})_B + (C_{N\alpha})_F]$

Fig. 2.18a Flare increases static stability.

b)
Example of Static Margin for THAAD (Statically Unstable Missile)



- ◆ $(C_{N_{\alpha}})_F = 2 [(18.7 / 14.6)^2 - 1] = 2 [(1.28)^2 - 1] = 1.28$ per rad
- ◆ $(x_{ac})_F = 230.9 + 0.33 (12.0) [2 (18.7 / 14.6) + 1] / [(18.7 / 14.6) + 1] = 237.1$ in
- ◆ $(x_{ac})_B = 0.63 (91.5) = 57.7$ in
- ◆ $x_{CGLaunch} = 146.9$ in
- ◆ $(x_{AC} - x_{CG})_{Launch} / d = - \{ 2 \{ [146.9 - 57.7] / 14.6 \} + 1.28 \{ [146.9 - 237.1] / 14.6 \} \} / [2 + 1.28] = -0.41$

Fig. 2.18b Flare increases static stability (Continued).

Figure 2.18b further illustrates the process of calculating static margin for a body-flare configuration, using the THAAD missile as an example. The flare somewhat compensates for the statically unstable body. As shown in the example, for a flare-to-body diameter ratio of $d_F/d = 18.7 \text{ in}/14.6 \text{ in} = 1.28$, THAAD is predicted to be statically unstable at launch, with a static margin of

$$(x_{AC} - x_{CG})/d = -0.41 \tag{2.32}$$

Type Stabilizer	Drag	Span	Heating	$\Delta C_{N_{\alpha}}$	Tail Control
Flare (e.g., THAAD)					
Tails (e.g., Standard Missile)					

Note: Superior Good Average Poor

Fig. 2.19 Tail stabilizers have lower drag while flares have lower aerodynamic heating and changes in stability.

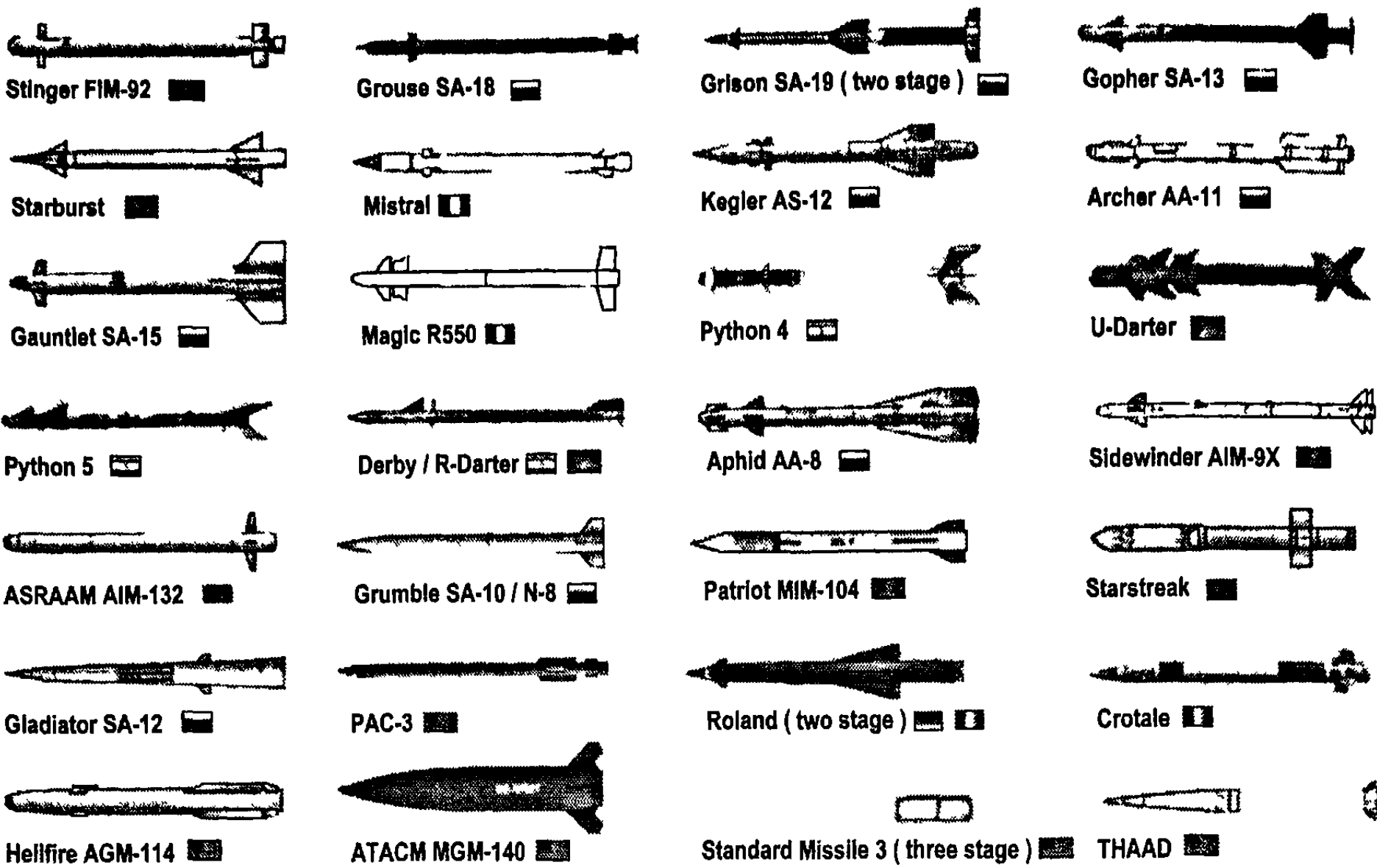
The flare half-angle of THAAD is 9.7 deg to avoid flow separation (and the separation pressure drag) in front of the flare.

Figure 2.19 is a comparison of a flare stabilizer with conventional tail stabilizers. A flare-stabilized missile, such as THAAD, has advantages of lower required span for the stabilizer, lower aerodynamic heating, and less variation in its aerodynamics with Mach number. Reduced aerodynamic heating is particularly beneficial for a hypersonic missile such as THAAD. Reentry vehicles also use flares to provide static stability while minimizing aerodynamic heating. An example of a missile that uses conventional tail surfaces for static stability is the Standard Missile. Compared to flares, conventional tails have benefits of lower drag and the availability of the tail surfaces for flight control in addition to stability. THAAD relies entirely on thrust vector control (TVC) and reaction jet control. Compared to aerodynamic control, a disadvantage of TVC and reaction jet control is a weight penalty because of the additional weight of propellant.

2.9 Wings Versus No Wings

There are many considerations in the tradeoff of a small wing/strake/no wing versus a larger wing. A small wing, a strake, or no wing has advantages of range in high supersonic flight, range in high dynamic pressure, maximum angle of attack (stability and control is better at high angles of attack, with less induced roll and delayed stall), launch platform compatibility, lower radar cross section, and volume and weight available for propellant/fuel in a volume- or weight-limited missile. A larger wing has advantages of range in subsonic flight, range in low dynamic pressure, lower guidance time constant, normal acceleration, high-altitude intercept, body stiffness, seeker tracking (less dome error slope error from lower angle of attack), less wipe velocity for the warhead at impact (lower angle of attack), and lower gimbal angle requirement for the seeker. Note that some of the disadvantages of a wingless missile can be alleviated by unconventional flight control (e.g., TVC, reaction jet/jet interaction control). It is possible for a wingless missile to achieve small guidance time constant, high normal acceleration or maneuverability, and a high-altitude intercept capability at low dynamic pressure through unconventional flight control.

Because supersonic missiles generally operate at relatively high dynamic pressure, body lift may be sufficient for the required aerodynamic efficiency (L/D) and maneuverability. Shown in Fig. 2.20 are examples of supersonic missiles that do not have wings. Many of these missiles are highly maneuverable and are capable of operating at high angle of attack. Wingless missiles that use canard control may have enhanced maneuverability from the additional lift of the canard surface. The majority of the wingless missiles (the first 15 of 28) in the figure use canard control, while the remaining 13 missiles have either tail control or thrust vector control. Unconventional flight control may also be used if additional maneuverability is required to supplement the body lift. Examples of supersonic wingless missiles that use TVC are THAAD, Archer AA-11 (TVC combined with canard control), Sidewinder AIM-9X (TVC combined with tail control), and Standard Missile (TVC combined with tail control).



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Fig. 2.20 Most supersonic missiles are wingless. Figure also in color section.

2.10 Normal Force Prediction for Surfaces
(e.g., Wing, Tail, Canard)

Wing, tail, and canard surfaces may be characterized by their location relative to the center of gravity, cg. Wing surfaces are located near the cg. Tail surfaces are located aft of the cg. Canard surfaces are located forward of the cg. Strake surfaces have very low aspect ratio ($A < 1$). Strakes may be located forward, aft, or near the center of gravity. Wing, tail, and canard surfaces may be fixed or movable (i.e., control surfaces). The methodology described here to predict the normal force on the wing is also applicable to other surfaces, such as tails and canards. As shown in the equations of Fig. 2.21, the surface normal force coefficient $(C_N)_{\text{Surface}}$ is a function of Mach number, local angle of attack, aspect ratio, and the surface planform area. $(C_N)_{\text{Surface}}$, which is based on the missile reference area, decreases with increasing supersonic Mach number and increases with angle of attack and the surface area. The prediction is based on the method of Ref. 9. In the figure, linear wing theory plus Newtonian impact theory are applied at high supersonic Mach number, with the requirement that

$$M^2 > 1 + [8/(\pi A)]^2 \tag{2.33}$$

The linear wing theory plus Newtonian impact theory equation is

$$|(C_N)_{\text{Surface}}| = [4|\sin\alpha' \cos\alpha'|/(M^2 - 1)^{1/2} + 2\sin^2\alpha'](S_{\text{Surface}}/S_{\text{Ref}}) \tag{2.34}$$

Note that $(C_N)_{\text{Surface}}$ based on linear wing theory is independent of aspect ratio. For the rocket baseline missile wing with an aspect ratio of $A_w = 2.82$, linear

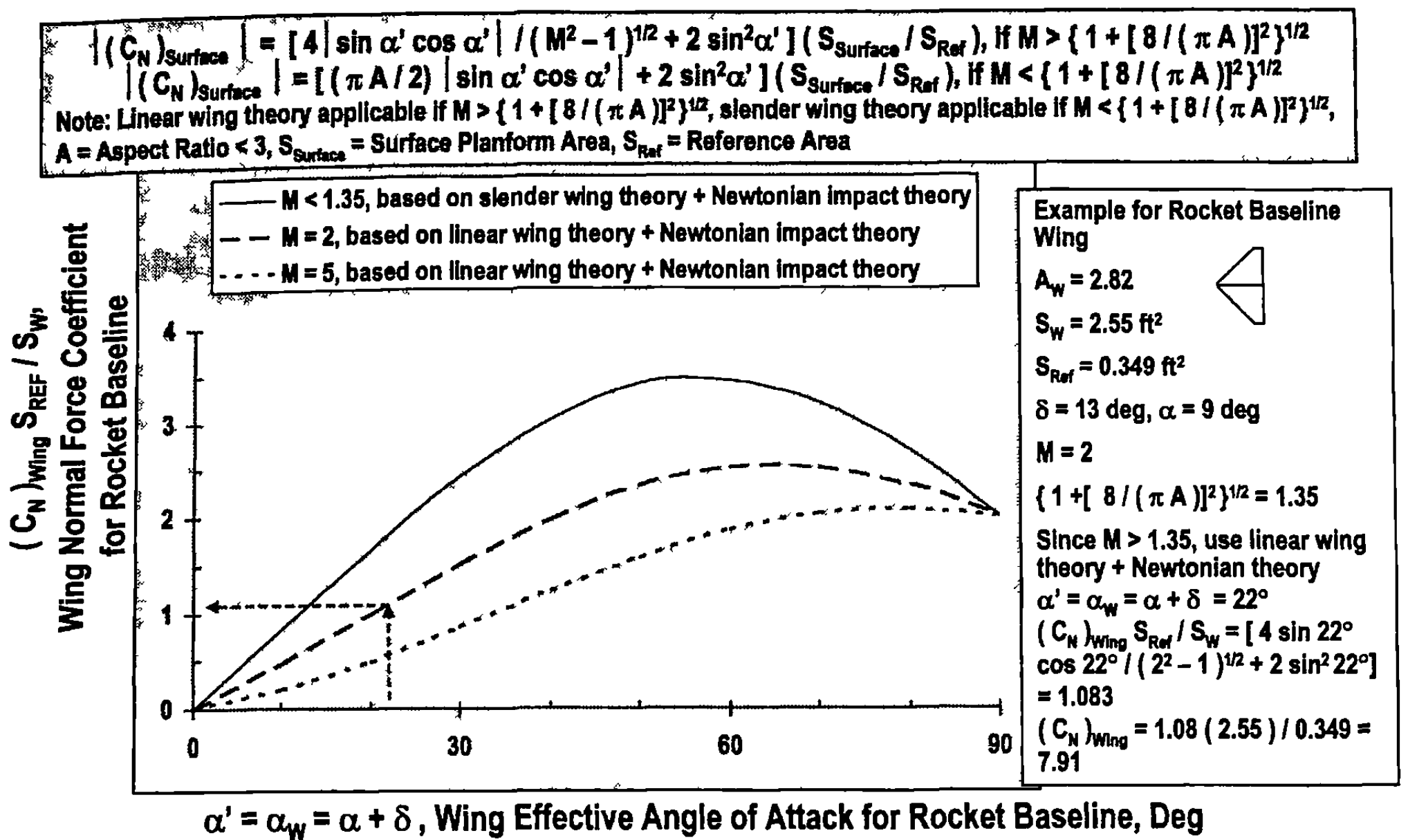


Fig. 2.21 Wings, tails, and canards with large area and at high angle of attack have high normal force.

wing theory is applicable for Mach numbers greater than 1.35. An alternative approach, slender wing theory plus Newtonian impact theory, is more applicable at subsonic and low supersonic Mach number, with the requirement that

$$M^2 < 1 + [8 / (\pi A)]^2 \quad (2.35)$$

The slender wing theory plus Newtonian impact theory equation is

$$|(C_N)_{\text{Surface}}| = [(\pi A / 2) |\sin \alpha' \cos \alpha'| + 2 \sin^2 \alpha'] (S_{\text{Surface}} / S_{\text{Ref}}) \quad (2.36)$$

Note that $(C_N)_{\text{Surface}}$, based on slender wing theory, is independent of Mach number. Slender wing theory is accurate for low aspect ratio wings ($A < \approx 3$). As a comparison with lifting surface theory, which is most applicable to high aspect ratio surfaces, at an aspect ratio of $A = 3$, slender wing theory overpredicts normal force by about 25% compared to lifting surface theory. For the rocket baseline missile wing ($A_W = 2.82$), slender wing theory is applicable for Mach numbers less than 1.35.

As an example, we shall evaluate the rocket baseline missile wing at Mach 2. Because the Mach number is greater than 1.35, the prediction is based on linear wing theory plus Newtonian impact theory. The rocket baseline missile wing is limited to a maximum local angle of attack of 22 deg because of stall of the wing. For a typical cg location, this results in a maximum angle of attack of 9.4 deg for a maximum wing control deflection of 12.6 deg. For this condition, the normal force coefficient of the wing, based on wing area $S_W = 2.55 \text{ ft}^2$, is equal to 1.08. Finally, the normal coefficient of the wing,

based on the body reference cross-sectional area $S_{\text{Ref}} = 0.349 \text{ ft}^2$, is computed to be $(C_N)_{\text{Wing}} = 7.91$. Because the rocket baseline missile has a relatively large wing, most of the normal force for the missile comes from the wing.

The rocket baseline has fixed tail surfaces with an exposed tail aspect ratio $A_T = 2.59$ and a tail planform area $S_T = 1.54 \text{ ft}^2$. At Mach 2 and an angle of attack $\alpha = 9.4 \text{ deg}$, the normal force coefficient of the tail is computed to be $(C_N)_{\text{Tail}} = 0.425$, based on the tail surface planform area. Based on the missile reference area, $(C_N)_{\text{Tail}} = 1.88$. Note that for the trimmed flight condition of Mach 2 and $\alpha = 9.4 \text{ deg}$, the tail normal force is relatively small compared to that of the deflected wing $[(C_N)_{\text{Tail}}/(C_N)_{\text{Wing}} = 1.88/7.91 = 0.24]$.

The total normal force coefficient of the wing-tail-body configuration is assumed to be the sum of the contributions from the wing plus the tail plus the body. As shown previously, the body normal force coefficient is a function of angle of attack, fineness ratio, and cross section geometry. The rocket baseline has a circular cross section ($a/b = 1$) and a fineness ratio $l/d = 17.99$. At an angle of attack of $\alpha = 9.4 \text{ deg}$, the body normal force coefficient is computed to be

$$(C_N)_{\text{Body}} = \sin(2\alpha) \cos(\alpha/2) + 2(l/d) \sin^2 \alpha = 1.28 \quad (2.37)$$

Note that the body normal force coefficient is only 16% of that provided by the deflected wing. Substituting, the total trimmed normal force coefficient at Mach 2, $\alpha = 9.4 \text{ deg}$, and $\delta = 12.6 \text{ deg}$ is

$$C_N = (C_N)_{\text{Wing}} + (C_N)_{\text{Tail}} + (C_N)_{\text{Body}} = 7.91 + 1.88 + 1.28 = 11.1 \quad (2.38)$$

A similar approach can be used to determine the wing normal force curve slope from angle of attack. $(C_{N\alpha})_{\text{Surface}}$ is used in sizing the tail to meet the static stability requirement. It is the derivative of the equation for surface normal force coefficient. At low angle of attack, $(C_{N\alpha})_{\text{Surface}} = 4/(M^2 - 1)^{1/2}$, if the parameter $M^2 > 1 + [8/(\pi A)]^2$. If the parameter $M^2 < 1 + [8/(\pi A)]^2$, then the appropriate equation is $(C_{N\alpha})_{\text{Surface}} = \pi A/2$. Note that $(C_{N\alpha})_{\text{Surface}}$ in these equations is based on the surface planform area and has the units per rad.

The total normal force curve slope from angle of attack is assumed to be the sum of the contributions from the wing plus the tail plus the body. As shown previously, the body normal force curve slope at low angle of attack is $(C_{N\alpha})_{\text{Body}} = 2[(a/b)\cos^2 \varphi + (b/a)\sin^2 \varphi]$. For the rocket baseline with a circular cross section ($a/b = 1$), $(C_{N\alpha})_{\text{Body}} = 2$ per radian. At Mach 2, the total normal force curve slope at low angle of attack, based on body cross-sectional reference area is

$$\begin{aligned} C_{N\alpha} &= (C_{N\alpha})_{\text{Wing}}(S_W/S_{\text{Ref}}) + (C_{N\alpha})_{\text{Tail}}(S_T/S_{\text{Ref}}) + (C_{N\alpha})_{\text{Body}} \\ &= 4/(2^2 - 1)^{1/2}(2.55/0.349) + 4/(2^2 - 1)^{1/2}(1.54/0.349) + 2 \\ &= 16.9 + 10.2 + 2 = 29.1 \text{ per rad} \end{aligned} \quad (2.39)$$

As a comparison, the total normal force curve slope from angle of attack (Chapter 7), which is based on wind tunnel data, is $C_{N\alpha} = 34.4 \text{ per rad}$. The predicted value of $C_{N\alpha}$ has an error of 14%.

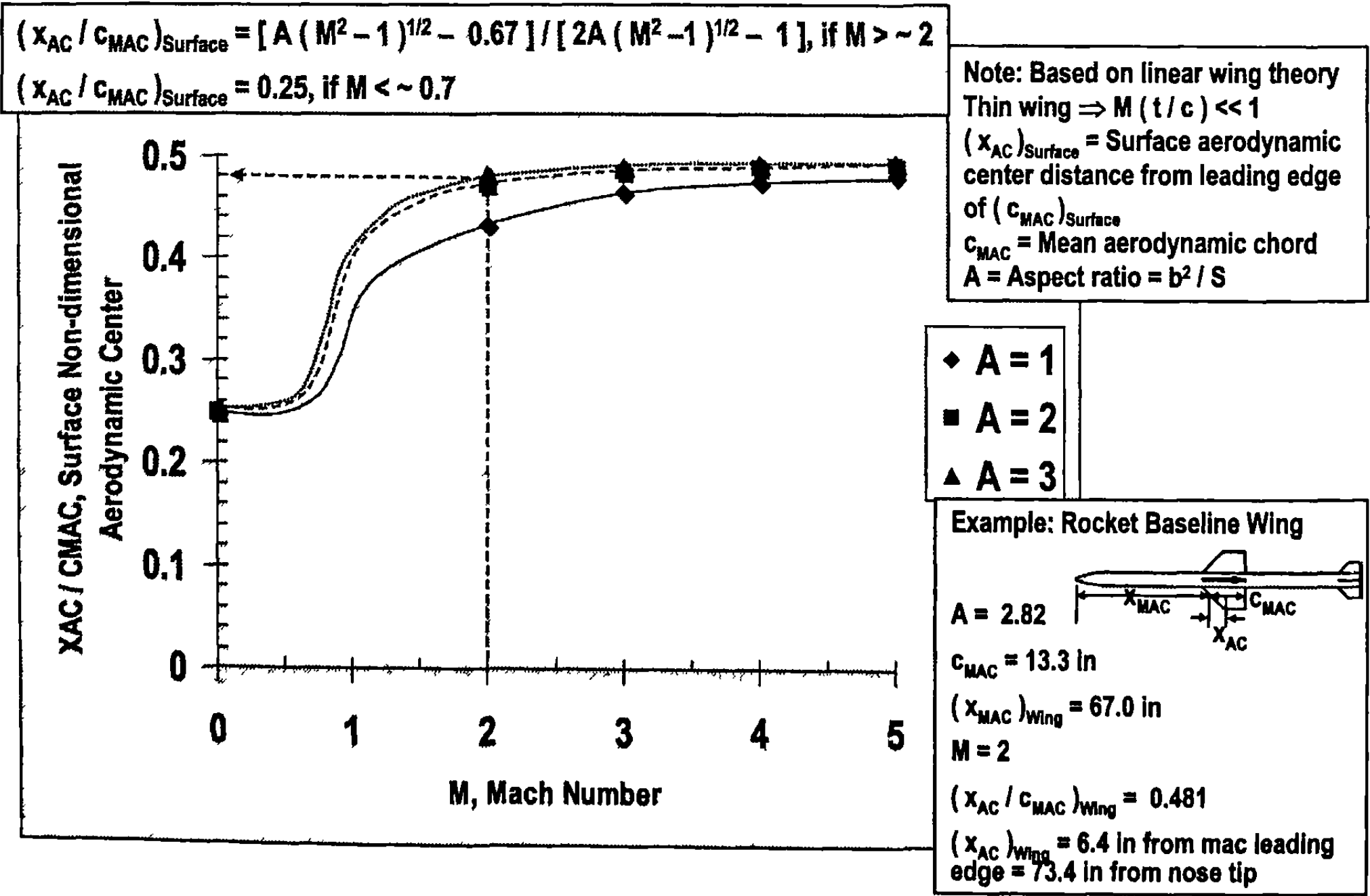
The normal force prediction is likely to be optimistic for very low Mach number (e.g., $M < 0.2$) and for very high altitude (e.g., $h > 80,000$ ft). For these flight conditions, boundary-layer transition from laminar to turbulent is delayed, resulting in a thicker boundary layer. The boundary-layer thickness on the aft body may blanket a significant portion of the tails, reducing the stability and control effectiveness of the tails.

2.11 Aerodynamic Center Prediction for Surfaces
(e.g., Wing, Tail, Canard)

Shown in Fig. 2.22 is the variation of aerodynamic center $(x_{AC})_{\text{Surface}}$ of a thin surface (e.g., wings, tails, canards) with Mach number and aspect ratio. Results are applicable at low angles of attack ($\alpha < 10$ deg). At low Mach number, $(x_{AC})_{\text{Surface}}$ is located near 25% of the mean aerodynamic chord C_{MAC} . At hypersonic Mach number $(x_{AC})_{\text{Surface}}$ shifts to 50% of C_{MAC} . Compressed carriage missiles with relatively long chord surfaces, such as AMRAAM AIM-120C, usually have a disadvantage of a larger supersonic hinge moment. The prediction at supersonic Mach number and low angle of attack is based on linear wing theory⁵, giving

$$(x_{AC}/C_{MAC})_{\text{Surface}} = [A(M^2 - 1)^{1/2} - 0.67]/[2A(M^2 - 1)^{1/2} - 1] \quad (2.40)$$

The rocket baseline missile wing has an exposed aspect ratio of 2.82, a mean aerodynamic chord length of 13.3 in, and a location of the leading edge of the



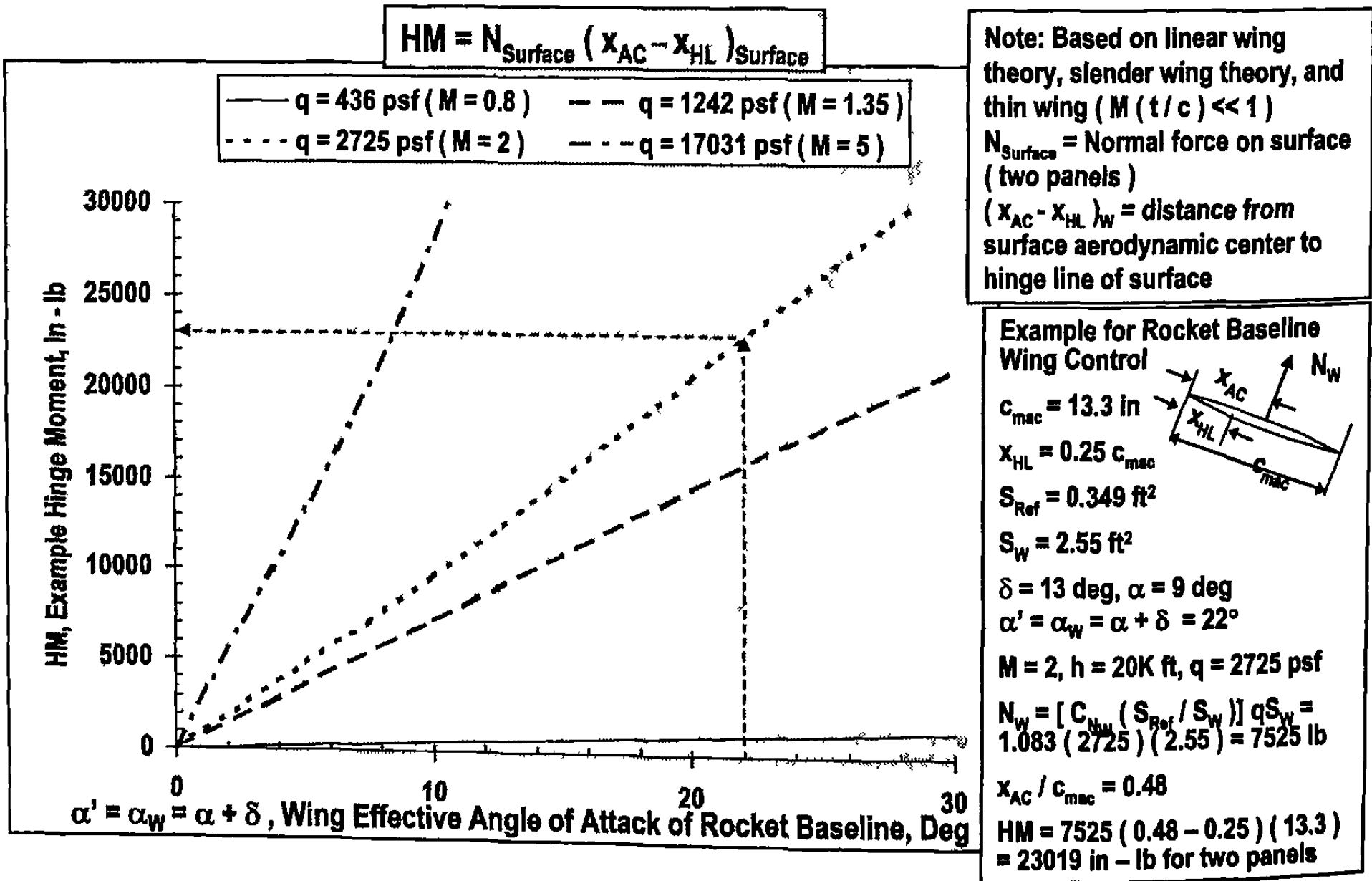
mean aerodynamic chord at 67.0 in from the nose tip. As shown in the figure, at Mach 2 the predicted aerodynamic center of the exposed wing is located at 48% of the length of the mean aerodynamic chord, providing a location of $(x_{AC})_{Wing} = 73.4$ in from the nose tip. At subsonic Mach number the wing aerodynamic center is located at approximately 25% of the mean aerodynamic chord, providing a location of $(x_{AC})_{Wing} = 70.3$ in from the nose tip. The center-of-gravity location of the rocket baseline missile is 84.6 in from the nose tip at launch and 76.2 in at burnout. Because the distance between the wing aerodynamic center and the missile center of gravity is always relatively small (less than 14.3 in) for all flight conditions, the deflection of the wing control causes a relatively small change in missile angle of attack. If either tail control or canard control were used for the rocket baseline missile instead of wing control, the change in angle of attack with control surface deflection would be larger because of the larger moment arm of a tail or canard control.

2.12 Hinge Moment Prediction

The hinge moment (HM) of an aerodynamic control surface is a function of the normal force $N_{Surface}$ on a surface times the moment arm between the aerodynamic center and the hinge line of the control surface $(x_{AC} - x_{HL})_{Surface}$. The equation is

$$HM = N_{Surface}(x_{AC} - x_{HL})_{Surface} \tag{2.41}$$

Shown in Fig. 2.23 is the variation of the HM with dynamic pressure and effective angle of attack for the rocket baseline missile. An assumed flight altitude is



$h = 20,000$ ft altitude. As previously, the prediction of N_{Surface} is based on linear wing theory at high supersonic Mach number and on slender wing theory at subsonic and low supersonic Mach number. It is also assumed that the surface section is thin [$M(t/c) \ll 1$]. At low subsonic Mach number, the aerodynamic center is located at the quarter chord of the mean aerodynamic center ($x_{AC} = 0.25C_{\text{mac}}$). For this example, the hinge line of the control surface of the rocket baseline is also assumed to be located at the quarter chord of the mean aerodynamic center ($x_{HL} = 0.25C_{\text{mac}}$). The hinge must be located either at or forward of the aerodynamic center to avoid unstable feedback through the flight control system.

An example is shown for the rocket baseline wing control at a maximum effective angle of attack $\alpha' = \alpha_w = 22$ deg. For this condition, the control deflection $\delta = 13$ deg and the body angle of attack $\alpha = 9$ deg. The total normal force on the two panels that comprise the wing is $N_{\text{Wing}} = [C_N S_{\text{Ref}}/S_w]qS_w = 1.083(2725)(2.55) = 7525$ lb. Also, from a previous calculation, the aerodynamic center is located at 48% of the mean aerodynamic center ($x_{AC} = 0.48 C_{\text{mac}}$). Finally, calculating the HM of the rocket baseline wing control at Mach 2, 20,000 ft altitude with maximum wing control deflection, $HM = 7525(0.48 - 0.25)(13.3) = 23,019$ in-lb for the two panels that comprise the wing. Each actuator of the two panels has an individual HM of 11,510 in-lb. Control surface actuators are sized by the highest HM flight condition, which usually occurs at the maximum maneuverability requirement for the highest dynamic pressure flight condition.

2.13 Thin Surface (e.g., Wing, Tail, Canard) Drag Prediction

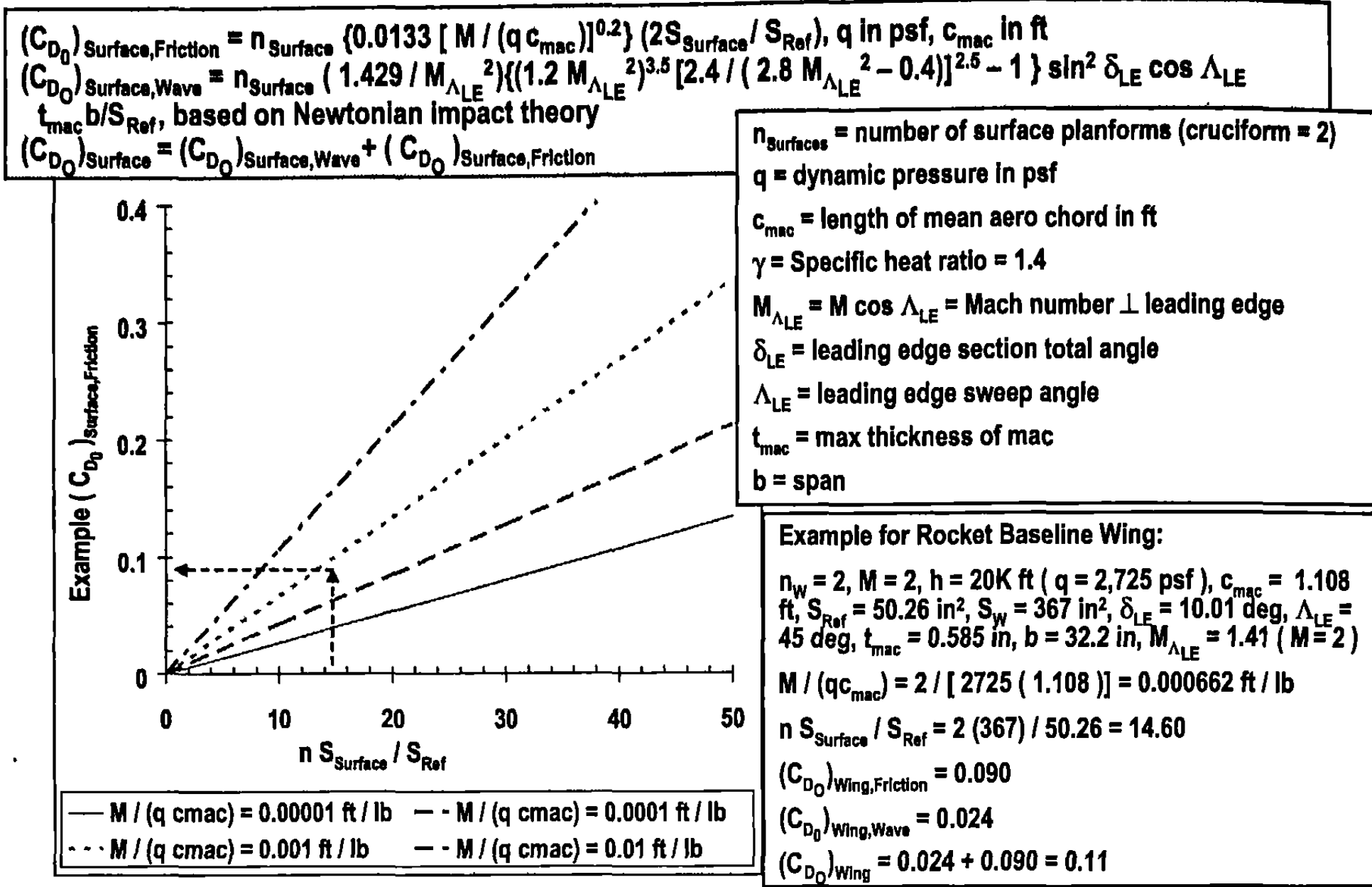
Equations for predicting thin surface drag coefficient are shown in Fig. 2.24. The surface skin friction drag prediction, based on Ref. 4, is the equation

$$(C_{D_0})_{\text{Surface,Friction}} = n_{\text{Surface}}[0.0133[M/(qC_{\text{mac}})]^{0.2}](2S_{\text{Surface}}/S_{\text{Ref}}) \quad (2.42)$$

In the equation the units of dynamic pressure q and mean aerodynamic chord C_{mac} are psf and ft respectively. Major contributors to the skin friction drag coefficient are the number of surface planforms N_{Surface} and the surface planform area S_{Surface} . The total wetted area for surface skin friction drag is equal to $2N_{\text{Surface}}S_{\text{Surface}}$. Surface skin friction drag coefficient is a weak function of Mach number, dynamic pressure, and the length of the mean aerodynamic chord. Assumptions for the equation are that the variation in the free stream speed of sound and viscosity with altitude is relatively small, and a turbulent boundary layer.

A second contributor to surface drag is wave drag. The wave drag prediction is based on modified Newtonian theory, described in Ref. 8, with the equation

$$\begin{aligned} (C_{D_0})_{\text{Surface,Wave}} = n_{\text{Surface}}[2/(\gamma M_{\text{ALE}}^2)]\{ \{[(\gamma + 1)M_{\text{ALE}}^2]/2\}^{\gamma/(\gamma-1)} \\ \times \{(\gamma + 1)/[2\gamma M_{\text{ALE}}^2 - (\gamma - 1)]\}^{1/(\gamma-1)} - 1\} \\ \times \sin^2 \delta_{\text{LE}} \cos \Lambda_{\text{LE}} t_{\text{mac}} b/S_{\text{Ref}} \text{ if } M_{\text{ALE}} > 1 \end{aligned} \quad (2.43)$$



contribution to zero-lift drag, based on two wings with four panels, is $(C_{D_0})_{\text{Wing,Friction}} = 0.090$. The wing wave drag contribution to zero-lift drag, based on two wings with four panels, is $(C_{D_0})_{\text{Wing,Wave}} = 0.024$. The total wing zero-lift drag coefficient is

$$(C_{D_0})_{\text{Wing}} = 0.09 + 0.02 = 0.11 \quad (2.45)$$

Note from above that 82% of the wing zero-lift drag is because of skin friction. Also note that the rocket baseline missile wing has a zero-lift drag coefficient that is only 27% of the body coast zero-lift drag coefficient (0.11 vs 0.41).

The rocket baseline has cruciform tail surfaces ($n_T = 2$), tail leading edge sweep angle of $\Lambda_{LE} = 57^\circ$, exposed tail planform area of $S_T = 1.54 \text{ ft}^2$, leading edge section total angle $\delta_{LE} = 6.17^\circ$, mean aerodynamic chord $C_{mac} = 1.025 \text{ ft}$, maximum thickness of the mean aerodynamic chord $t_{mac} = 0.33 \text{ in}$, and exposed span $b = 24.0 \text{ in}$. At Mach 2 and an altitude of 20,000 ft, the zero-lift drag coefficient of the tail surfaces, based on the missile reference area, is computed to be

$$(C_{D_0})_{\text{Tail}} = (C_{D_0})_{\text{Tail,Wave}} + (C_{D_0})_{\text{Tail,Friction}} = 0.003 + 0.048 = 0.051 \quad (2.46)$$

The zero-lift drag of the tail is much smaller than that of the wing. This is because of the smaller leading edge section angle (6.17° vs 10.01°) and the smaller planform area. Note that for the flight condition of Mach 2 and 20,000 ft altitude, the tail provides only 12% of the zero-lift drag compared to that of the body.

Finally, the total zero-lift drag coefficient is assumed to be the sum of the contributions from the body plus the wing plus the tail. Substituting, the total zero-lift drag coefficient of the rocket baseline during coasting flight at Mach 2 and 20,000 ft altitude is computed to be

$$C_{D_0} = (C_{D_0})_{\text{Body}} + (C_{D_0})_{\text{Wing}} + (C_{D_0})_{\text{Tail}} = 0.41 + 0.11 + 0.05 = 0.57 \quad (2.47)$$

2.14 Surface Geometry Tradeoffs

Shown in Fig. 2.25 are examples of alternative surface panel geometries for missile tails, canards, and wings. The figure is a comparison of a triangular (delta), conventional trapezoidal with an aft swept leading edge, “bow tie” trapezoidal with a forward swept leading edge and an aft swept trailing edge, a double swept leading edge, and a rectangular surface planform geometry. The comparison is based on the same planform area because roughly the same area is required to provide the same static stability and maneuverability. The comparison also assumes that there is a span constraint, which is often the case for launch platform compatibility. Note from the figure that the double swept leading edge geometry has a major advantage of minimizing the shift in aerodynamic center. As a wing, the double swept leading edge geometry has more nearly constant static margin. As a control surface, the double swept leading edge geometry has reduced HM, allowing lighter weight, smaller size, and lower cost actuators. In addition to a

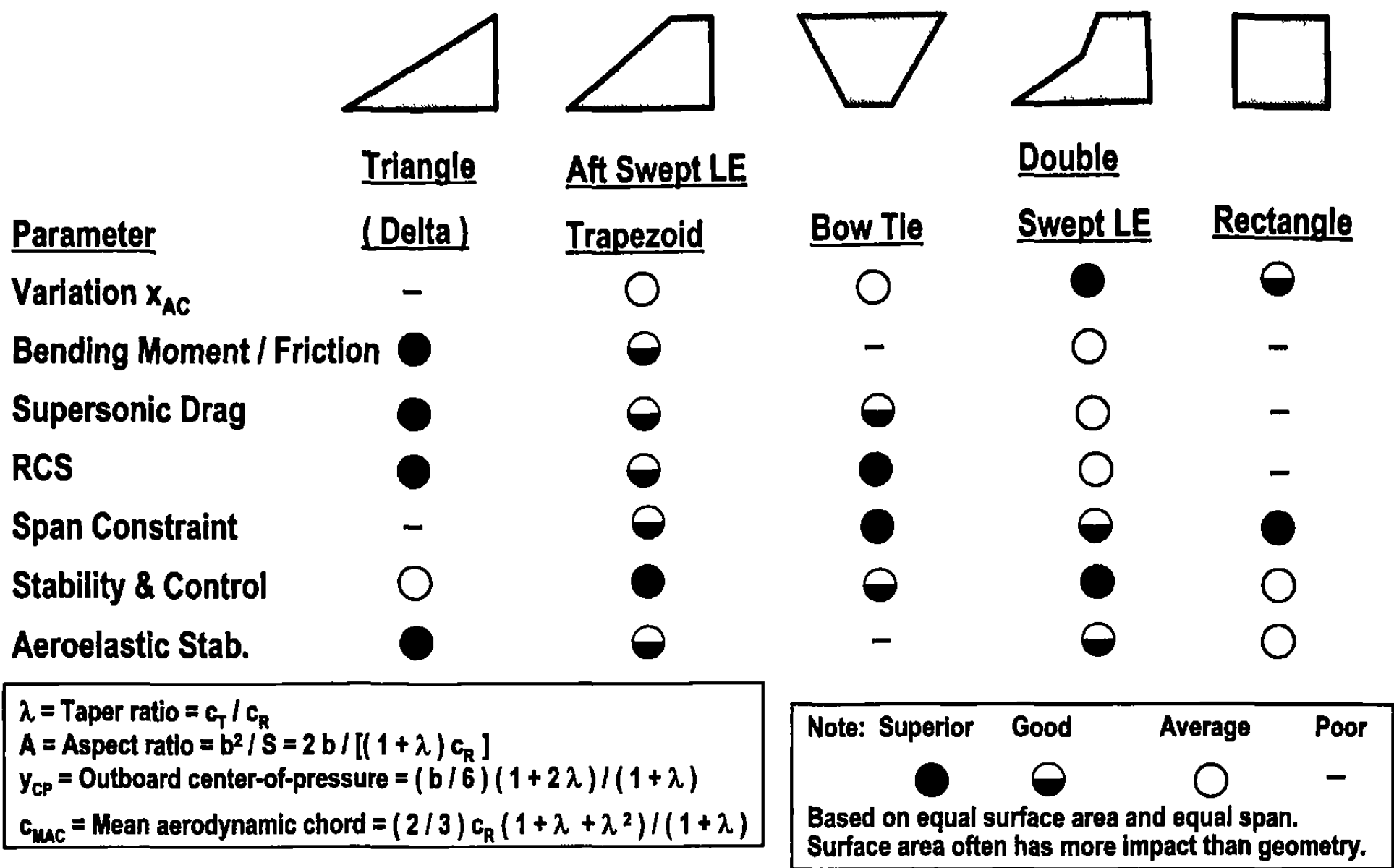


Fig. 2.25 Examples of wing, tail, and canard panel geometry alternatives.

low variation in the location of aerodynamic center, the double swept leading edge geometry has superior control effectiveness (especially at subsonic Mach number), good aeroelastic stability, and no major weaknesses in the other design considerations. Standard Missile and HARM are examples of missiles with double delta wings. Both are high-speed missiles that benefit from the reduced static margin shift of a double swept leading edge wing. Examples of missile that have control surfaces with double swept leading edges are Super 530, Standard Missile, Crotale, AIM-91, Magic, and U-Darter. All are high speed missiles that benefit from the reduced HM and lighter weight actuators of a double swept leading edge. Another example of a relatively new aeromechanics technology is the bow tie geometry, with a forward swept leading edge and an aft swept trailing edge. This is beneficial for missiles that require low radar cross section (RCS) or have small span requirements for aircraft compatibility. The smaller span also provides a smaller roll moment of inertia, allowing higher roll acceleration. A bow tie surface has low RCS because of the combination of the wing sweep and the attenuation of backscatter bouncing off the adjacent body. In addition to a low RCS and a smaller span requirement, the bow tie geometry has low supersonic drag and good control effectiveness. An inherent disadvantage of a bow tie is the increased potential for aeroelastic instability. Composite structure is synergistic with a bow tie surface because the higher stiffness of composites mitigates aeroelastic instability and facilitates aeroelastic tailoring of the wing. Composite material may also be used in radar absorbing structure. Another disadvantage of a bow tie surface is a larger bending moment, which increases shaft friction and requires more actuator torque to maintain bandwidth. The U.S. AGM-129 Advanced Cruise Missile and the Russian AA-10 are examples of missiles with forward swept wings. A third

alternative of a triangular (delta) surface geometry is a candidate if the span can be accommodated within the launch platform span constraint. A delta surface has strengths of low bending moment/less required thickness for structural integrity (which results in lighter weight), low supersonic drag (because of the larger sweep angle), low radar cross section, and high structural stiffness. A disadvantage of delta geometry is a large variation in aerodynamic center. AGM-65 and AIM-54 are examples of missiles with triangular wings. A fourth alternative, a trapezoidal surface with an aft swept leading edge, is often selected for missile wing, canard, and tail surfaces because it has high control effectiveness and does not have any serious weaknesses in the other attributes. Patriot and AIM-9X are examples. Finally, rectangular surfaces have many disadvantages and are used less often, but a superior characteristic of a rectangular surface is that it has the largest surface area that can be accommodated within launch platform chord and span constraints. A rectangular planform also has a small shift in aerodynamic center because of the relatively small required chord. The AGM-130, AGM-65, and AIM-54 are examples of missiles with rectangular tails.

The surface planform geometry parameters shown in the figure include taper ratio, aspect ratio, outboard location of the center of pressure, and the mean aerodynamic chord. The planform geometry is based on two panels joined at the root chord. Taper ratio is the ratio of the tip chord to the root chord ($\lambda = C_T/C_R$). Aspect ratio is defined as the square of the surface span divided by the surface area ($A = b^2/S$). For a conventional planform with straight leading and trailing edges,

$$A = 2b/[(1 + \lambda)C_R] \quad (2.48)$$

The outboard location of the mean aerodynamic chord, or center-of-pressure location, of a straight leading/trailing edge planform is given by

$$y_{CP} = (b/6)(1 + 2\lambda)/(1 + \lambda) \quad (2.49)$$

Note from the equation for y_{CP} that $\lambda = 0$ (delta wing) provides the benefit of a small value of the lateral center of pressure ($y_{CP} = b/6$), resulting in lower bending moment, while $\lambda = 1$ (rectangular wing) gives the adverse effect of a larger value of the lateral center-of-pressure location ($y_{CP} = b/4$) and a larger bending moment.

The equation for the length of the mean aerodynamic chord of a trapezoidal planform is

$$C_{MAC} = (2/3)C_R(1 + \lambda + \lambda^2)/(1 + \lambda) \quad (2.50)$$

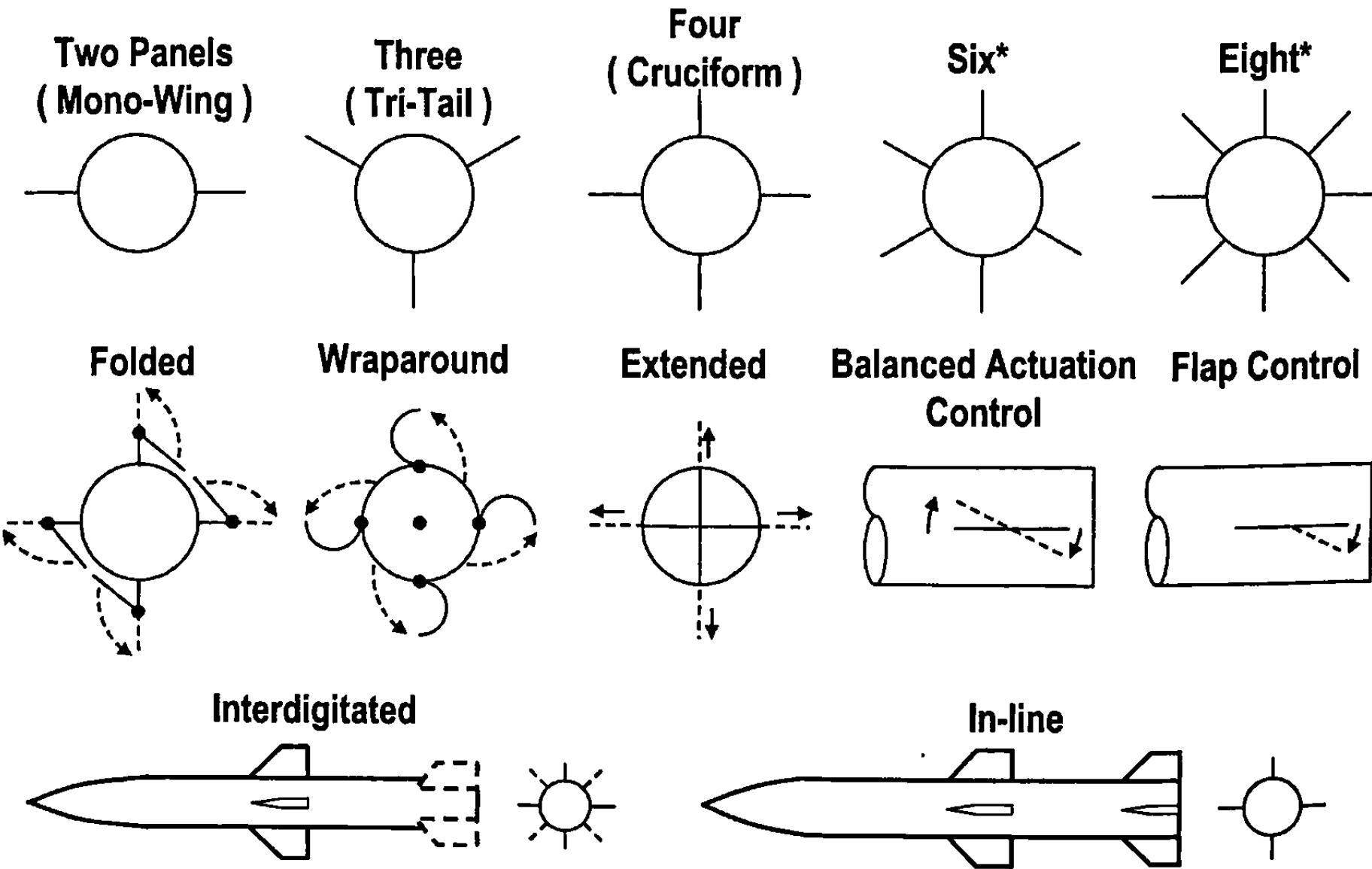
The aerodynamic center, which is the same as the center of pressure for a surface with a symmetrical noncambered cross section, is located on the mean aerodynamic chord.

Surface area usually has a greater impact on aerodynamic stability, control, and performance than surface geometry. Because missiles are usually span limited, they usually cannot take advantage of high aspect ratio geometry.

Wing, stabilizer, and control surface integration alternatives are shown in Fig. 2.26 for a number of surfaces (two to eight), folding arrangements (folded, wraparound, extended), control surface deflection (balanced actuation control, flap control), and tandem surface orientation (interdigitated, inline).

At least three tail surfaces are required for static stability. An example of a missile with three tails is the AGM-69 SRAM missile. For most missiles, the wings, tails, and canards are comprised of four surfaces (cruciform). This works best with skid-to-turn maneuvering. For bank-to-turn maneuvering missiles, two wing surfaces are usually used, with at least three tail surfaces. If a missile has more than four tail surfaces, the reasons are often either a span limit imposed by the launch platform or an attempt to reduce the rolling moment from vortices. A high angle of attack missile may have six or more tail surfaces that are uncoupled from the airframe and are on bearings that are free to roll. The free-to-roll tails provide the required static stability, but with low induced roll.

Surfaces may be folded to reduce the span requirement for compressed carriage with the launch platform. A folded surface usually has either a ballistic actuator or a spring that deploys the surface after separation from the launch platform. An example of a missile with folded tail surfaces is the AGM-86 cruise missile. Another approach to compressed carriage is wraparound surfaces. These are particularly suited for tube launch. A third approach is surfaces that are extended laterally after launch, such as switchblade wings and diamondback wings. The BGM-109 Tomahawk cruise missile has a switchblade wing. Enhanced range JDAM has a diamondback wing. Finally, reduced span surfaces provide compressed carriage. An example is compressed carriage AMRAAM.



Note: More than four tails are usually free-to-roll pitch / yaw stabilizers, for low induced roll.

Fig. 2.26 Examples of surface arrangement and aerodynamic control alternatives.

Alternatives for the location of the hinge line/control shaft of flight control actuators are either near the aerodynamic center of the surface (balanced actuation control) or near the trailing edge of the surface (flap control). Balanced actuation control, using an all movable surface, has advantages of lower HM and lower drag. Flap control has advantages of a stable HM and generally has a lower radar cross section.

Tandem surfaces include canard-tail and wing-tail planforms. Alternative orientation arrangements of the surfaces are either inline or interdigitated. Interdigitated surfaces may have advantages of less induced roll ($C_{l\phi}$) and launch platform compatibility. However, inline surfaces are usually preferred because of lower drag and RCS.

2.15 Flight Control Alternatives

Most missiles use four aerodynamic control surfaces to provide combined integrated control of all three axes (pitch/yaw/roll). However, a few missiles use only two control surfaces to fly with combined two-axis control (either pitch/yaw or pitch/roll) and a few missiles use more than four control surfaces to fly with independent, dedicated control of each axis. Other missiles have combined control of the pitch/yaw axes with independent roll control. Figure 2.27 is a comparison of the control integration approaches. Advantages of missiles that use only two-axis control (pitch/yaw, pitch/roll) are lower cost of the flight control actuators (only two actuators required) and ease of packaging only two actuators. Stinger is a free-to-roll missile with two

Control Integ	Control Surfaces	Example	Control Effect	Cost	Packaging
◆Pitch / Yaw	2	Stinger FIM-92 	—	●	●
◆Pitch / Roll	2	ALCM AGM-86 	—	●	●
◆Pitch / Yaw / Roll	3	SRAM 	—	◐	◐
◆Pitch / Yaw / Roll	4	Adder AA-12 	○	○	○
◆Pitch + Yaw + Roll	5	Kitchen AS-4 	●	—	—
◆Pitch / Yaw + Roll	6	Derby / R-Darter 	◐	—	—

Note: ● Superior ◐ Good ○ Average — Poor

Fig. 2.27 Most missiles use four control surfaces with combined pitch/yaw/roll control integration.

control surfaces for combined pitch/yaw control, but no control of the roll axis. The ALCM cruise missile is a bank-to-turn missile with combined pitch/roll control, but no control of the yaw axis. A disadvantage of two-axis control is that it does not provide precision maneuvering. Combined three-axis control requires at least three or four surfaces. The control effectiveness is typically less for a missile with three control surfaces. Another disadvantage of a combined control missile with three control surfaces, such as SRAM, is a larger induced rolling moment. Also, it is more difficult to design an efficient autopilot. Another potential approach, to use independent control surfaces for each axis, is uncommon for a missile. In this case, the missile flight control is similar to an aircraft. The Russian Kitchen AS-4, which has five control surfaces (two for pitch, two for roll, one for yaw) is an example of independent axis control. Although independent axis control has high control effectiveness and a relatively simple autopilot, it is more expensive because of the larger number of actuators, and it requires more volume for the actuators. Another example of flight control integration that is most appropriate for high angle of attack, highly maneuverable missiles, is the approach used by R-Darter. It has six control surfaces (four surfaces for combined pitch/yaw control plus two surfaces for dedicated roll control). Because vortex shedding and induced roll are problems for high angle of attack missiles, dedicated roll control provides more roll control effectiveness at high angle of attack.

Exo-atmospheric interceptors such as SM-3 and THAAD have dedicated control for each axis. Four divert thrusters (up, down, left, right) provide direct forces to change the flight trajectory, while eight attitude control thrusters [pitch up, pitch down, yaw right, yaw left, roll clockwise (two thrusters), roll counterclockwise (two thrusters)] provide independent stabilization in the three axes.

The types of flight control (tail, canard, wing, thrust vector/reaction jet control) and the alternative approaches for design and integration with fixed surfaces are shown in Fig. 2.28. For tail control, the control surface design alternatives include the number of tails (usually four) and the launch platform integration (no compressed carriage, folded, wraparound, extended). A tail control missile may also have forward surfaces (fixed wings, strakes, canards). The forward fixed surfaces may be either inline or interdigitated with the tail controls. The forward surfaces of a tail control missile may consist of two surfaces (usually used with bank-to-turn maneuvering), three surfaces (usually used with a tri-tail), or four (cruciform) surfaces.

For canard control, there are similar control alternatives. Canard control has an additional design alternative of two control surfaces for a rolling airframe missile. Using only two control surfaces reduces cost and subsystem volume. Examples of missiles with two control surfaces are the 5-in-diameter Rolling Airframe Missile (RAM) and the 2.75-in-diameter Stinger missile. The number of tail surfaces for a canard control missile may be tri-tail (three), cruciform (four), or more (six, eight) surfaces. The missile may have free-to-roll tails to minimize the rolling moment at high angle of attack. An example of a highly maneuverable canard control missile with free-to-roll tails is the Python 4.

Wing control requires three or more tail surfaces for static stability, similar to canard control. Although a wing control could have fixed forward surfaces

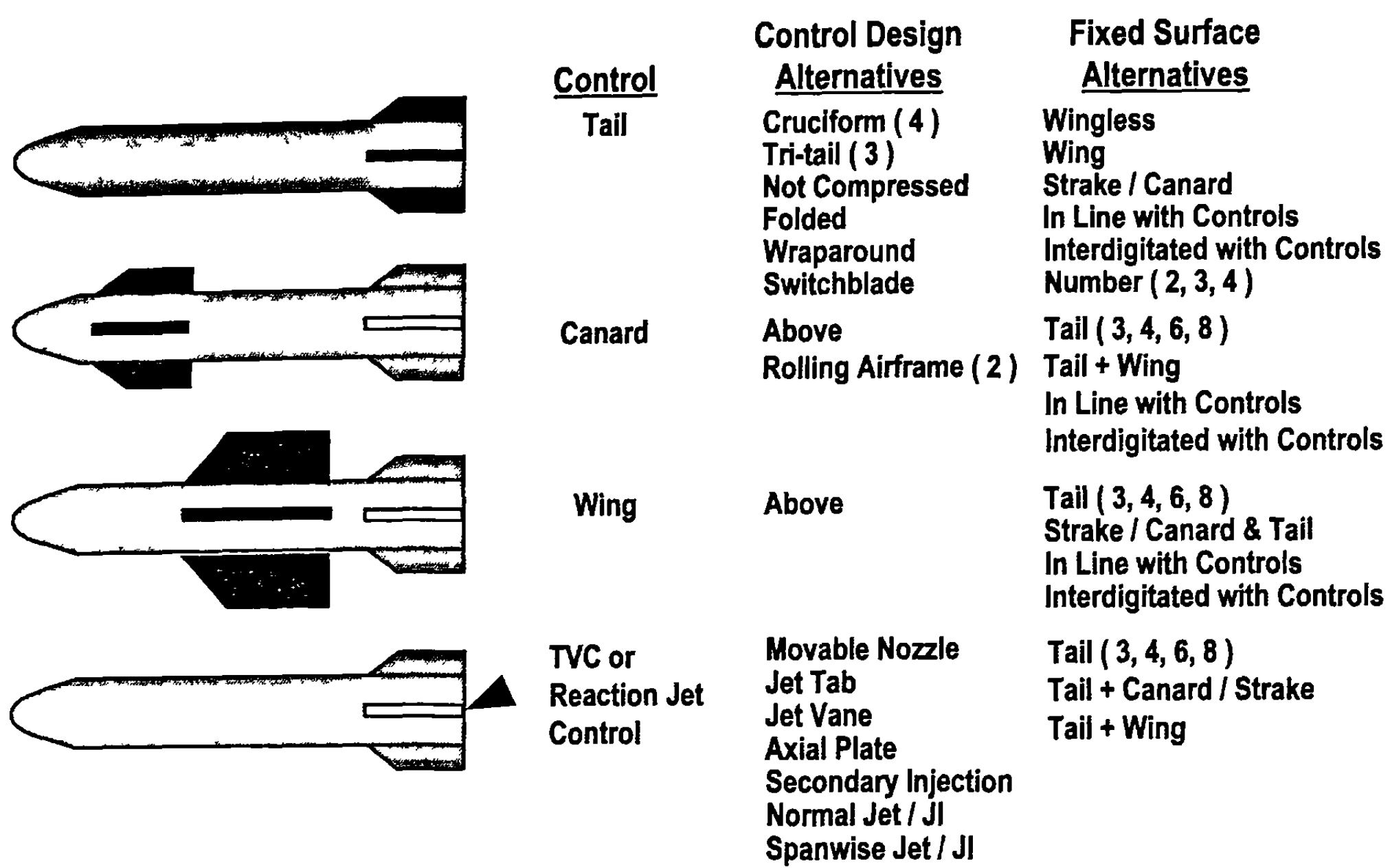


Fig. 2.28 There are many flight control alternatives.

(canards/strakes), it has not been used because of problems at high angle of attack from vortex shedding.

Finally, unconventional flight control includes TVC and jet interaction/reaction jet control. There are a number of approaches to unconventional flight control, with considerations including performance, cost, weight, packaging, and the capability for roll control. Thrust vector/reaction jet control is used when aerodynamic control has either insufficient control effectiveness or an excessive response time. Unconventional flight control is usually augmented by aerodynamic control for flight at higher dynamic pressure and long duration. Fixed surfaces used with thrust vector/reaction jet control include tails (three or more), tails plus canards/strakes, and tails plus wings.

Advantages and disadvantages of tail control are illustrated in Fig. 2.29. Typical advantages of tail control include an efficient packaging of the subsystems around a blast tube, a low HM and a low actuator torque because of the small local angle of attack, low induced rolling moment because the vortices shed by tail deflection trail behind the missile, and good efficiency at high angle of attack. Note from the figure that for a statically stable tail control missile, the control surface deflection to trim the missile reduces the local angle of attack at the control surface. A disadvantage of tail control is decreased normal force and lift for a statically stable missile because of the negative incremental normal force/lift from the control surface deflection. As shown in the figure, for a statically stable tail control missile, the control deflection force is opposite that of the force on the body. Conversely, for a statically unstable missile, the control deflection force adds to the force on the body.

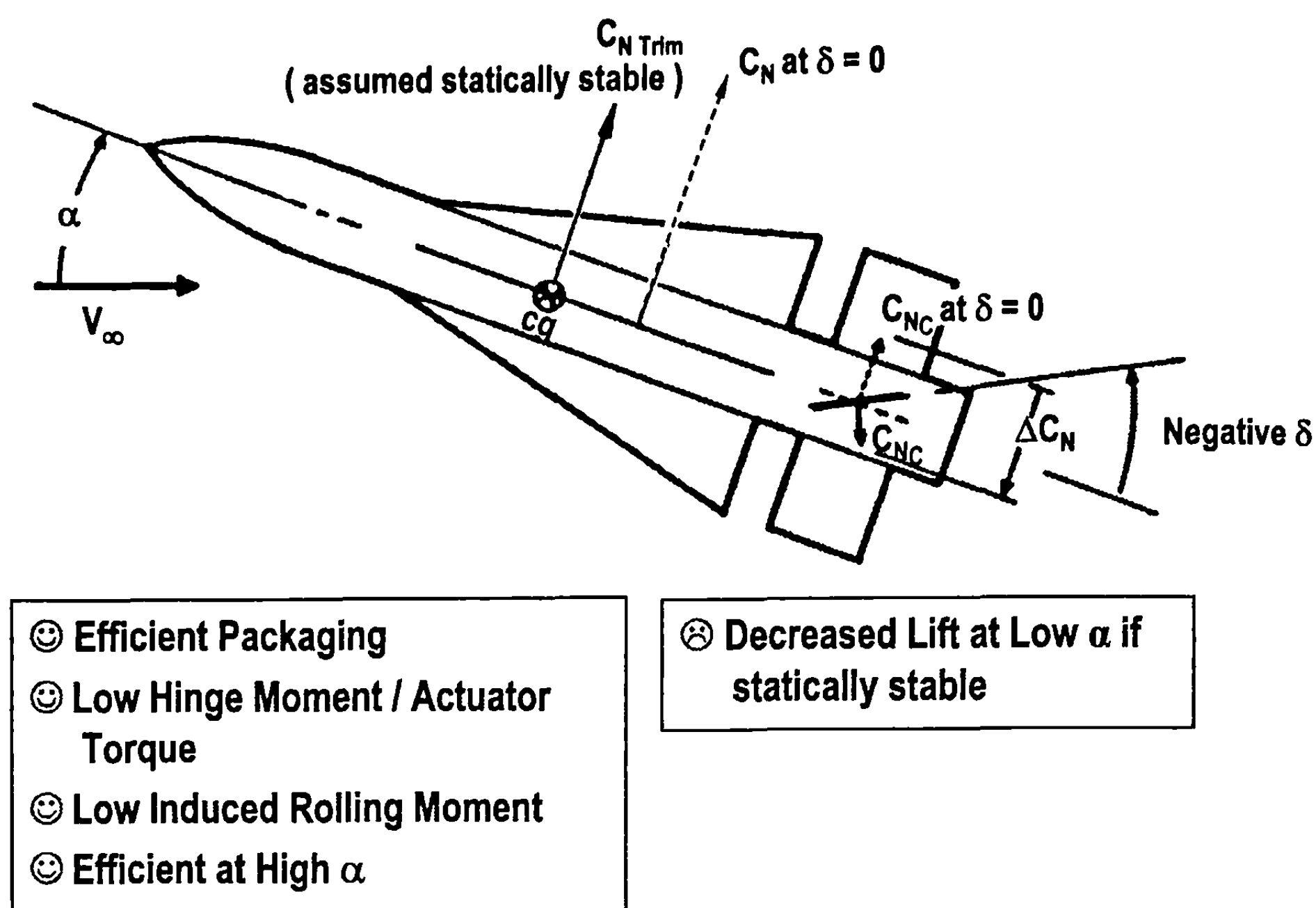


Fig. 2.29 Tail control is efficient at high angle of attack.

For a statically stable missile, tail control is usually more effective than conventional canard control at high angle of attack. As shown in Fig. 2.30, the deflection of the tail to trim the missile reduces the local angle of attack at the tail. As a result the tail can be deflected to a high deflection angle without stall. In the case of conventional canard control, the deflection of the canard to trim the missile increases the local angle of attack at the canard. The canard can be deflected only to relatively small angles before stalling. Also, the vortices shed by the canard can interact with the tail, resulting in a loss of roll control and large induced roll.

Note that a fixed forward surface in front of the movable canard (such as the split canards of Python 4) alleviates the problems of stall, loss of roll control, and induced roll at high angle of attack. Also, free-to-roll tails (such as the free-to-roll tails of Python 4) alleviate the problems of loss of roll control and induced roll at high angle of attack. The Python 4 free-to-roll tails are mounted on roll bearings that are decoupled from the airframe. Another approach to improve roll control effectiveness at high angle of attack is dedicated roll control surfaces, such as the Derby and R-Darter missiles. Dedicated roll control surfaces avoid the problem of control surface deflection saturation and simplify the autopilot design. Finally, a statically unstable missile alleviates the problem of canard stall. The required deflection of the canard for trim is opposite the angle of attack, reducing the local angle of attack at the canard.

About 70% of tail control missiles have low aspect ratio wings. Wings can provide greater aerodynamic efficiency (L/D) and higher maneuverability. Shown in Fig. 2.31 are examples of 24 subsonic and supersonic missiles that

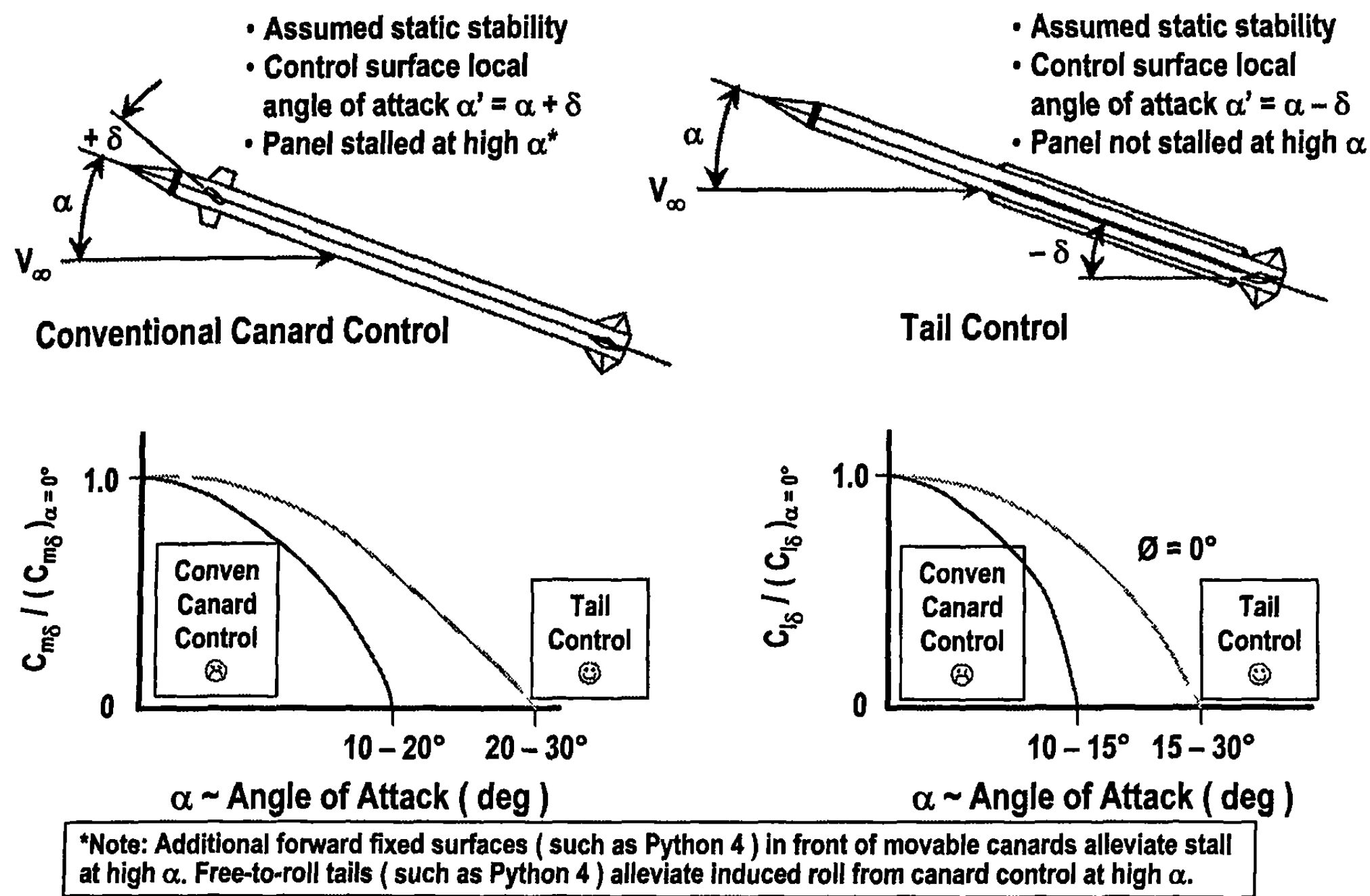
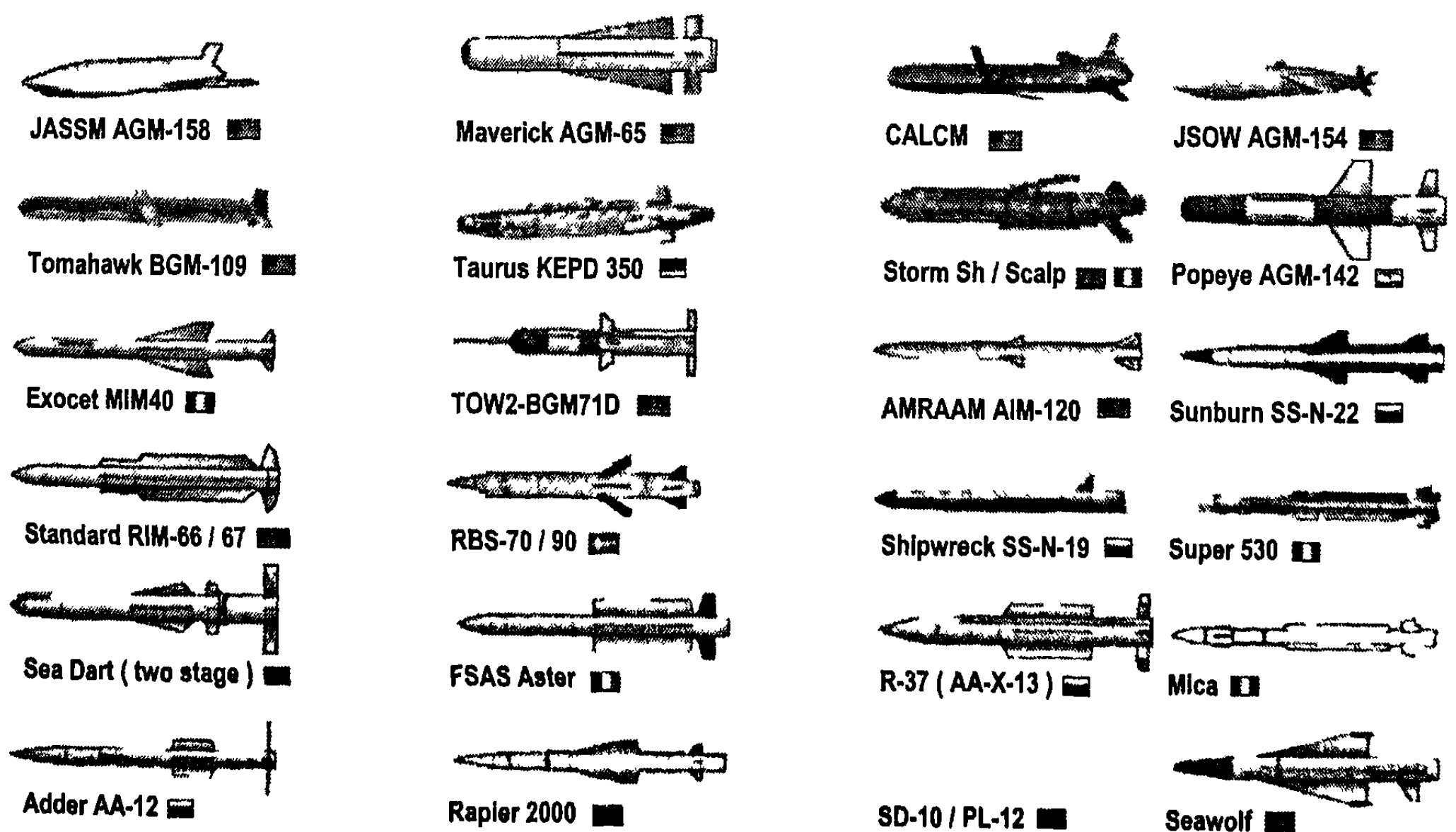


Fig. 2.30 Tail control is more effective than conventional canard control at high angle of attack.



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Fig. 2.31 About 70% of tail control missiles also have wings. Figure also in color section.

have wings and use tail control. The first 10 missiles are subsonic missiles. Subsonic missiles with wings, such as cruise missiles, generally have higher wing aspect ratio. The remaining 14 missiles in the figure are supersonic missiles. For supersonic missiles, the wing aspect ratio is generally less than 2.

An example of a new aeromechanics technology is lattice tail control. Lattice fins have advantages of lower HM and higher control effectiveness. Figure 2.32 shows a comparison of lattice tail control with two conventional approaches to tail control, balanced actuation fin and flap control. Except for RCS, lattice tail control has good-to-superior performance. RCS is particularly high when there is a resonance of a lattice dimension with the radar wavelength. For example, a lattice with individual compartments of 1 × 1×1 in will have high RCS if it is illuminated by an X-band (10 GHz frequency or 1-in wavelength) radar. Also shown in the figure are examples of missiles with the tail control alternatives of lattice tail control (Adder AA-12/R-77), conventional balanced actuation fin control (ASRAAM AIM-132), and flap tail control (Hellfire AGM-114). The vertical arrangement of surfaces and the smaller chord length of lattice control result in greater control effectiveness and less variation in the location of the center of pressure of the tail, leading to a lower HM.

Lattice fins, such as the fins shown in the top right section of Fig. 2.33, are most appropriate for either subsonic or high supersonic missiles. The bottom right section of the figure shows the flow through the lattice at a subsonic, transonic, a low supersonic, and a high supersonic Mach number. At subsonic Mach number the drag of lattice fins is comparable to that of traditional fins. At transonic Mach number, lattice fins have higher drag and lower control

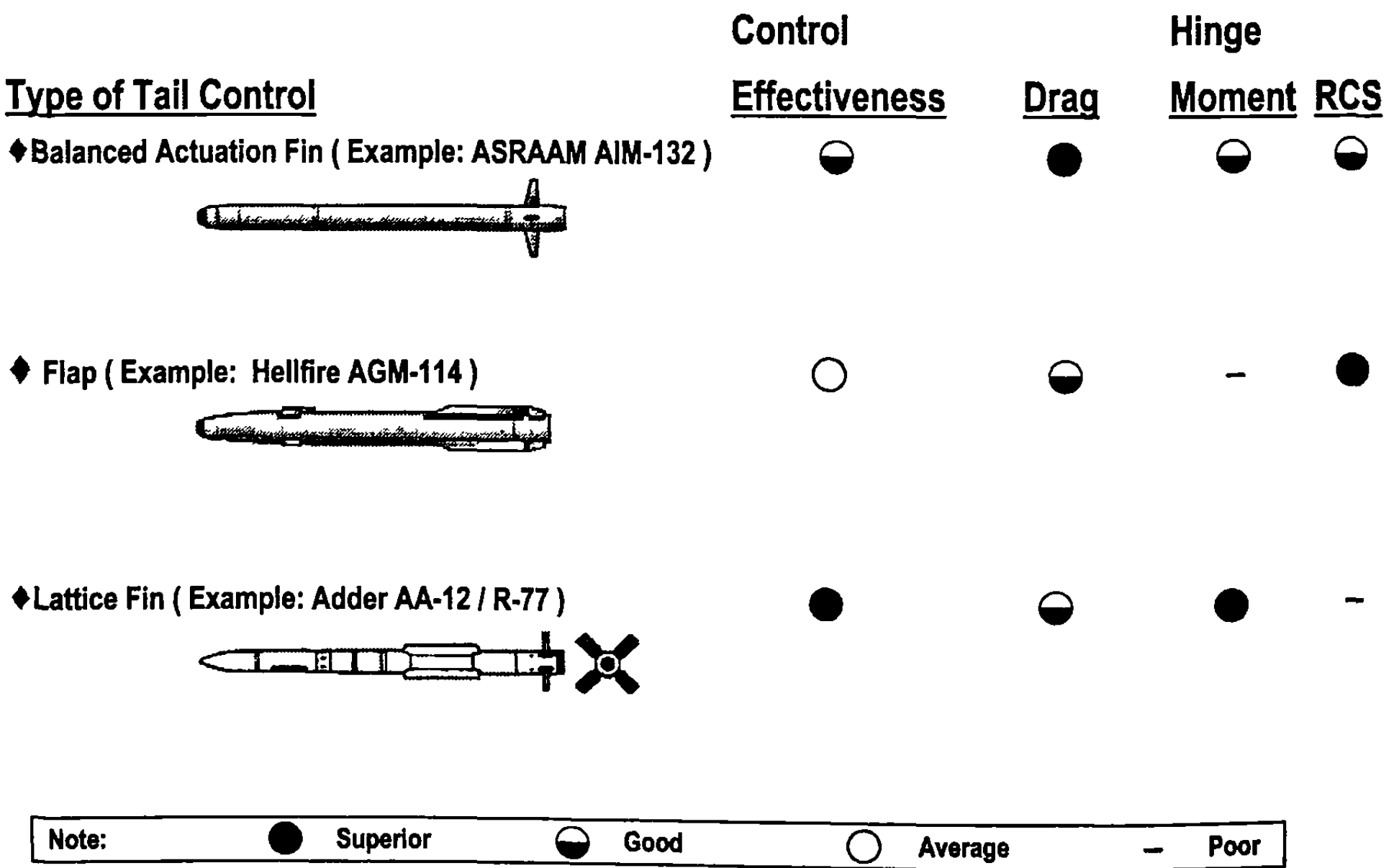


Fig. 2.32 Tail control alternatives: conventional balanced actuation fin, flap, and lattice fin.

◆Advantages

◆High control effectiveness at low subsonic and high supersonic Mach number

◆Low hinge moment

◆Short chord length

◆Disadvantages

◆High RCS (cavities, normal leading edges)

◆High drag at transonic Mach number (choked flow)

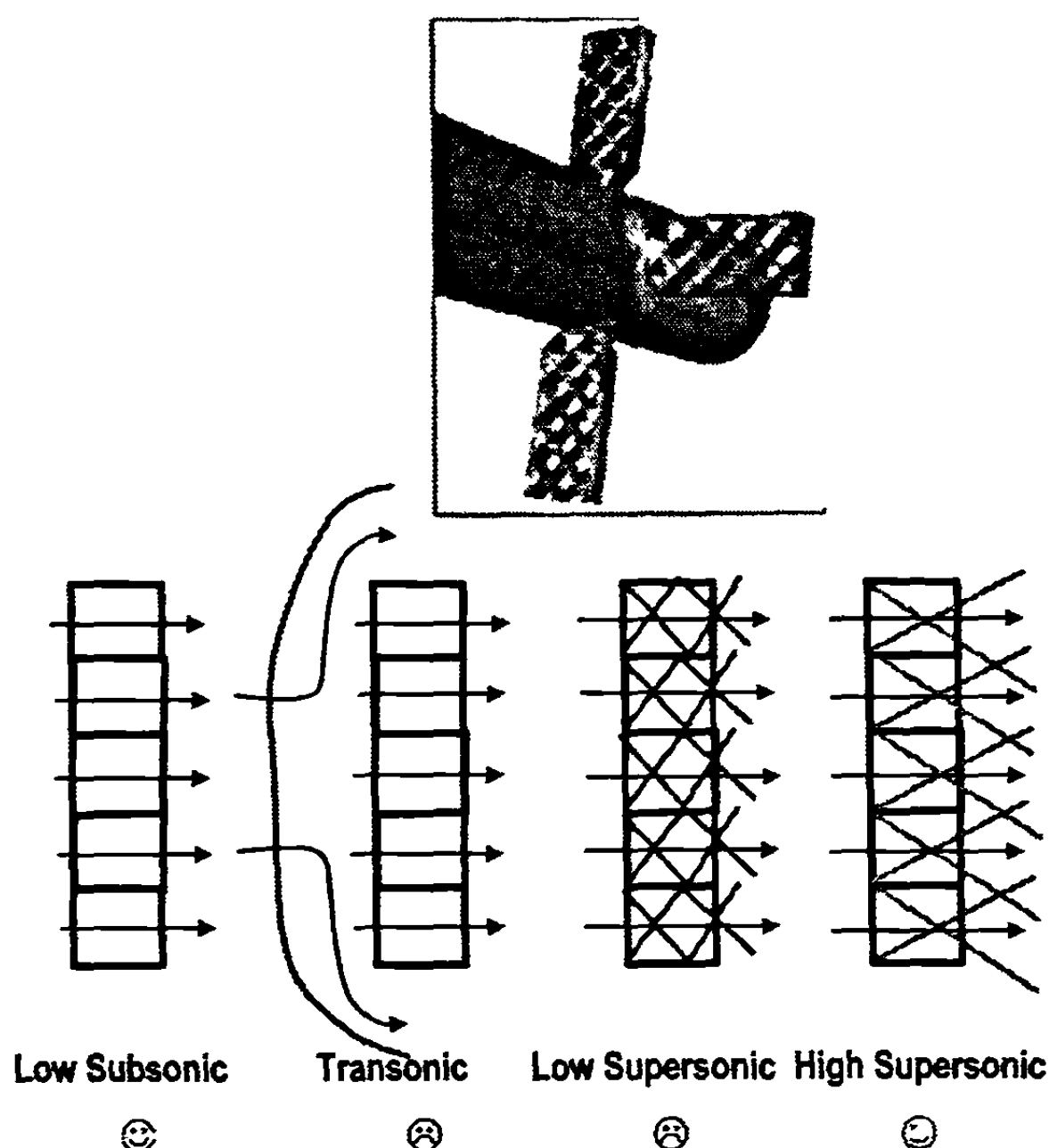


Fig. 2.33 Lattice fins have advantages for low subsonic and high supersonic missiles. Figure also in color section.

effectiveness than traditional fins. At a low transonic freestream Mach number less than 1, the local flow through the lattice can accelerate to Mach 1 because of the thickness of the surface and the thickness of the boundary layer. For a transonic freestream Mach number slightly greater than 1, the flow through the lattice remains choked. A detached, normal shock wave in front of the lattice spills excess air flow around the lattice. The lattice remains choked until the Mach number is sufficiently high to allow the lattice to swallow the shock. An oblique shock is then formed on each leading edge of the surfaces of the lattice. At low supersonic Mach number the oblique shock angle of each leading edge is large, and the oblique shocks are reflected downstream off the lattice surfaces. The large number of reflected shocks results in a relatively high drag. At higher Mach number the oblique shock angle of each leading edge is smaller, passing through the lattice without intersecting a lattice surface, resulting in low drag. The leading edge angles of the lattice can be small because of the efficiency of the box structure. The small angle and thickness provide low drag.

A disadvantage of lattice fins is high RCS. There are numerous cavities, with dimensions comparable to the threat radar wavelength (e.g., 1 in), providing resonance. Also, the numerous leading edges are not swept, resulting in large backscatter.

In summary, lattice fins have advantages of high control effectiveness, low HM, and short chord length. Their best application is at low subsonic or high supersonic Mach number, where the drag is comparable to conventional fins. A disadvantage of lattice fins is high RCS.

Figure 2.34 shows the advantages and disadvantages of conventional canard control. Advantages of canard control include 1) efficient packaging of the

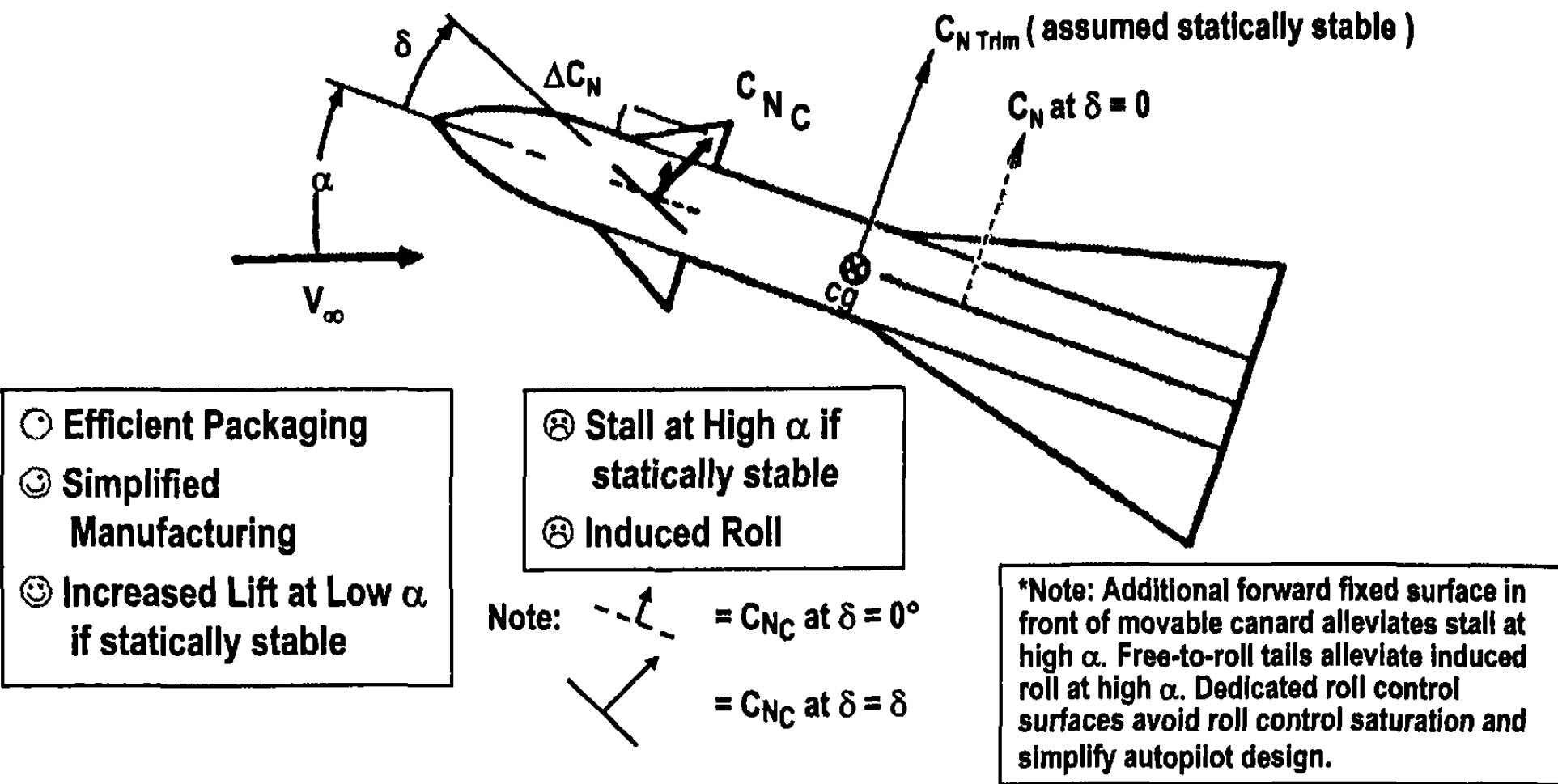


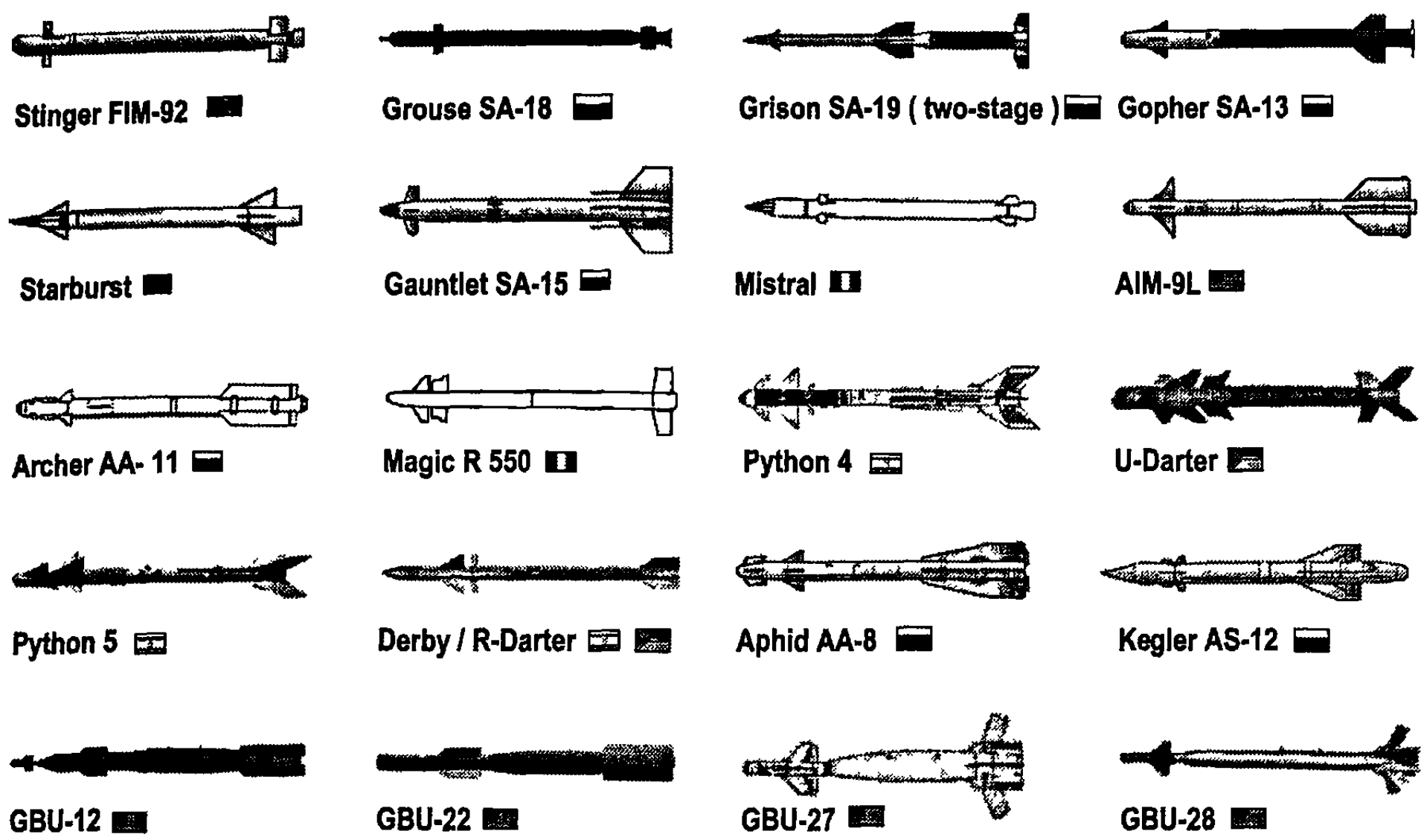
Fig. 2.34 Conventional canard control is efficient at low angle of attack but stalls at high alpha. Figure also in color section.

guidance, navigation, and control subsystems in the same location; 2) simplified manufacturing and ease of integration of the guidance, navigation, and control subsystems; and 3) for a statically stable missile, increased lift at low angle of attack. The figure illustrates the incremental normal force from the canard to trim the missile, providing additional lift. Disadvantages of conventional canard control include 1) a tendency to stall at high angles of attack for a statically stable missile and 2) a large induced roll at high angles of attack. As noted previously, these problems are alleviated by the use of split canards, free-to-roll tails, and dedicated roll control surfaces.

Shown in Fig. 2.35 are examples of operational missiles that have canard control. Canard control missiles do not have wings because the induced rolling moment from the vortex shedding from the canard makes it difficult to integrate a wing. Also, the additional surface area of the canard and the larger tail required for stability usually provide sufficient aerodynamic efficiency (L/D) and maneuverability. Because location of the canard is forward of the center of gravity, the canard is destabilizing, requiring a larger tail for stability. Shown in the figure are 20 operational wingless missiles that use canard control. Fifteen of the 20 missiles are air intercept missiles (SAM or AAM). Sixteen of the 20 missiles, including the Kegler AS-12, are supersonic missiles. Four are subsonic laser-guided bombs (GBU-12, 22, 27, and 28).

A split canard has a smaller local angle of attack, enhancing the maneuverability at high angle of attack. Modern highly maneuverable missiles use split canards for flight control. Split canard control has a fixed surface in front of the movable canard. Shown in Fig. 2.36 is a schematic of the local flow that illustrates the advantage of split canards. The incremental normal force coefficient (ΔC_N) in the figure is the difference between the normal force coefficient of the deflected control surface and the normal force coefficient of an undeflected control surface. Note that the forward surface reduces the local effective angle

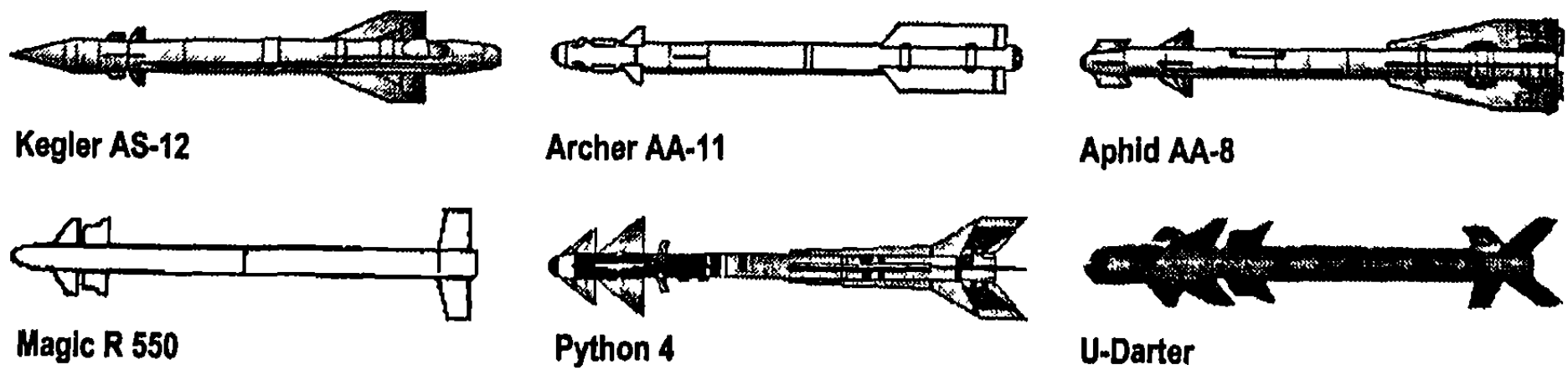
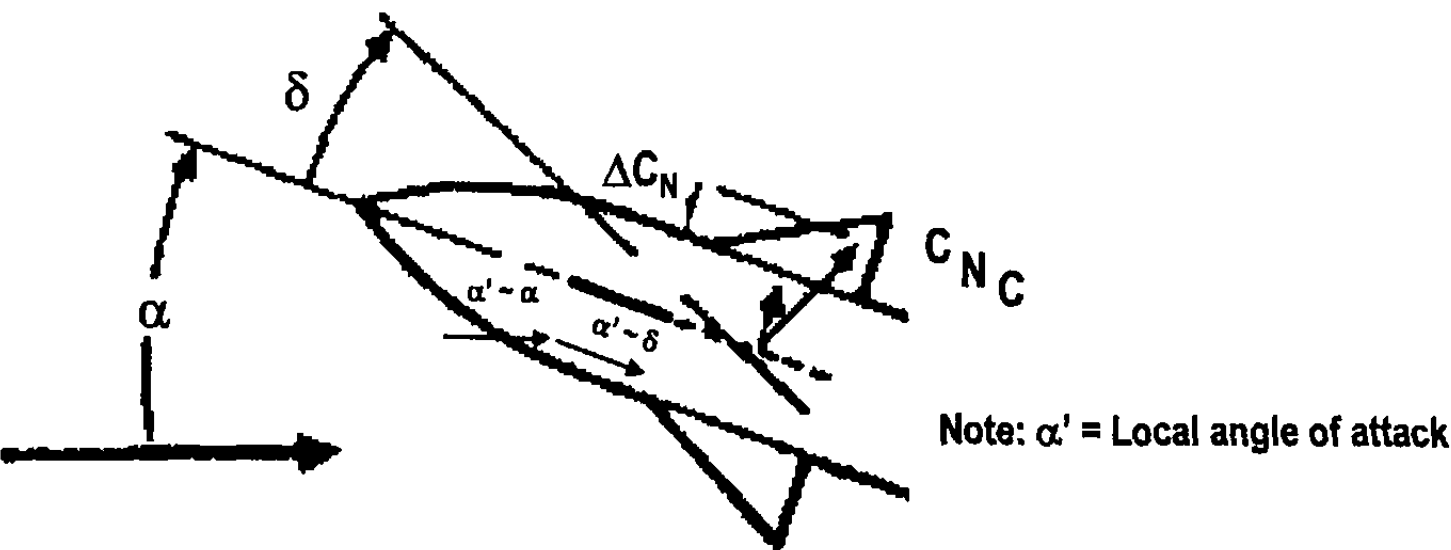
AERODYNAMIC CONSIDERATIONS IN TACTICAL MISSILE DESIGN 63



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Fig. 2.35 Canard control missiles are wingless and most are supersonic. Figure also in color section.

of attack α' . Because the trailing canard control surface has a smaller local angle of attack, it is more effective at high control surface deflection δ and high angle of attack α , operating without stall. Another benefit of a fixed forward surface is that the location of the vortex shedding is fixed, reducing the uncertainty in the



Note: Forward fixed surface reduces local angle of attack for movable canard, providing higher stall angle of attack. Forward surface also provides a fixed, symmetrical location for vortex shedding from the body.
Python 4 also has free-to-roll tails and separate roll control allerons.

Fig. 2.36 Missiles with split canards have enhanced maneuverability at high angle of attack.

aerodynamics. In addition to a fixed forward surface, some canard control missiles (e.g., AA-11) also have a weather-cocking surface to sense angle of attack. The measured angle of attack is fed into the autopilot to improve the missile maneuverability. Most modern canard control missiles use split canard control, including the Kegler AS-12, Archer AA-11, Aphid AA-8, Magic R-550, Python 4, and U-Darter. In addition to split canards for high angle-of-attack maneuverability, Python 4 has dedicated roll control surfaces and free-to-roll tails for enhanced maneuverability.

Figure 2.37 shows advantages and disadvantages of wing control. The advantages of wing control are few and the disadvantages are many. One advantage of wing control is that only a small body rotation is required to maneuver. This results in smaller seeker tracking error because of the lower dome error slope. A second advantage is fast response for maneuverability, particularly for skid-to-turn maneuvers. As shown in the figure, most of the missile lift is derived from deflecting the wing. Because the wing can usually be deflected faster than body rotation, the time required to achieve maneuver g is shorter.

A major disadvantage of wing control is the limited space available for packaging of the large actuators. For a high-performance missile, the wing may be located over the motor case. Another disadvantage is the larger HM required to deflect the wing. A third disadvantage is that the wing must be larger; it must be sized for maneuverability in addition to aerodynamic efficiency (L/D). A fourth disadvantage is the large induced roll produced by the wing. Finally, a wing-control missile usually has a higher aspect ratio, contributing to a wing stall at a relatively low angle of attack.

The strong vortex shedding for a wing control missile often has adverse effects on missile stability and control. The top right portion of Fig. 2.38 from Ref. 10 illustrates wing and body vortex shedding for a body-wing-tail configuration at

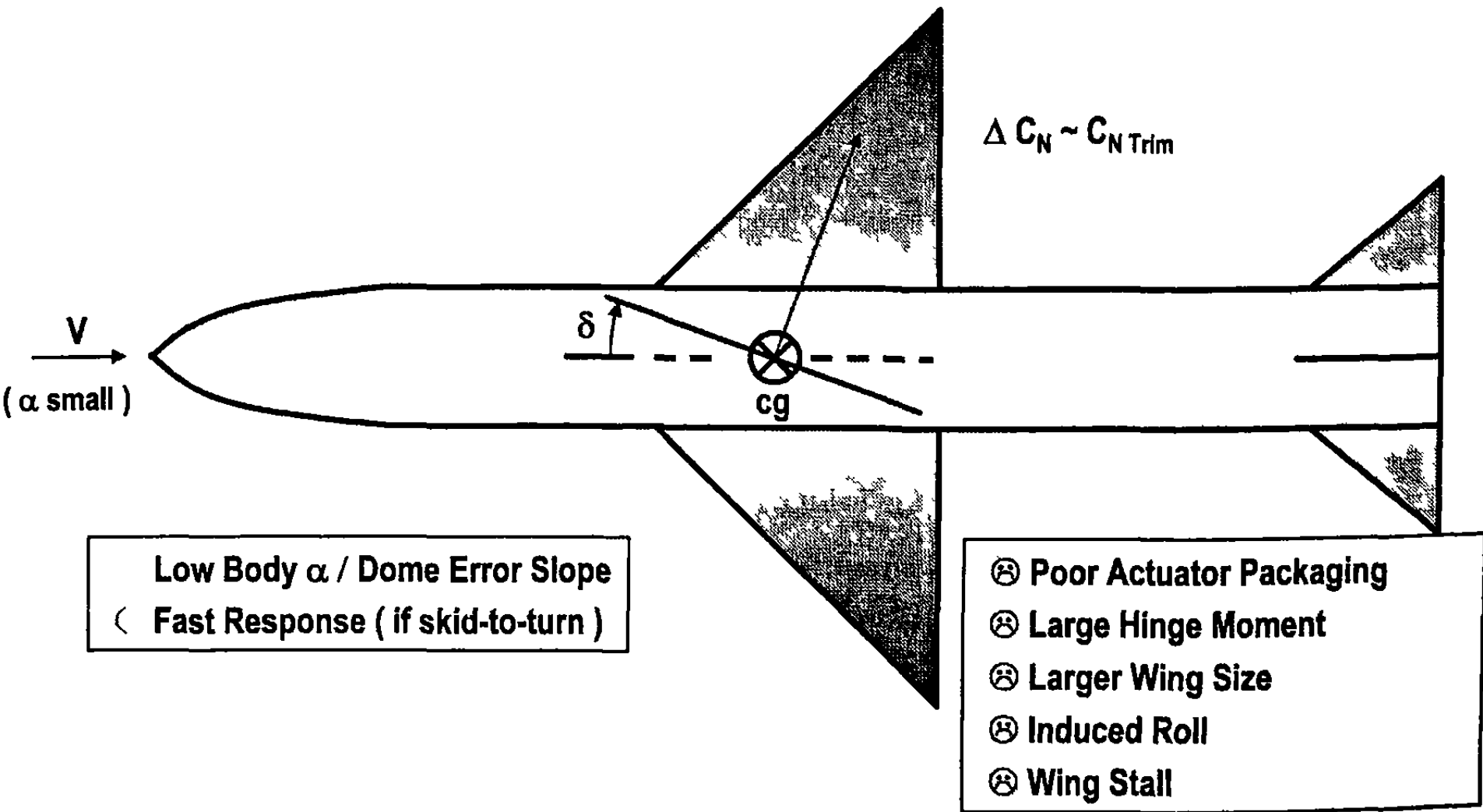
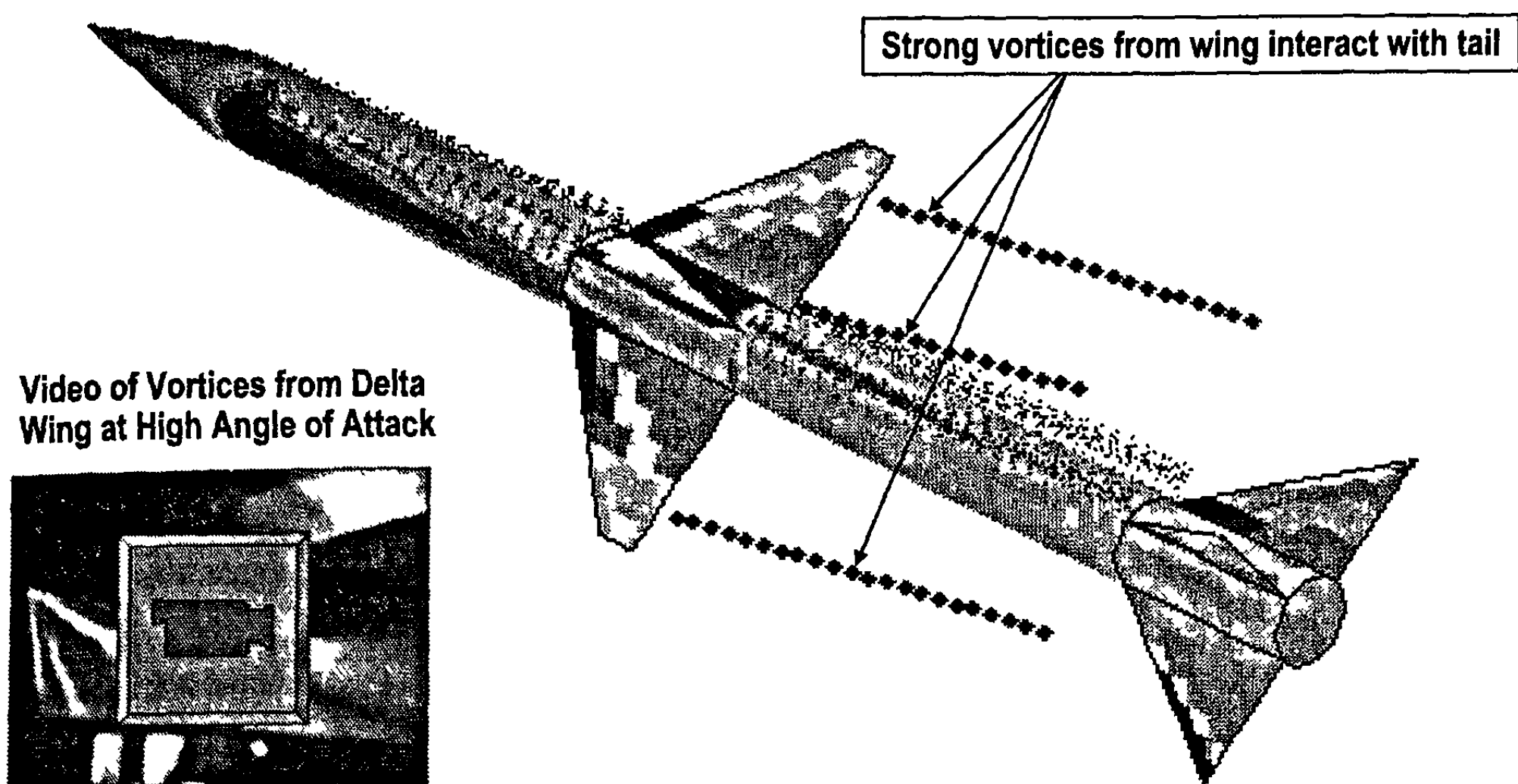


Fig. 2.37 Wing control requires less body rotation but exhibits high hinge moment, induced roll, and wing stall.



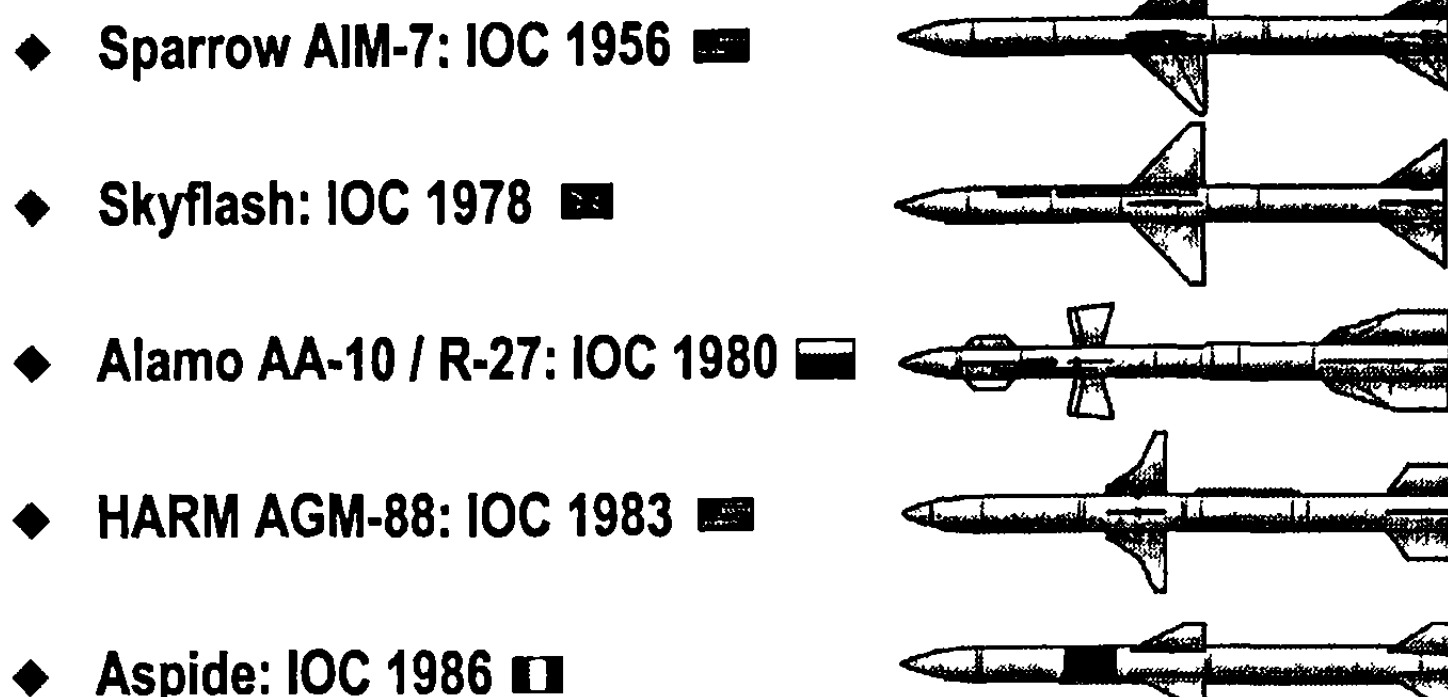
Source: University of Notre Dame web site:
<http://www.nd.edu/~ame/facilities/SubsonicTunnels.html>

Source: Nielsen Engineering & Research (NEAR) web site:
<http://www.nearinc.com/near/project/MISDL.htm>

Fig. 2.38 Wing control missile susceptible to high vortex shedding. Figure also in color section. Video is in the CD included with this textbook.

high angle of attack. As shown in the figure, wing vortices are shed near the wing tips, trailing back with the free stream. Body vortices are shed near the nose center of pressure, approximately two-thirds of the length of the nose. At high angles of attack, additional body vortices may be shed from the cylindrical body downstream of the nose. Surfaces aft of the wing, particularly aft surfaces with a span comparable to that of the wing, are susceptible to the strong vortices shed from the wing, and the missile may have large adverse induced roll, a loss in stability, and a loss in control effectiveness. Shown in the bottom left portion of the figure is a wind tunnel video of a highly swept wing pitching over an angle of attack range from 0 to 90 deg. The source of the video was the University of Notre Dame School of Aerospace and Mechanical Engineering. Note that the vortices are relatively weak at low angle of attack. As the angle of attack increases, the vortices increase in strength up to an angle of attack of about 60 deg. For angles of attack greater than about 60 deg, the flow detaches from the leading edge of the wing and there is a wake over the entire rear surface of the wing.

Examples of operational missiles that use wing control are shown in Fig. 2.39. Note that the wings are generally large and located near the center of gravity. The configurations of Skyflash, HARM, and Aspide are similar to that of the Sparrow missile, which was developed in 1956. The AA-10/R-27 Alamo configuration is different from Sparrow. It has a fixed forward strake surface and a "bow tie" forward swept movable wing. The Alamo geometry of a fixed, forward strake may have more predictable vortex shedding at high angle of attack because it tends to fix the location of vortex shedding. Also, the bow tie wing requires less wing span, which enhances the fitment in exterior carriage. The HARM missile wing is somewhat different from the Sparrow wing because of a



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Fig. 2.39 Wing control missiles are old technology. Figure also in color section.

double swept leading edge. The double swept leading edge reduces the variation in aerodynamic center, providing a lower HM.

Because of deficiencies such as actuator packaging, HM, wing size, and angle of attack limitations from wing stall/induced roll, no wing control missile has been developed in recent years. Modern missiles use tail, canard, or unconventional flight control. Although the wing control Sparrow has been largely replaced by the tail control AMRAAM as a preferred medium-range air-to-air missile, the Sparrow was selected as a rocket baseline for this text because of the availability of unclassified data for the Sparrow missile.

TVC and reaction jet control alternatives are illustrated in Fig. 2.40. The figure is a schematic of seven different types of unconventional flight control. It also shows a relative deflection of the thrust for each of the flight control types. Liquid injection TVC uses a fluid such as freon, injected normal to the local stream, which causes a bow shock wave inside the nozzle. Deflection is of the order of ± 7 deg. Another type is hot gas injection TVC, which ducts hot gas products from the motor chamber into the nozzle. The injection of the hot gas into the nozzle causes a bow shock, similar to that of liquid injection TVC. Hot gas injection has high deflection, although there may be problems for the high temperature valves, which must operate with gas products up to 5500°F . A third type is jet vanes, which are control surfaces inside the nozzle. Jet vanes have an advantage in that differential deflection also provides roll control. Jet vanes can also share common actuators with aerodynamic control. However, jet vanes are continually exposed to the rocket exhaust plume, which results in about 3–5% loss in specific impulse even if they are not deflected. Axial plate thrust vector control is a fourth type, consisting of a plate that extends back and forth. When the plate is fully extended, the plume attaches to the plate because of the Coanda effect. A fifth type is jet tabs, which are simple devices that switch in and out of the plume. The jet tab sets up a bow shock inside the nozzle, similar to that of liquid injection TVC. A sixth type of TVC is movable nozzle. Although movable nozzle TVC is often used with

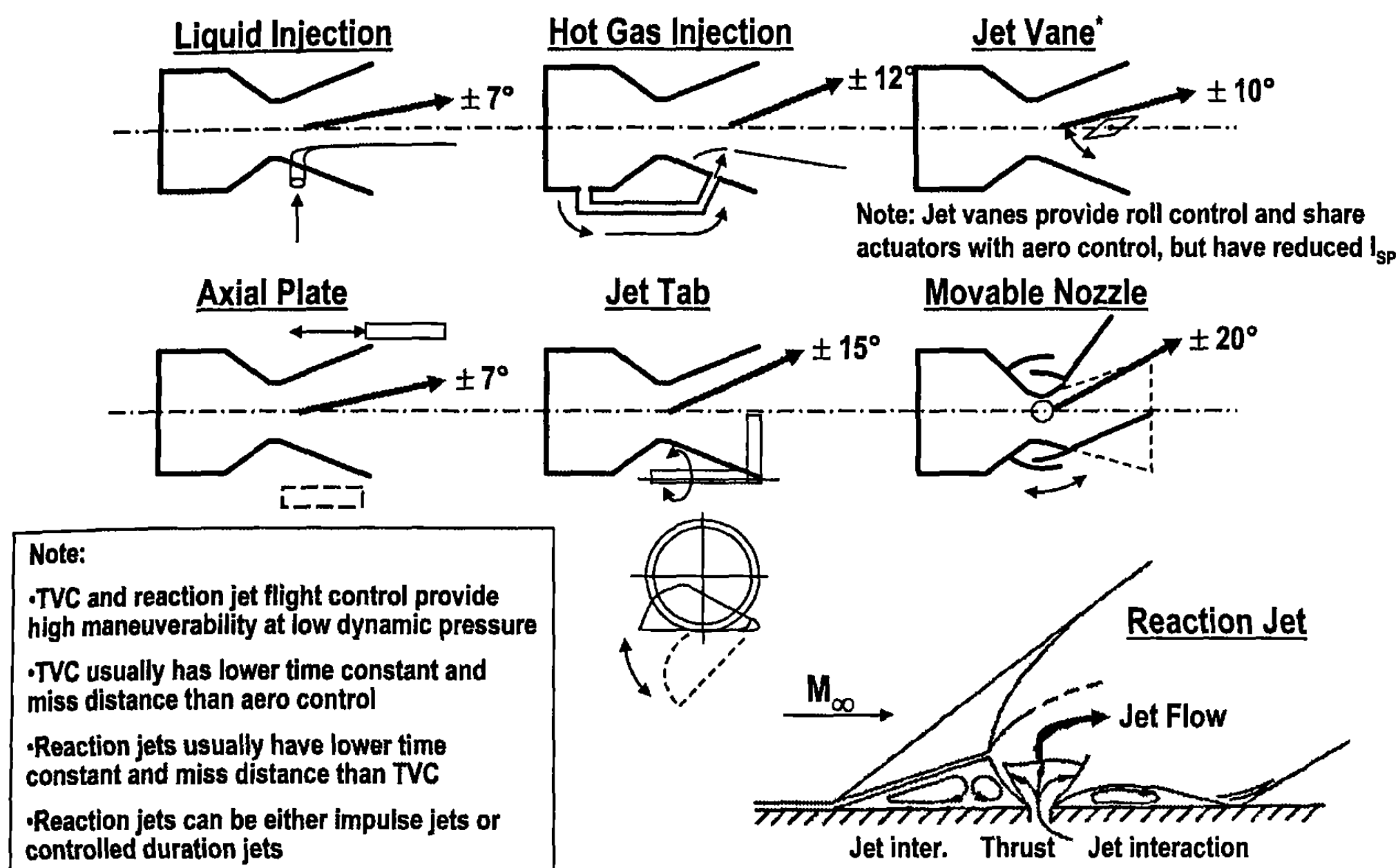


Fig. 2.40 TVC and reaction jet flight control alternatives.

strategic ballistic missiles, its application to tactical missiles, because of its high cost, is primarily limited to expensive, high-performance missiles such as Standard Missile and THAAD. A seventh type of unconventional flight control is reaction jet control. Reaction jet control can be based on either short-duration impulse, "shotgun shell-type" jets similar to PAC-3, or longer duration controlled jets similar to SM-3 or THAAD. During endo-atmospheric flight, reaction jet control interacts with the free stream air. The total force is a combination of the thrust force and the jet interaction force. The jet interaction force results from the pressure distribution on the airframe surface. As illustrated in the figure, the free stream flow separates from the airframe surface because of the obstruction of the jet. This causes a higher local pressure in the separated flow region in front of the nozzle and a lower local pressure in the separated flow region behind the nozzle. Although the jet interaction force usually augments the thrust force, there are relatively high uncertainty and risk in the magnitude of the jet interaction force.

Most unconventional flight control usually provides higher maneuverability and faster response than aerodynamic control. In general, reaction jet control has faster response and smaller miss distance than TVC, enabling hit-to-kill guidance. Unconventional flight control is especially suited for short duration operation to enhance the maneuverability and response of aerodynamic control. Cost, weight, subsystem packaging, risk, and the capability for roll control are design considerations.

Shown in Fig. 2.41 are examples of operational tactical missiles that use either TVC or reaction jets for flight control. Jet vane plus aerodynamic control is the most popular type of TVC. As mentioned previously, jet vanes can provide roll

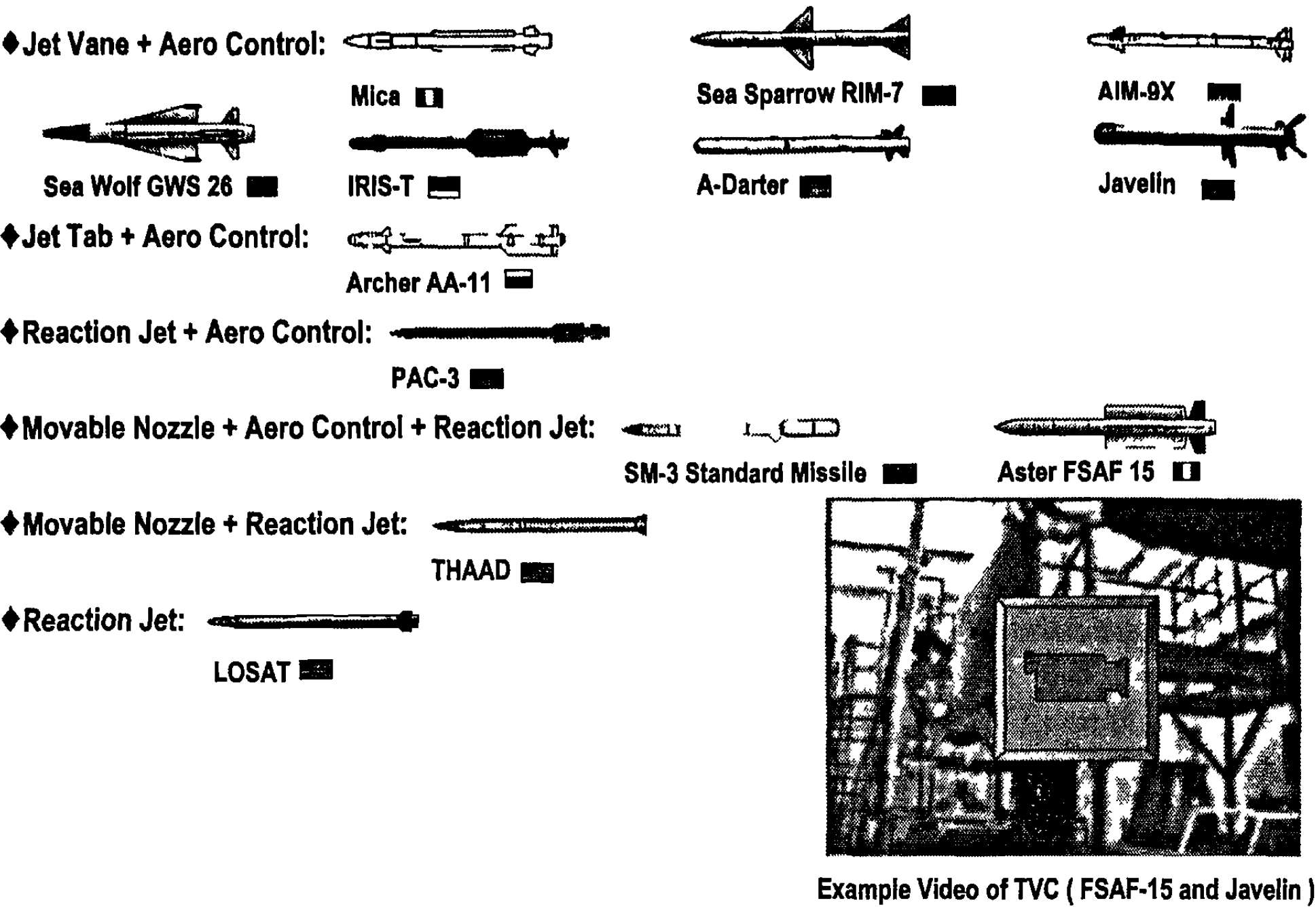


Fig. 2.41 Most tactical missiles with TVC or reaction jet control also use aerodynamic control. Figure also in color section. Video is in the CD included with this textbook.

control and can also share actuators with aerodynamic control. Examples of missiles that use jet vane plus aerodynamic control include the French Mica; the U.S. Sea Sparrow RIM-7, Sidewinder AIM-9X, and Javelin; the U.K. Seawolf GWS 26; the German IRIS-T; and the South African A-Darter. A disadvantage of jet vanes is weight. Because jet vanes operate in the high temperature nozzle plume, they must be constructed from materials with high temperature capability, such as tungsten, molybdenum, or carbon-carbon. For a missile flight control system that consists of combined jet vane plus aerodynamic control, the jet vanes provide the large off-boresight maneuvering that is often required after launch, while aerodynamic control provides the required trim and maneuvering after motor burnout.

An example of a missile that uses jet tab plus aerodynamic control is the Russian Archer AA-11. Jet tabs have advantages of higher pitch/yaw vectoring and lighter weight than jet vanes, but have a disadvantage in that they cannot provide roll control.

An example of missiles that use reaction jet plus aerodynamic control is the U.S. PAC-3 surface-to-air missile defense missile. Reaction jets are usually required for hit-to-kill anti-missile missiles because their smaller time constant ($\sim$ ms) is necessary for the smaller required miss distance.

Examples of complex missiles that combine movable nozzle TVC, aerodynamic control, and reaction jet control are the U.S. SM-3 Standard Missile and the French Aster FSAF 15. Movable nozzle TVC is used for launch stabilization

and off-boresight maneuvering following launch. Following launch, aerodynamic control provides midcourse trim and maneuvering. Divert and attitude control thruster reaction jets provide terminal maneuvering during the hit-to-kill intercept.

The last two missiles in the figure are examples of missiles that do not use any aerodynamic control. THAAD has movable nozzle TVC for stabilization and off-boresight maneuverability during launch and midcourse. Similar to SM-3, THAAD uses divert and attitude control thruster reaction jets for the exo-atmospheric hit-to-kill intercept. Shown at the bottom left of the figure is LOSAT. LOSAT uses only reaction jet control to provide the fast response time necessary for hit-to-kill accuracy of the hypersonic missile. The weight of the LOSAT reaction jet control system is relatively low because of the short time of flight.

Shown in the bottom right of the figure is a video of examples of TVC (FSAF-15 and Javelin). FSAF-15 has sophisticated dual TVC nozzles, which allow full axis control (roll control in addition to pitch/yaw control). A mission for FSAF-15 is defense against low-altitude sea-skimming missiles. Because its active radar seeker is not effective at a low grazing angle, FSAF-15 flies a lofted flight trajectory, impacting the target from above. Also shown in the video is a Javelin lofted trajectory flight test against a T-72 tank at 2.5 km range. Javelin has the option of either a lofted trajectory for a top impact or a line-of-sight trajectory for a horizontal impact. Note that the missile has high maneuverability and top impact kill, even at short range.

2.16 Maneuver Alternatives

Figure 2.42 compares missile maneuver alternatives of skid-to-turn maneuvering, bank-to-turn maneuvering, and rolling airframe maneuvering. Bank-to-turn maneuvering is a relatively new technology for tactical missiles. Missiles using bank-to-turn will first roll until the wings (or the major axis of a lifting body) are oriented perpendicular to the target line of sight. Following the roll maneuver, the missile then conducts a pitch maneuver, maintaining the preferred roll orientation. A benefit of bank-to-turn maneuvering is higher maneuverability for a lifting body, which has a noncircular cross section, or for a missile with wings. Another benefit is a smaller sideslip angle for missiles with inlets. Bank-to-turn is particularly suited for midcourse guidance maneuvers prior to seeker lock-on to the target. A disadvantage of bank-to-turn maneuvering is slower response in terminal maneuvers that could degrade guidance accuracy, increasing the missile miss distance. Alternative approaches to alleviate this problem include faster actuators for roll control, faceted or multilens dome to decrease dome error slope, and switching from bank-to-turn maneuvering to skid-to-turn maneuvering for terminal flight. Bank-to-turn missiles often have high greater miss distance because of dome error slope. The seeker looks through a dome with a large variation in radial angle in addition to the variation in off-boresight angle.

Most missiles use skid-to-turn maneuvering. Skid-to-turn is particularly applicable to axisymmetric, cruciform missiles with low aspect ratio surfaces.

- ◆ **Skid-to-Turn (STT)**
 - Advantage: Fast response
 - Features
 - Does not require roll commands from autopilot
 - Works best for axisymmetric cruciform missiles
- ◆ **Bank-to-Turn (BTT)**
 - Advantage: Provides higher maneuverability for planar wing, noncircular / lifting bodies, and airbreathers
 - Disadvantages
 - Time to roll
 - Requires fast roll rate
 - May have higher dome error slope
 - Features
 - Roll attitude commands from autopilot
 - Small sideslip
- ◆ **Rolling Airframe (RA)**
 - Advantage: Requires only two sets of gyros / accelerometers / actuators (packaging for small missile)
 - Disadvantages for aero control
 - Reduced maneuverability for aero control
 - Requires high rate gyros / actuators
 - Requires precision geometry and thrust alignment
 - Features
 - Bias roll rate and roll moment
 - Can use impulse steering (e.g., PAC-3, LOSAT)
 - Compensates for thrust offset

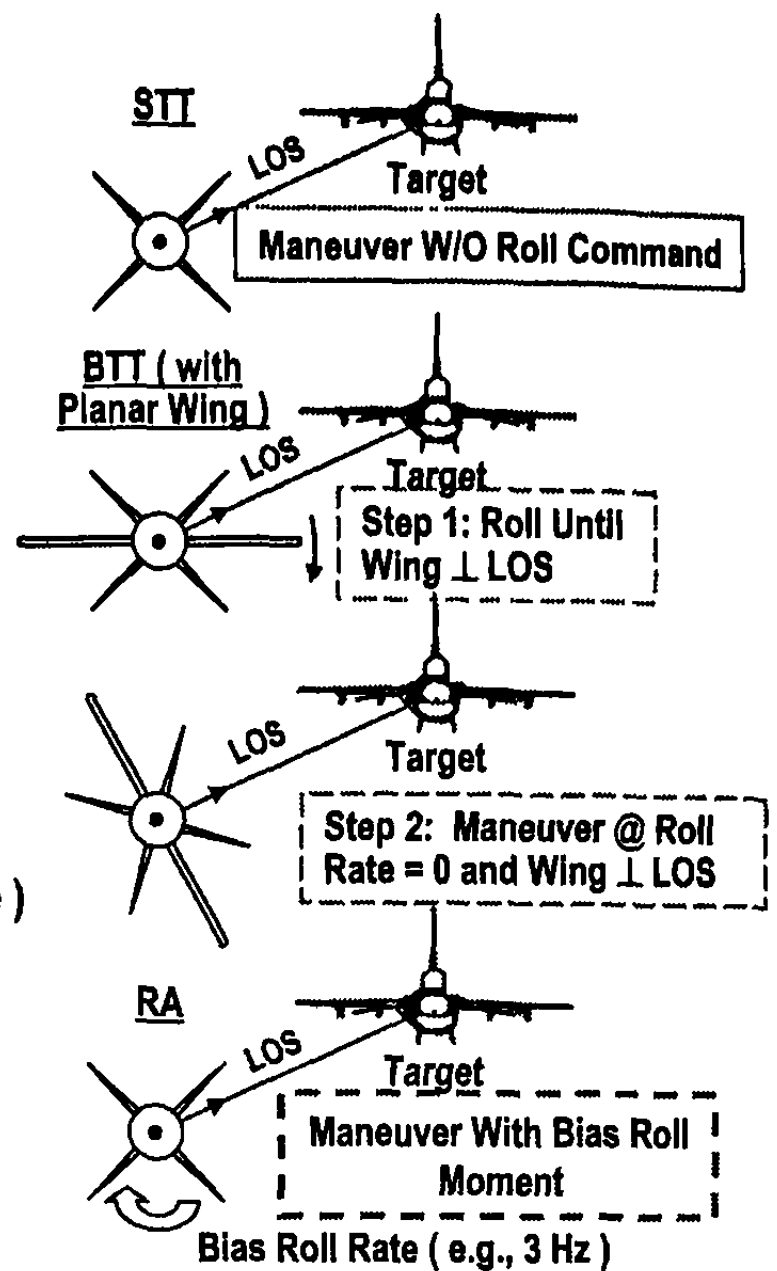


Fig. 2.42 Skid-to-turn is the most common maneuver law.

As shown in the figure, a skid-to-turn maneuver is a commanded maneuver along the line of sight of the seeker, without a roll command. An advantage of skid-to-turn is fast response because the maneuver is immediately made along the line of sight without delay.

A third type of maneuver is a rolling airframe maneuver. Rolling airframe maneuvering is similar to skid-to-turn, except the missile also has a rolling moment bias input to provide a bias roll rate. The rolling moment bias may be provided either by thrust offset, tail incidence, canard incidence, or a combination of approaches. Advantages of rolling airframe missiles include lower cost because of fewer flight control gyros, accelerometers, and actuators; lower volume to package less flight control hardware; compensation for thrust misalignment; and compatibility with reaction jet impulse control. The 5-in-diameter Rolling Airframe Missile (RAM) and the 2.75-in-diameter Stinger missile are examples of small-diameter rolling airframe missiles. RAM and Stinger use two-axis flight control, requiring less flight control hardware packaging within the small diameter. PAC-3 and LOSAT are examples of hypersonic rolling airframe missiles that use impulse reaction jet control. Disadvantages of an aerodynamic controlled rolling airframe include reduced maneuverability from using fewer controls, a requirement for high rate gyros and actuators, and a requirement for precision geometry with precision thrust alignment. The maximum roll rate of a rolling airframe missile is about 3 Hz because most actuators are limited to a maximum rate of about 1000 deg/s. Another potential problem is susceptibility to roll-pitch-yaw cross-coupling instability. Roll-pitch-yaw cross coupling tends to occur from roll resonance when the roll rate

is comparable to the missile natural frequency in pitch or yaw. Designing a rolling airframe missile requires consideration of the precession and nutation modes of motion. Note that the miss distance from the seeker dome error slope may be reduced because of the averaging effect of roll rate. However, the miss distance may also be increased because of susceptibility to roll resonance.

Twin and single inlet air-breathing missiles usually require bank-to-turn maneuvering to avoid loss of inlet efficiency from sideslip. Figure 2.43 shows examples of air-breathing missiles that use bank-to-turn maneuvering. The French ASMP is an example of an operational ramjet missile with twin side inlets that are separated by a radial angle of 180 deg. Also shown are two developmental missiles, HSAD and Meteor, with twin cheek inlets. The cheek inlets are in tandem with the bottom cruciform tails and are separated by a radial angle of 90 deg. A major difference between the U.S. HSAD and the U.K. Meteor is the propulsion system. The HSAD is a variable flow solid hydrocarbon ducted rocket. An advantage of a hydrocarbon ducted rocket is a low observable plume. The Meteor engine is a variable flow boron fuel ducted rocket. Boron fuel provides longer range. Compared to liquid fuel ramjets, ducted rockets have simpler logistics and higher acceleration capability. Shown in the bottom of the figure are examples of bank-to-turn missiles with a single inlet. An example of a developmental ramjet missile with a chin inlet is the U.S. Advanced Strategic Air Launched Missile (ASALM). ASALM uses angle of attack α and angle of sideslip β feedback in the flight control system to provide a coordinated turn in bank-to-turn maneuvering. The ASALM was selected as a ramjet baseline for this text because of the availability of unclassified data and its high performance. The BGM-109 Tomahawk and the Storm Shadow/Scalp are examples of bank-to-turn turbojets with a single bottom inlet. Finally, the AGM-86 ALCM

◆Examples of Twin Inlet Missiles with Bank-to-Turn

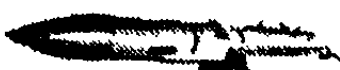
◆Twin Side Inlets Ramjet: ASMP 



◆Twin Cheek Inlets Ducted Rocket: HSAD 

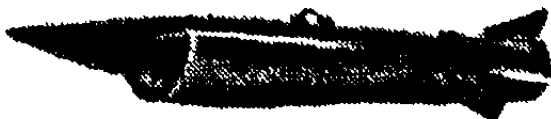


◆Twin Cheek Inlets Ducted Rocket: Meteor 



◆Examples of Single Inlet Missiles with Bank-to-Turn

◆Chin Inlet Ramjet: ASALM 



◆Bottom Inlet Turbojet: BGM-109 Tomahawk 



◆Bottom Inlet Turbojet: Storm Shadow / Scalp  



◆Top Inlet Turbofan: AGM-86 ALCM 



Note: Bank-to-turn maneuvering maintains low sideslip for better inlet efficiency.

Fig. 2.43 Asymmetric inlets require bank-to-turn maneuvering. Figure also in color section.

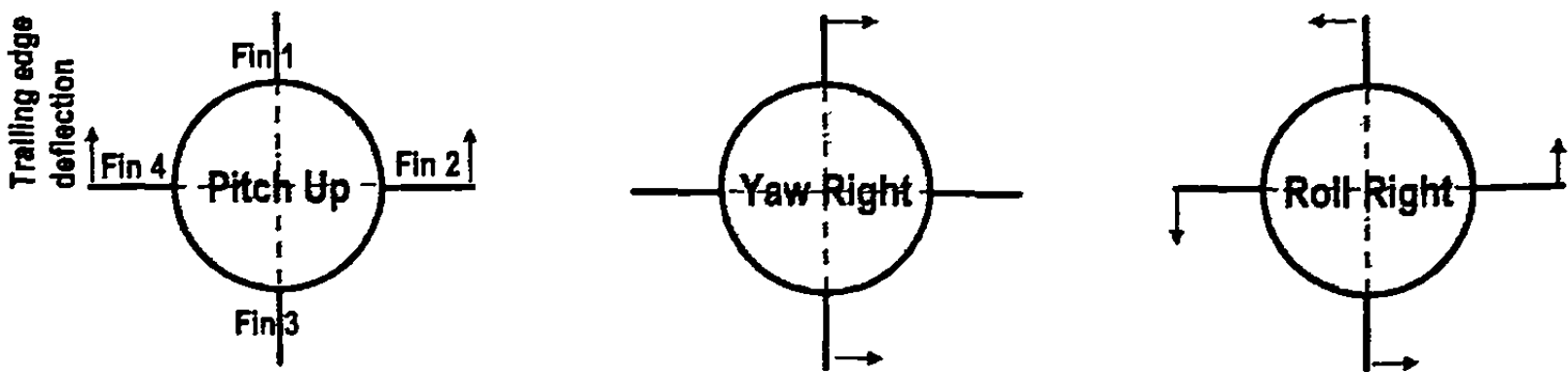
is an example of a single top inlet. An advantage of locating the inlet on top is reduced RCS against ground-based radars because of the shielding of the body. A disadvantage is reduced specific impulse at high angles of attack because of inlet flow distortion. Inlet flow distortion occurs at high angle of attack because of crossflow, a thicker boundary, and a tendency for flow separation.

2.17 Roll Orientation

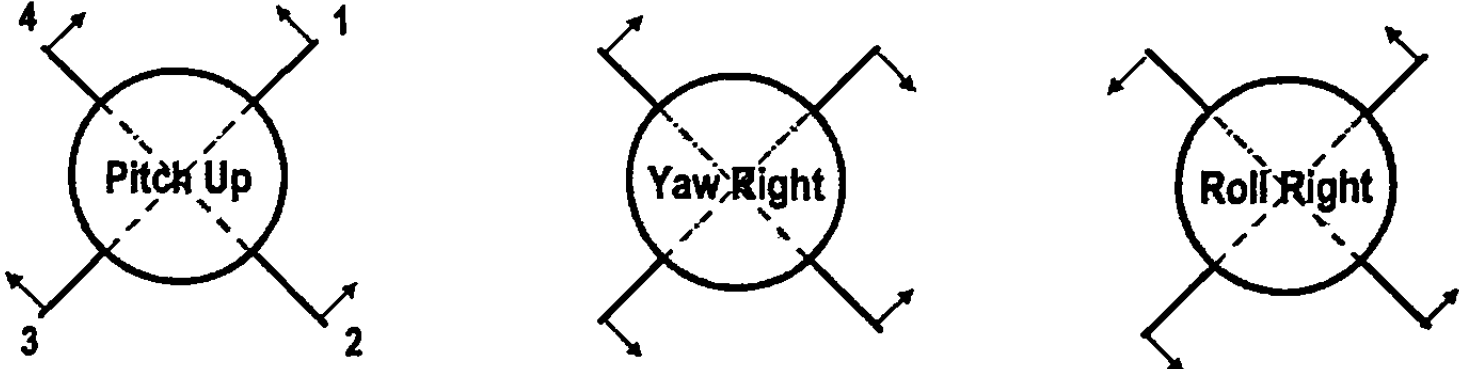
Figure 2.44 compares roll attitude and maneuver control deflection alternatives for a cruciform missile. Approaches to symmetric roll orientation during missile flight, with no induced roll, are + roll orientation and x roll orientation. Each has advantages and disadvantages. The + orientation has the simplest control mechanization. For a pitch command using conventional control with four tail surfaces, the trailing edges of the two horizontal surfaces are deflected upward to cause a positive pitch-up moment. Similarly for a yaw command using conventional tail control, the trailing edges of the two vertical surfaces are deflected to the right to cause a positive yaw right moment. For a roll command, the trailing edges of all four surfaces may be deflected to provide either positive clockwise (roll right) or negative counterclockwise (roll left) roll. The + roll configuration usually has an advantage of lower trim drag. A disadvantage is that it usually has a statically unstable rolling moment derivative ($C_{l\phi} > 0$). The + roll configuration has less static stability and less control effectiveness in pitch and yaw.

An alternative approach of the x roll orientation during missile flight is somewhat more complex in its control mechanization. For a pitch command, all four surfaces are deflected to provide a pitching moment without side force. For a yaw

+ Roll Orientation with Four Tail Surfaces Control of Pitch / Yaw / Roll, Looking Forward from Base



X Roll Orientation with Four Tail Surfaces Control of Pitch / Yaw / Roll, Looking Forward from Base



Note: + roll orientation usually has lower trim drag, less static stability and control effectiveness in pitch and yaw, and statically unstable roll moment derivative ($C_{l\phi} > 0$)
X roll orientation has better launch platform compatibility, higher L / D, higher static stability and control effectiveness in pitch and yaw, and statically stable roll moment derivative ($C_{l\phi} < 0$)

Fig. 2.44 X roll orientation is usually better than + roll orientation.

command, all four surfaces are deflected to provide a yawing moment without normal force. The roll command is similar to that of the $+$ orientation. For roll control, all four surfaces are deflected to provide either clockwise roll or counter-clockwise roll. The x roll orientation often has advantages of better fit for launch platform compatibility, higher aerodynamic efficiency (L/D), and a statically stable rolling moment derivative ($C_{l\phi} < 0$). Also, the x roll orientation has higher static stability and control effectiveness in pitch and yaw. Generally, the x orientation is preferred over the $+$ orientation.

2.18 Trim

Trim is defined at zero pitching moment. As shown in Fig. 2.45, the normal force at a trimmed flight condition is lower for either a statically stable missile using tail control or a statically unstable missile using canard control. Conversely, the trimmed normal force is higher for a statically unstable missile using tail control or a statically stable missile using canard control. A rule of thumb for conceptual design of a tail or canard control missile is that the change in angle of attack from control deflection should be greater than 1 ($|\alpha/\delta| > 1$). Most control surfaces have deflection limits less than $+/-30$ deg. $|\alpha/\delta| > 1$ usually allows a trim angle of attack up to about 30 deg.

Relaxed static margin allows the missile to trim at higher angle of attack, providing higher maneuverability. Shown in Fig. 2.46 is the benefit of relaxed static stability for the rocket baseline missile. The rocket baseline missile has excessive static margin (e.g., 0.88 diameter at Mach 2), resulting in a relatively low maximum angle of attack ($\alpha = 9.4$ deg) and a relatively low maximum trimmed normal force ($C_N = 9$). Reducing the static margin increases the maximum angle of attack and the maximum trimmed normal force. In the

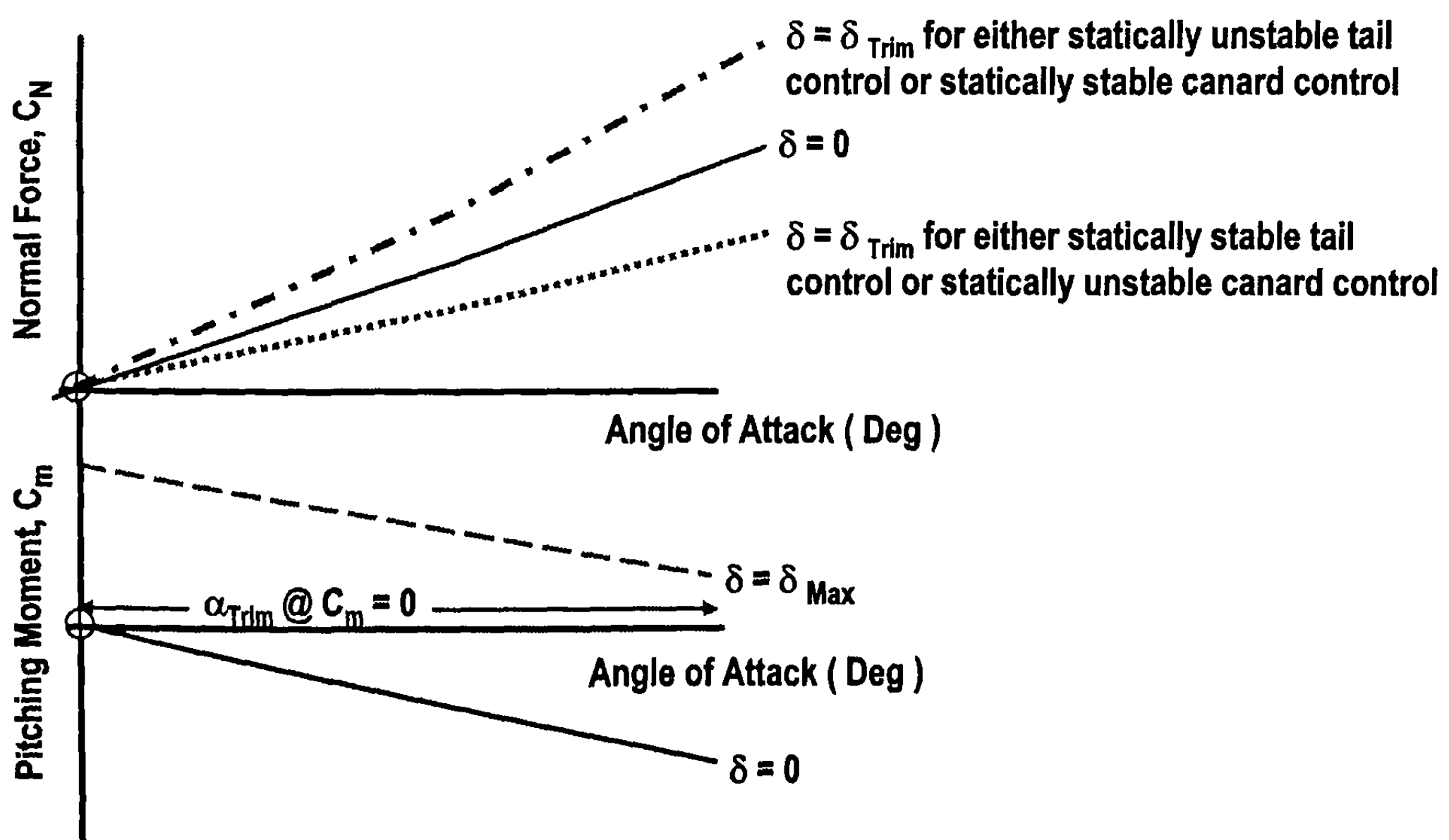


Fig. 2.45 Trimmed normal force is defined at zero pitching moment.

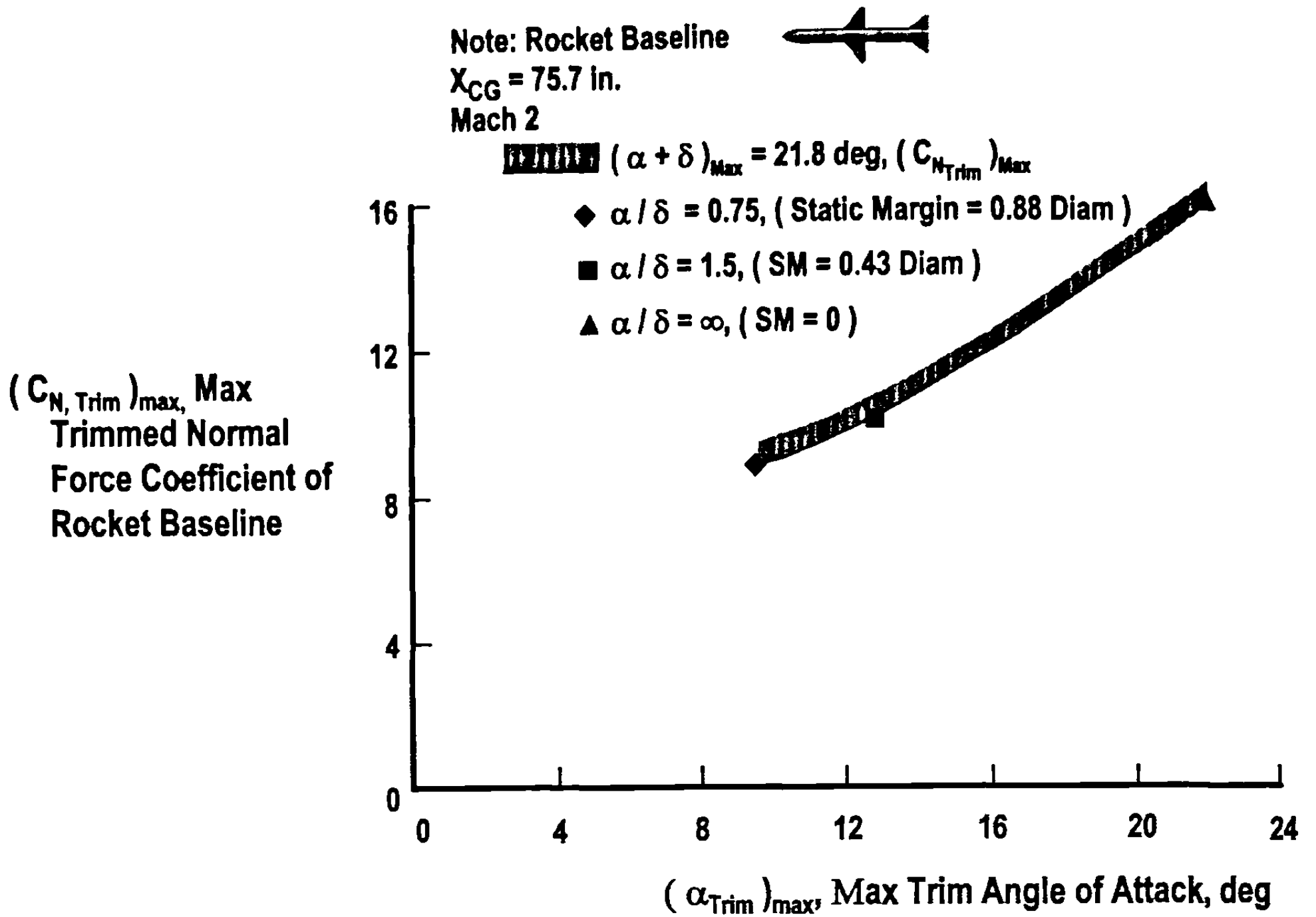


Fig. 2.46 Relaxed static margin allows higher trim angle of attack and higher normal force.

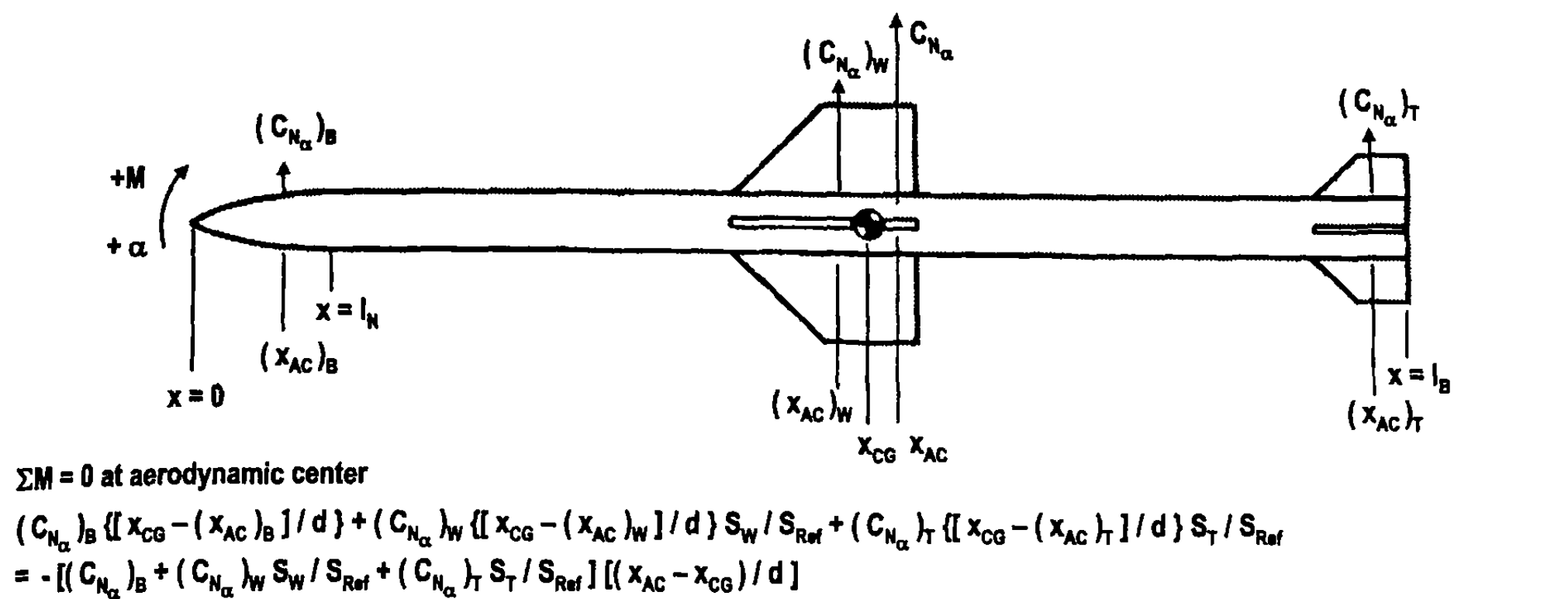
example, a neutrally stable missile ($\alpha/\delta = \infty$ or static margin = 0) has a 130% increase in maximum angle of attack ($\alpha = 9.4$ to 21.8 deg) and an 80% increase in maximum available normal force ($C_N = 9$ to 16).

2.19 Tail Area Sizing

Much of the conceptual design configuration sizing process is oriented toward tail sizing. Because missiles are usually volume limited and the subsystems are of comparable density, the center-of-gravity location is usually near 50% of the length of the missile. Because the nose and any forward surfaces (strakes/canards) are destabilizing, the tails must be sized to provide static stability. Figure 2.47 illustrates the process for tail sizing. The tail contribution to pitching moment stability, provided by the tail normal force effectiveness $(C_{N\alpha})_T(S_T/S_{Ref})$ times its moment arm to the center of gravity $[(x_{AC})_T - x_{CG}]$, must balance the contributions to pitching moment from the nose and wing. Shown in the figure is the equation for the static margin, given a specified tail area. Also shown in the figure is the equation for the required tail area, given a specified static margin.

Of particular interest is the required tail area for neutral stability. The equation is

$$\begin{aligned}
 S_T = & \{ (C_{N\alpha})_B [x_{CG} - (x_{AC})_B] / d \\
 & + (C_{N\alpha})_W \{ [x_{CG} - (x_{AC})_W] / d \} (S_W / S_{Ref}) \} \\
 & \times \{ d / [(x_{AC})_T - x_{CG}] \} S_{Ref} / (C_{N\alpha})_T
 \end{aligned}
 \tag{2.51}$$



Static margin for a specified tail area is

$$(x_{AC} - x_{CG}) / d = - \{ (C_{N\alpha})_B \{ [x_{CG} - (x_{AC})_B] / d \} + (C_{N\alpha})_W \{ [x_{CG} - (x_{AC})_W] / d \} S_W / S_{Ref} + (C_{N\alpha})_T \{ [x_{CG} - (x_{AC})_T] / d \} (S_T / S_{Ref}) \} / [(C_{N\alpha})_B + (C_{N\alpha})_W S_W / S_{Ref} + (C_{N\alpha})_T S_T / S_{Ref}]$$

Required tail area for a specified static margin is

$$S_T / S_{Ref} = \{ (C_{N\alpha})_B \{ [x_{CG} - (x_{AC})_B] / d \} + (C_{N\alpha})_W \{ [x_{CG} - (x_{AC})_W] / d \} (S_W / S_{Ref}) + \{ [(C_{N\alpha})_B + (C_{N\alpha})_W S_W / S_{Ref}] [(x_{AC} - x_{CG}) / d] \} / \{ (C_{N\alpha})_T \{ [(x_{AC})_T - x_{CG}] / d - (x_{AC} - x_{CG}) / d \} \}$$

Fig. 2.47 Tails are sized for desired static margin.

Figure 2.48 shows the rocket baseline required tail area for neutral static margin as a function of Mach number and wing location. It is based on predicted aerodynamics. Note that the required tail area for neutral static stability must be larger as the Mach number increases toward hypersonic. The tail loses aerodynamic

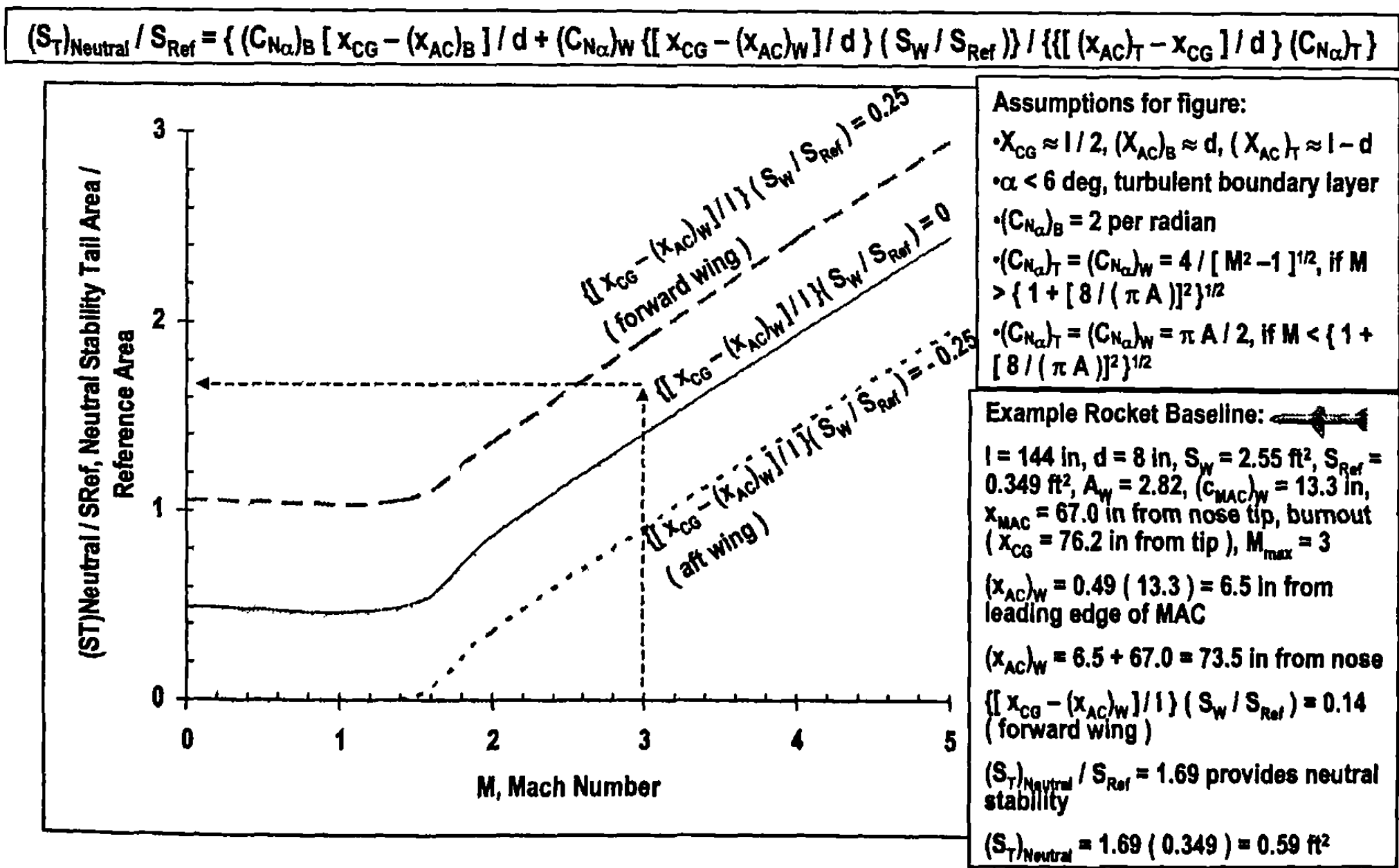


Fig. 2.48 A Larger tail area is required for neutral stability at high Mach number.

efficiency as Mach number increases. Also note that placing the wing such that its aerodynamic center is forward of the center-of-gravity location is destabilizing, requiring a larger tail area for compensation. The rocket baseline wing aerodynamic center is forward of the center of gravity, and therefore the rocket baseline requires a larger tail. Placing the wing aft of the center of gravity is stabilizing, allowing a smaller tail area to meet the static margin requirement. Note that for a typical missile wing without camber, the location of the wing aerodynamic center is the same as the wing center-of-pressure location.

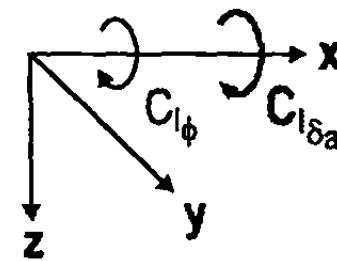
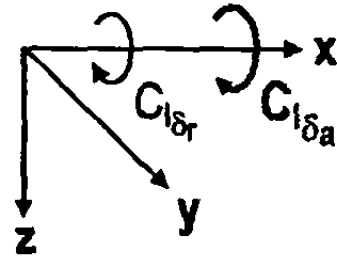
The figure is based on analytical prediction of aerodynamics from slender body theory $[(C_{N\alpha})_B = 2 \text{ per rad and } (x_{CP})_B \approx d]$ for the body. For the wing and tail, linear wing theory $[(C_{N\alpha})_W = (C_{N\alpha})_T = 4/[M^2 - 1]^{1/2}]$ is used at high Mach number and slender wing theory $[(C_{N\alpha})_W = \pi A_W/2, (C_{N\alpha})_T = \pi A_T/2]$ is used at subsonic and low supersonic Mach numbers. It is assumed that the moment arm from the tail aerodynamic center location to the missile center of gravity is approximately equal to the moment arm from the body (nose) aerodynamic center to the center of gravity. Another assumption is low angle of attack ($\alpha < 6 \text{ deg}$). Still another assumption is a turbulent boundary layer with small boundary layer thickness. Note that for flight conditions with a laminar boundary layer over much of the body (e.g., $M < 0.2$ or $h > 80,000 \text{ ft}$), the body boundary-layer thickness at the tails may blanket a significant portion of the tails, reducing the tail effectiveness.

In the example for the rocket baseline at Mach 3, the predicted tail area must be 0.59 ft^2 for neutral static stability. The predicted tail area is 1.69 times the reference body cross-sectional area. The actual rocket baseline missile has a tail area that is more than four times the reference body cross-sectional area. This is an indication that the rocket baseline missile has excessive static margin from a tail that is unnecessarily large. Also, the wing aerodynamic center location of the rocket baseline is in front of the missile center of gravity, requiring a larger tail area.

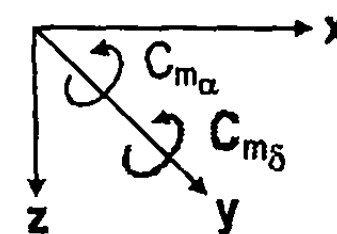
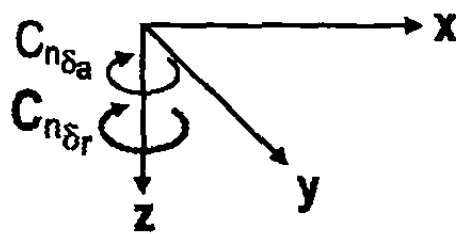
2.20 Stability and Control Conceptual Design Criteria

Stability and control have an impact on the aerodynamic configuration design, particularly tail sizing, and should be considered early in conceptual design. Because tactical missiles tend to have near-neutral static stability and use combined pitch/yaw/roll controls, it may be difficult to accurately predict static margin and control effectiveness. Although the actual stability and control derivatives prediction may have large error, it is often possible to size the canard, wing, or tail surfaces with less than 10% error in the surface area. Predicting stability and control derivatives usually requires iteration as the surfaces are resized and the center-of-gravity location changes during the design convergence. It is helpful to start with a baseline missile, such as the rocket and ramjet baselines in this text, which have a balance of system engineering in their stability and control derivatives. It is useful to compare stability and control derivatives with wind tunnel data to assess the accuracy of prediction methods. Conceptual design criteria are also useful, providing a “sanity check” and facilitating design convergence. Figure 2.49 shows conceptual design criteria for roll cross coupling (rolling moment from rudder deflection $C_{l\delta_r}$), yaw cross coupling (yawing

$$|C_{l_{\delta_r}} / \mathbf{C_{l_{\delta_a}}}| < 0.3 \text{ (Roll Due to Rudder Deflection)} \quad |C_{l_{\phi}} / \mathbf{C_{l_{\delta_a}}}| < 0.5 \text{ (Roll Due to Roll Angle)}$$

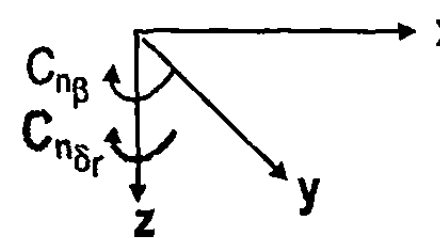
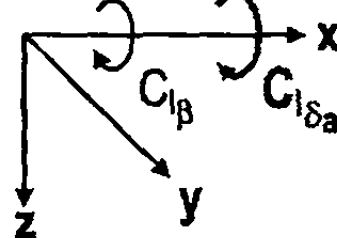


$$|C_{n_{\delta_a}} / \mathbf{C_{n_{\delta_r}}}| < 0.2 \text{ (Yaw Due to Aileron Deflection)} \quad |C_{m_{\alpha}} / \mathbf{C_{m_{\delta}}}| < 1 \text{ (Pitch Due to } \alpha \text{)}$$



$$|C_{l_{\beta}} / \mathbf{C_{l_{\delta_a}}}| < 0.3 \text{ (Roll Due to Sideslip)}$$

$$|C_{n_{\beta}} / \mathbf{C_{n_{\delta_r}}}| < 1 \text{ (Yaw Due to Sideslip)}$$



Note: The primary control derivative (larger bold font) should be larger than the undesirable stability and control derivative.

Fig. 2.49 Stability and control derivatives conceptual design criteria. Figure also in color section.

moment from aileron deflection $C_{n_{\delta_a}}$, roll control effectiveness $C_{l_{\delta_a}}$ to compensate for the rolling moment from sideslip $C_{l_{\beta}}$, roll control effectiveness $C_{l_{\delta_a}}$ to compensate for induced roll from roll attitude $C_{l_{\phi}}$, pitch control effectiveness $C_{m_{\delta}}$, and yaw control effectiveness $C_{n_{\delta_r}}$. Stability and control cross coupling or out-of-plane moments should generally be less than 30% of the “pure” in-plane moments, to maximize the control power and minimize the time constant for maneuvers. Examples of stability and control conceptual design criteria are:

- 1) $|C_{l_{\delta_r}}| < 0.3 |C_{l_{\delta_a}}|$, rolling moment from rudder deflection should be less than 30% of the aileron control effectiveness to have adequate control margin
- 2) $|C_{n_{\delta_a}}| < 0.2 |C_{n_{\delta_r}}|$, yawing moment from aileron deflection should be less than 20% of the rudder control effectiveness to have adequate control margin
- 3) $|C_{l_{\beta}}| < 0.3 |C_{l_{\delta_a}}|$, rolling moment from sideslip should be less than 30% of the aileron control effectiveness to have adequate control margin
- 4) $|C_{l_{\phi}}| < 0.5 |C_{l_{\delta_a}}|$, rolling moment from roll orientation should be less than 50% of the aileron control effectiveness to have adequate control margin
- 5) $|C_{m_{\alpha}}| < |C_{m_{\delta}}|$, pitching moment from angle of attack should be less than the elevator control effectiveness to have adequate control margin
- 6) $|C_{n_{\beta}}| < |C_{n_{\delta_r}}|$, yawing moment from sideslip should be less than the rudder control effectiveness to have adequate control margin

2.21 Body Buildup

A first-order estimate of total normal force, suitable for conceptual design, is to assume a body buildup of the configuration, which consists of the contribution from the body plus the contribution from each surface. As an example, for a

wing-body-tail conceptual design, it is assumed that the total normal force is the summation of that of an isolated body + wing (exposed) + tail (exposed). The effects of wing-body, body-wing, tail-body, body-tail, and wing-tail interference are assumed to be small when compared to the contributions from the isolated configuration geometries. For a wing-body-tail configuration the equation is

$$(C_N)_{Total} = (C_N)_{Wing-Body-Tail} \approx (C_N)_{Body} + (C_N)_{Wing} + (C_N)_{Tail} \tag{2.52}$$

Figure 2.50 shows a body buildup of the normal force coefficient of the rocket baseline missile. The body buildup in the figure is based on undeflected control surfaces. Note that for the rocket baseline missile most of the normal force is provided by the wing contribution. Deflecting the wing control provides an additional contribution to normal force. The second largest contributor to normal force is from the tail. As mentioned previously, the rocket baseline has a large tail and a large static margin. Finally, the body provides a relatively small contribution to normal force. This has an implication in the design for maneuverability. The maneuverability requirements of the rocket baseline missile are satisfied through sizing the wing.

The body buildup approach based on summing the isolated contributions from the body plus each surface is used in the conceptual design prediction of normal force, drag, and pitching moment. For preliminary design, more accurate methods based on including the effects from wing-body, body-wing, tail-body, body-tail, and wing-tail interface are used. Body buildup wind tunnel tests are also normally conducted during preliminary design. For the conceptual design prediction methods of this text, the conceptual design predictions without

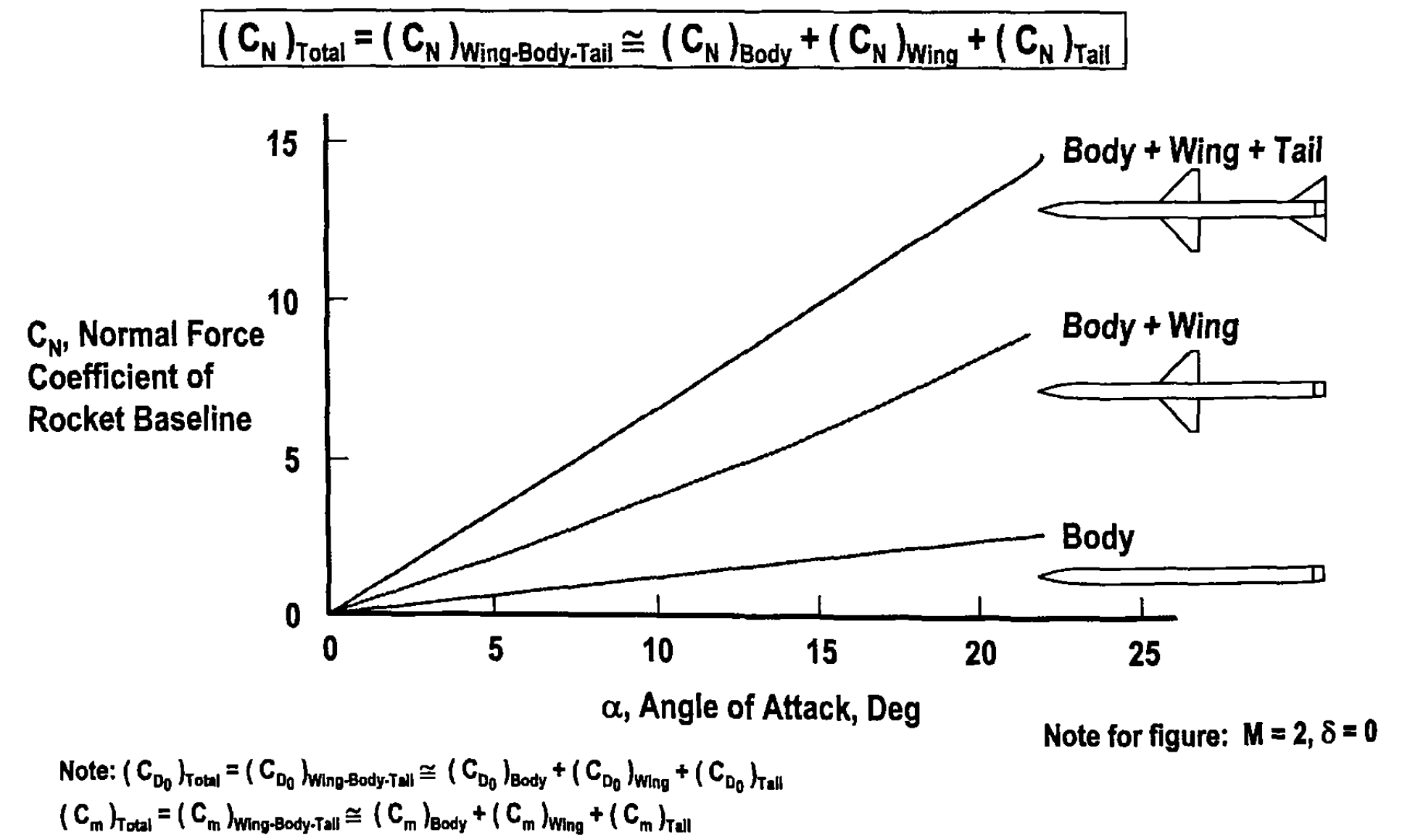


Fig. 2.50 Most of the rocket baseline body buildup normal force is provided by the wing.

interference effects are corrected by the baseline missile aerodynamics, which include interference effects.

2.22 Summary

Chapter 2 provided conceptual design methods, design tradeoffs, design criteria, and technologies for the aerodynamics of tactical missiles. The emphasis was on the aerodynamics of low aspect ratio wing and wingless missiles. The conceptual design methods of this text addressed the prediction of the normal force coefficient, the drag force coefficient, and the aerodynamic center/pitching moment coefficient of bodies and surfaces and the hinge moment of surfaces. Design tradeoffs addressed missile diameter, nose fineness, boattail, lifting body versus axisymmetric body, wings versus no wings, tails versus flares, surface planform geometry, flight control alternatives, maneuver alternatives, roll orientation, and static margin. A process was given for tail area sizing. Stability and control design criteria were provided for static stability, control effectiveness, and cross coupling. A process to compute the total forces and moments based on the summation of the individual contributions (e.g., body, wing, tail) was demonstrated. New aerodynamic technologies were identified that have high payoff for tactical missiles. These include faceted/window/multi-lens domes, bank-to-turn maneuvering, lifting body airframe, forward swept surfaces, neutral static margin, lattice tails, split canard control, and free-to-roll tails.

Problems

- 2.1. Missile diameter tradeoffs include consideration of seeker range, warhead lethality, structural mode frequency, and d_____.
- 2.2. Benefits of a high-fineness nose include lower supersonic drag and lower r_____ c_____ s_____.
- 2.3. Three contributors to drag are base drag, wave drag, and s_____ f_____ drag.
- 2.4. To avoid flow separation, a boattail or flare angle should be less than_____ deg.
- 2.5. A lifting body is most efficient at a d_____ p_____ of about 700 psf.
- 2.6. At low angle of attack the aerodynamic center of the body is on the n_____.
- 2.7. Subsonic missiles often have w_____ for enhanced range.
- 2.8. The aerodynamic center of the wing is between 25% and 50% of the m_____ a_____ c_____.

- 2.9. Hinge moment increases with the local flow angle from control surface deflection and the a_____ o_ a_____.
- 2.10. Increasing the surface area increases the s_____ f_____ d_____.
- 2.11. Leading edge sweep reduces drag and r_____ c_____ s_____.
- 2.12. A missile with six control surfaces, four surfaces providing combined pitch/yaw control plus two surfaces providing roll control, has an advantage of good c_____ e_____.
- 2.13. A missile with two control surfaces providing only combined pitch/yaw control has advantages of lower c_____ and good p_____.
- 2.14. A tail control missile has larger trim normal force if it is statically u_____.
- 2.15. Lattice fins have low h_____ m_____.
- 2.16. Split canards allow higher maximum angle of attack and higher m_____.
- 2.17. Two types of unconventional control are thrust vector control and r_____ j_____ control.
- 2.18. The most common type of TVC for tactical missiles is j_____ v_____ control.
- 2.19. Three maneuver laws are skid to turn, bank to turn, and r_____ a_____.
- 2.20. Bank-to-turn maneuvering is usually required for missiles with a single wing or with a_____ inlets.
- 2.21. A missile is statically stable if the aero center is behind the c_____ o_ g_____.
- 2.22. Tail stabilizers have low drag while a f_____ stabilizer has low aero heating and a relatively small shift in static stability.
- 2.23. If the moments on the missile are zero the missile is in t_____.
- 2.24. Total normal force on the missile is approximately the sum of the normal forces on the surfaces (e.g., wing, tail, canard) plus normal force on the b_____.
- 2.25. Increasing the tail area increases the s_____ m_____.